STPM Physics · Semester 1
Mechanics & Heat

FULL NOTES

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12 Heat Transfer

PhysiQ · ellyelias.github.io/physiq · Worked examples, exercises with answers and the simulations from the app.

Chapter 1 · Semester 1

PHYSICAL QUANTITIES AND MEASUREMENTS

Notes by Cikgu Suhaili

29 worked examples 1 simulation Exercises with answers

1.1 Dimensions of physical quantities

Physical quantity

is a quantity that can be **observed and measured**.

Basic (base) quantity

is a quantity that **cannot be derived from any other quantity**.

Derived quantity

is a quantity that is **derived from a combination of base quantities** by mathematical operations (multiplication and division).

SI units

- In 1960 the General Conference of Weights and Measures recommended one metric system of measurement for everyone: the International System of Units, abbreviated **SI** in all languages.
- SI comes from its French name, **Système International**.
- **Prefixes** represent larger or smaller values.

Table 1.1 SI base units

Base quantity	Unit	Symbol
length	metre	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

Table 1.2 Prefixes for powers of ten

Power	Prefix	Symbol
10^{-12}	pico	p
10^{-9}	nano	n
10^{-6}	micro	μ
10^{-3}	milli	m
10^{-2}	centi	c
10^{-1}	deci	d
10^{3}	kilo	k
10^{6}	mega	M
10^{9}	giga	G

Table 1.3 Derived units

Quantity	Unit	In terms of base units
Velocity, v	m s^{-1}	m s^{-1}
Acceleration, a	m s^{-2}	m s^{-2}
Frequency, f	hertz, Hz	s^{-1}
Force, F	newton, N	kg m s^{-2}
Energy and work, W	joule, J	kg m^2 s^{-2}
Power, P	watt, W	kg m^2 s^{-3}
Pressure, p	pascal, Pa	kg m^{-1} s^{-2}

Unit conversion

Unit conversion

is a **procedure** of changing a physical quantity from one system of units to another. It is sometimes necessary, or more useful, to change from one set of units to another.

Conversions in the metric (mks) system you must be able to do:

$\text{km h}^{-1} \rightarrow \text{m s}^{-1}$ | $\text{mm}^3 \rightarrow \text{m}^3$ | $\text{g cm}^{-3} \rightarrow \text{kg m}^{-3}$ | $\text{K} \leftrightarrow \text{°C}$

Example 1.1

The speed of a car is **72 km/h**. What is the speed in $\textbf{m s}^{-1}$?

Solution

$$72\ \text{km h}^{-1} = \frac{72\ \text{km}}{1\ \text{hour}} = \frac{72\ (1000\ \text{m})}{3600\ \text{s}} = \mathbf{20\ m\,s^{-1}}$$

Example 1.2

The density of lead is 11.3 g cm^{-3}. What is the value in kilograms per cubic metre?

Solution

$$11.3\ \frac{\text{g}}{\text{cm}^3} = \frac{11.3 \times 10^{-3}\ \text{kg}}{(10^{-2}\ \text{m})^3} = \mathbf{1.13 \times 10^4\ kg\,m^{-3}}$$

Example 1.3

How many nanoseconds does it take for light to travel 1.00 km in vacuum?

Solution

$$\text{time} = \frac{\text{distance}}{\text{speed}} = \frac{10^3\ \text{m}}{3.00 \times 10^8\ \text{m s}^{-1}} = 3.33 \times 10^{-6}\ \text{s} = \mathbf{3.33 \times 10^3\ ns}$$

Exercises: unit conversion

1. A rectangular metal plate of thickness 2.5 mm is 50 mm long and 65 mm wide. Its mass is 60 g. Calculate (a) the distance round its edge, in metres, (b) the area of one of its flat faces, in m^2, (c) its volume, in m^3, (d) its density, in kg m^{-3}.

   Answer

   (a) 0.230 m (b) 3.25×10^{-3} m^2 (c) 8.13×10^{-6} m^3 (d) 7380 kg m^{-3}

2. A solid aluminium rod of length 0.2 m has a uniform diameter of 52 mm. The density of aluminium is 2700 kg m^{-3}. Calculate (a) its volume, in m^3, (b) its mass, in kg.

   Answer

   (a) 4.25×10^{-4} m^3 (b) 1.15 kg

3. (a) Calculate the volume of a steel sphere of diameter 12 mm. (b) Calculate the diameter of a steel sphere of volume 4.80 cm^3.

   Answer

   (a) 9.0×10^{-7} m^3 (b) 2.09×10^{-2} m

Dimensions

> **Dimension**
> of a physical quantity shows **how that quantity is related to the base quantities**.

- Symbol: [quantity] or [symbol of quantity].
- Example: [mass] = M or $[m]$ = M. We say "the dimension of mass is M".

Base quantity	Dimension
[length] or $[l]$	L
[mass] or $[m]$	M
[time] or $[t]$	T
[current] or $[I]$	I
[temperature] or $[T]$	θ
[luminous intensity]	J
[amount of substance]	N

Dimensionless quantities

are quantities that have **no dimension and no unit**. All numerical values (numbers) and coefficients have no dimensions.

$$[2] = [\pi] = [\mu] = 1$$

Example 1.4

What are the dimensions of volume and velocity?

Solution

[volume] = [length] × [width] × [height] = L × L × L = L^3

[velocity] = [displacement] ÷ [time] = $\mathrm{L\,T^{-1}}$

Exercise: dimensions

1. Find the dimensions of (a) density, (b) pressure, (c) work, (d) power.

Answer

(a) $\mathrm{M\,L^{-3}}$ **(b)** $\mathrm{M\,L^{-1}\,T^{-2}}$ **(c)** $\mathrm{M\,L^{2}\,T^{-2}}$ **(d)** $\mathrm{M\,L^{2}\,T^{-3}}$

Applications of dimensions

Dimensional analysis can be used to

- **(a) verify (check) the homogeneity of an equation**. However, an equation that is dimensionally correct is not necessarily correct.
- **(b) construct an equation** from the given quantities.
- **(c) determine the dimension and unit** of a physical quantity in an equation.

Teacher's note

Dimension on the left-hand side = dimension on the right-hand side, and **each term in the equation has the same dimension**.

Rules for adding, multiplying and dividing

Addition or subtraction: A + B = C − D

$$[A] = [B] = [C] = [D]$$

Multiplication: A = B × C

$$[A] = [B] \times [C]$$

[B] and [C] may be equal or different. Distance $x = vt$: $[x] = [v][t]$ although $[v] \neq [t]$.

Division: $A = \frac{B}{C} \Rightarrow [A] = \frac{[B]}{[C]}$, and in general [B] ≠ [C] ≠ [A].

Example 1.5

Which is correct: (a) 2 m + 3 m = 8 m − 3 m or (b) 2 s + 3 kg = 8 m − 3 cm?

Solution

> **(a)** is correct: every term is a length, so every term has the same dimension, L. In (b) a time is added to a mass, which is meaningless.

Example 1.6

Multiplication: $1\ m^2 = 1\ m \times 1\ m$, and distance $x = vt$.

Solution

> $[x] = [v][t] = (L\,T^{-1})(T) = L$, even though $[v] \neq [t]$.

Example 1.7

Division: (i) $1\ \text{m}^2 = \dfrac{1\ \text{m}^3}{1\ \text{m}}$ (ii) $1\ \text{m s}^{-1} = \dfrac{1\ \text{m}}{1\ \text{s}}$

Solution

> The dimension of a quotient is the quotient of the dimensions: $\dfrac{\text{L}^3}{\text{L}} = \text{L}^2$ and $\dfrac{\text{L}}{\text{T}} = \text{L T}^{-1}$.

(a) To verify (check) the homogeneity of an equation

Step 1. Identify all the physical quantities in the equation.

Step 2. Write the dimension of each known physical quantity.

Step 3. Equate the dimensions of the quantities on both sides of the equation.

Step 4. The equation is **homogeneous** if the dimensions on both sides are **equal**.

Example 1.8

Show that the equation $v^2 = u^2 + 2as$ is homogeneous.

Solution

> $[v]^2 = [\text{velocity}]^2 = \dfrac{[\text{displacement}]^2}{[\text{time}]^2} = \dfrac{\text{L}^2}{\text{T}^2} = \text{L}^2\,\text{T}^{-2}$
>
> $[u]^2 = [\text{velocity}]^2 = \text{L}^2\,\text{T}^{-2}$
>
> $[2][a][s] = (1)[\text{acceleration}][\text{displacement}] = \dfrac{[\text{velocity}]}{[\text{time}]}[\text{displacement}] = \dfrac{\text{L T}^{-1}}{\text{T}}\,\text{L} = \text{L}^2\,\text{T}^{-2}$
>
> $[v]^2 = [u]^2 = [2][a][s]$: each term on both sides has the same dimension, so the equation is **dimensionally correct (homogeneous)**.

(b) To construct an equation from given quantities

Step 1. Write a general equation that relates all the quantities.

Step 2. Identify the dimensionless quantities.

Step 3. Write the dimension of every quantity on both sides.

Step 4. Equate the indices of each dimension on both sides.

Step 5. Determine the values of the indices.

Step 6. Rewrite the equation with the indices.

Example 1.10

Construct an expression for the period t of a simple pendulum which shows how t is related to the mass m of the bob, the length l of the string and the acceleration due to gravity g.

Solution

Step 1: let $t \propto m^x l^y g^z$, so $t = k\, m^x l^y g^z$

Step 2: k is a dimensionless constant; x, y, z are unknowns.

Step 3: $[t] = [k][m]^x[l]^y[g]^z \Rightarrow \mathrm{T} = \mathrm{M}^x \mathrm{L}^y (\mathrm{L}\,\mathrm{T}^{-2})^z = \mathrm{M}^x \mathrm{L}^{y+z} \mathrm{T}^{-2z}$

Step 4: M: $0 = x$; L: $0 = y + z$; T: $1 = -2z$

Step 5: $z = -\frac{1}{2}$, $y = \frac{1}{2}$, $x = 0$

Step 6: $t = k\, m^0 l^{\frac{1}{2}} g^{-\frac{1}{2}}$, so $\boldsymbol{t = k\sqrt{\frac{l}{g}}}$

Example 1.11

The frequency of a simple pendulum depends only on its **length** and the **gravitational field strength**. Use dimensions to derive a possible form of the equation for the frequency.

Solution

Let $f \propto l^x g^y$, so $f = k\, l^x g^y$ (k dimensionless; x, y unknown)

$[f] = [k][l]^x[g]^y \Rightarrow \mathrm{T}^{-1} = (1)(\mathrm{L})^x(\mathrm{L\,T^{-2}})^y = \mathrm{L}^{x+y}\,\mathrm{T}^{-2y}$

T: $-1 = -2y$; L: $0 = x + y \Rightarrow y = \tfrac{1}{2}$, $x = -\tfrac{1}{2}$

$f = k\, l^{-1/2} g^{1/2}$, so $\boldsymbol{f = k\sqrt{\dfrac{g}{l}}}$

Exercises: homogeneity and constructing equations

1. The speed of a wave on a rope is $C = \sqrt{\dfrac{F}{m}}$. Check its homogeneity using dimensional analysis.

   Answer

   It is homogeneous when m is the mass per unit length (kg m^{-1}): $[F/m] = \dfrac{\mathrm{M\,L\,T^{-2}}}{\mathrm{M\,L^{-1}}} = \mathrm{L^2\,T^{-2}}$, so $[\sqrt{(F/m)}] = \mathrm{L\,T^{-1}} = [C]$.

2. Show whether each equation is homogeneous: $v = u + at^2$ and $s = ut + \tfrac{1}{2}at^2$.

   Answer

   $v = u + at^2$ is not homogeneous ($[at^2] = \mathrm{L}$, but $[v] = \mathrm{L\,T^{-1}}$). $s = ut + \tfrac{1}{2}at^2$ is homogeneous (every term L).

3. A student suggests that the force F on a vertical wall due to a horizontal jet of liquid striking it is $F = k\rho A v$, where k is a dimensionless constant and ρ, A and v are the density of the liquid, the cross-sectional area of the jet and the speed of the jet. Is the expression dimensionally correct?

   Answer

   No. $[\rho A v] = (\mathrm{M\,L^{-3}})(\mathrm{L^2})(\mathrm{L\,T^{-1}}) = \mathrm{M\,T^{-1}}$, but $[F] = \mathrm{M\,L\,T^{-2}}$. ($F = k\rho A v^2$ would be homogeneous.)

4. The time t taken by a body to fall freely from rest through a height h depends only on h and the acceleration due to gravity g. Using dimensions, find the equation relating t with h and g.

Answer

$$t = k\sqrt{\frac{h}{g}}$$

(c) To determine the dimension and unit of a quantity in an equation

Step 1. Identify all the physical quantities in the equation.

Step 2. Write the dimension of each **known** physical quantity.

Step 3. Solve the equation to get the dimension of the **unknown** physical quantity.

Example 1.9

Find the dimension and unit of acceleration in the equation $v = u + at$.

Solution

$[v] = [u] = \mathrm{L\,T^{-1}}$ and $[t] = \mathrm{T}$

$v = u + at \therefore a = \dfrac{v - u}{t}$, so $[a] = \dfrac{[v - u]}{[t]} = \dfrac{\mathrm{L\,T^{-1}}}{\mathrm{T}}$

$[a] = \mathrm{L\,T^{-2}}$; unit of a = **m s^{-2}**

Limitations of dimensional analysis

- It cannot determine numerical constants such as 2, π, ½ or e. For $T = 2\pi\sqrt{\dfrac{l}{g}}$ it gives only $T \propto \sqrt{\dfrac{l}{g}}$.
- It cannot tell whether terms are added or subtracted. In $s = ut + \frac{1}{2}at^2$, the factor ½ cannot be obtained.
- It cannot distinguish scalar quantities from vector quantities.
- It cannot handle trigonometric functions (sin θ, cos θ, tan θ).
- A dimensionally correct equation may still be physically incorrect.

Exam note: dimensional analysis only tests the homogeneity of an equation; it cannot prove that the equation is physically correct.

STPM Practice 1.1 · Question 1

A sphere moving through a fluid experiences a viscous drag force $F = 6\pi\eta r v$, where r is the radius, v the speed and η the coefficient of viscosity. Determine the dimensions and the SI base unit of η. [4 marks]

Solution

$[F] = \mathrm{M\,L\,T^{-2}}$, $[r] = \mathrm{L}$, $[v] = \mathrm{L\,T^{-1}}$ (6π is dimensionless)

$$[\eta] = \frac{[F]}{[r][v]} = \frac{\mathrm{M\,L\,T^{-2}}}{\mathrm{L \cdot L\,T^{-1}}} = \mathrm{M\,L^{-1}\,T^{-1}}$$

Dimension of η = $\mathrm{M\,L^{-1}\,T^{-1}}$; SI unit = **kg m^{-1} s^{-1}**

STPM Practice 1.1 · Question 2

Using dimensional analysis, show that $p = \rho g h + \tfrac{1}{2}\rho v^2$ is homogeneous, where p is pressure, ρ density, g the acceleration of free fall, h height and v speed. [4 marks]

Solution

$$[p] = \text{force/area} = \frac{\mathrm{M\,L\,T^{-2}}}{\mathrm{L^2}} = \mathrm{M\,L^{-1}\,T^{-2}}$$

$[\rho g h] = (\mathrm{M\,L^{-3}})(\mathrm{L\,T^{-2}})(\mathrm{L}) = \mathrm{M\,L^{-1}\,T^{-2}}$

$[\tfrac{1}{2}\rho v^2] = (\mathrm{M\,L^{-3}})(\mathrm{L\,T^{-1}})^2 = \mathrm{M\,L^{-1}\,T^{-2}}$

All three terms have the same dimension, $\mathrm{M\,L^{-1}\,T^{-2}}$, so the equation is **homogeneous**.

STPM Practice 1.1 · Question 3

The period T of a small liquid drop oscillating under surface tension may depend on its density ρ, radius r and surface tension σ (SI unit N m^{-1}). Use dimensional analysis to derive an expression for T.

[6 marks]

Solution

Let $T = k\rho^a r^b \sigma^c$ (k dimensionless)

$$[\sigma] = \text{N m}^{-1} = \frac{\text{M L T}^{-2}}{\text{L}} = \text{M T}^{-2}$$

$$\text{T} = (\text{M L}^{-3})^a(\text{L})^b(\text{M T}^{-2})^c = \text{M}^{a+c}\,\text{L}^{b-3a}\,\text{T}^{-2c}$$

M: $a + c = 0$; T: $-2c = 1$; L: $b - 3a = 0 \Rightarrow c = -\tfrac{1}{2}$, $a = \tfrac{1}{2}$, $b = 3/2$

$$T = k\rho^{1/2} r^{3/2} \sigma^{-1/2} = k\sqrt{\frac{\rho r^3}{\sigma}}$$

1.2 Scalars and vectors

Scalar quantity

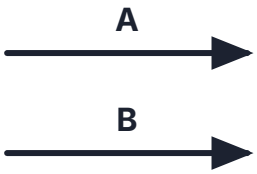
is a quantity which has **magnitude only**. Examples: mass, temperature, kinetic energy, time and density.

Vector quantity

is a quantity which has **both magnitude and direction**. Examples: displacement, velocity, acceleration, force and momentum.

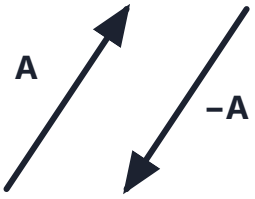
Properties of two vectors

(a) Equality of two vectors: two vectors are equal only when they have the **same magnitude** and the **same direction**.



$A = B$
(parallel)

(b) Negative of a vector: it has the **same magnitude** as the original vector but the **opposite direction**.



−A: same size, opposite direction
(antiparallel)

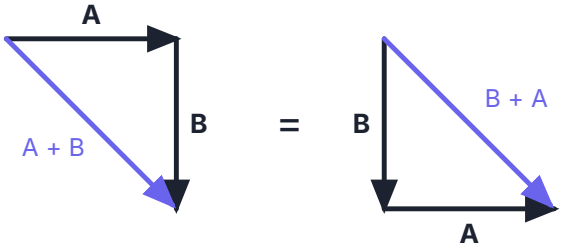
Vector addition

Two methods of finding the resultant vector:

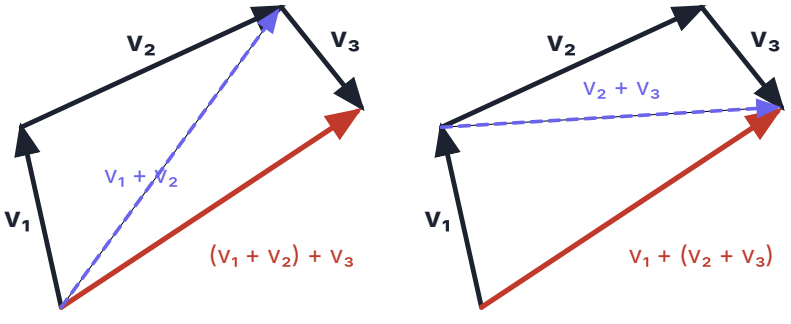
- **(a) Graphical method**: (i) tail to tip (triangle and polygon methods); (ii) parallelogram.
- **(b) Component method**: by calculation.
- The graphical (tail-to-tip) method obeys the **commutative**, **associative** and **distributive** laws.

Commutative law: $A + B = B + A$

The arrow drawn from the **tail** of the first vector to the **tip** of the second represents the sum (resultant). This is the **triangle method**.

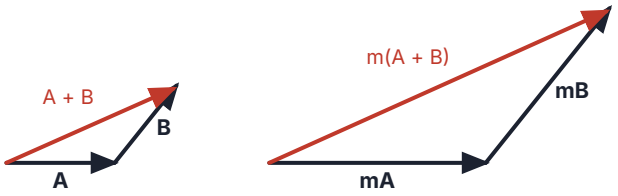


Associative law (more than two vectors): $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$. This is the **polygon method**.

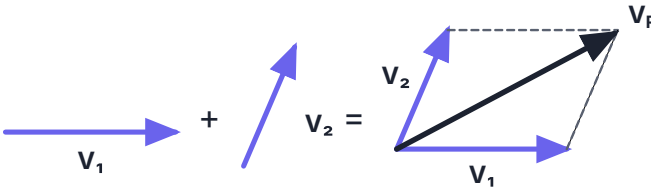


Either grouping gives the same resultant (red).

Distributive law: $m(A + B) = mA + mB$



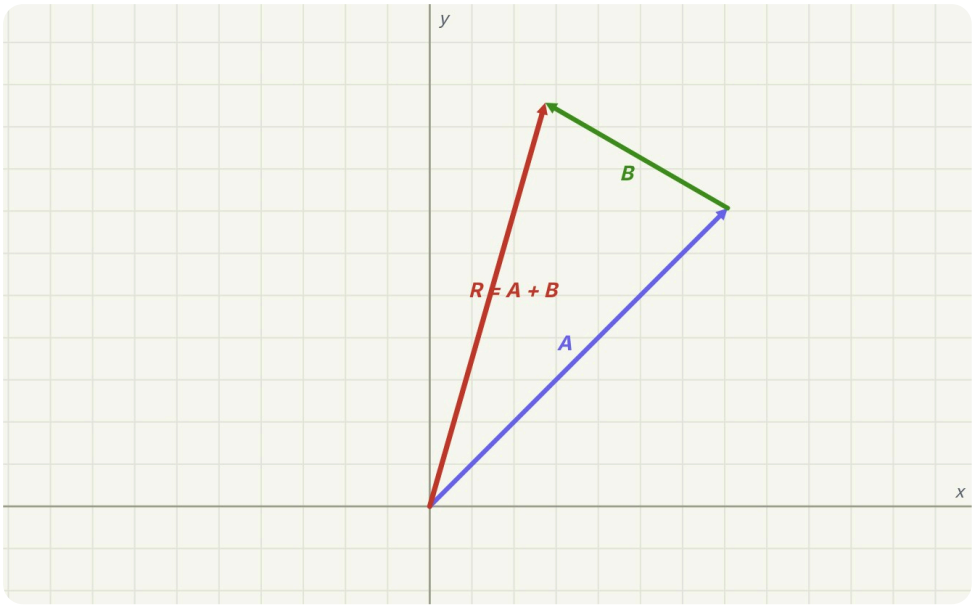
Parallelogram method (for two vectors): V_R is the **diagonal** of the parallelogram and represents the sum, or resultant, of the two vectors.



Simulation

Vector addition: tail to tip, parallelogram and components

Drag the sliders to change the two vectors. Switch between the triangle and parallelogram methods, and watch the component table add up to the same resultant.



$\lvert R\rvert = \sqrt{(R_x^2 + R_y^2)}$	direction from +x axis	angle between A and B	$A \cdot B = AB \cos\theta$
9.96	**74.0°**	**105°**	**−12.94**

$\lvert A \times B\rvert = AB \sin\theta$
48.30

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 1 to use the simulation.

Scalar (dot) product and vector (cross) product

Scalar (dot) product
of two vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ is a **scalar**.

$$\boldsymbol{A} \cdot \boldsymbol{B} = AB\cos\theta$$

θ is the angle between $\boldsymbol{A}$ and $\boldsymbol{B}$ (0° ≤ θ ≤ 180°). Component form: $\boldsymbol{A} \cdot \boldsymbol{B} = A_x B_x + A_y B_y$.

- $\boldsymbol{A} \cdot \boldsymbol{B} = \boldsymbol{B} \cdot \boldsymbol{A}$ (commutative)
- $\boldsymbol{A} \cdot \boldsymbol{B} = 0$ when $\boldsymbol{A} \perp \boldsymbol{B}$ (θ = 90°)
- Maximum when θ = 0° (parallel); $\boldsymbol{A} \cdot \boldsymbol{A} = A^2$

Vector (cross) product
of two vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ is a **vector**.

$$|\boldsymbol{A} \times \boldsymbol{B}| = AB\sin\theta$$

Its direction is perpendicular to the plane of $\boldsymbol{A}$ and $\boldsymbol{B}$, given by the right-hand rule. In the x-y plane: $\boldsymbol{A} \times \boldsymbol{B} = (A_x B_y - A_y B_x)\,\boldsymbol{k}$.

- $\boldsymbol{A} \times \boldsymbol{B} = -(\boldsymbol{B} \times \boldsymbol{A})$ (anti-commutative)
- $\boldsymbol{A} \times \boldsymbol{B} = 0$ when $\boldsymbol{A} \parallel \boldsymbol{B}$ (θ = 0° or 180°)
- Maximum when θ = 90°; $|\boldsymbol{A} \times \boldsymbol{B}|$ = area of the parallelogram

Worked example · scalar product

Two coplanar vectors $\boldsymbol{A}$ and $\boldsymbol{B}$ have magnitudes A = 5.0 N and B = 4.0 N. The angle between them is 60°. Find $\boldsymbol{A} \cdot \boldsymbol{B}$.

Solution

$\boldsymbol{A} \cdot \boldsymbol{B} = AB\cos\theta = (5.0)(4.0)\cos 60° = 20 \times 0.5 =$ **10** (a scalar)

Worked example · vector product

For the same two vectors (5.0 N and 4.0 N at 60°), find the magnitude of $\boldsymbol{A} \times \boldsymbol{B}$.

Solution

$|\boldsymbol{A} \times \boldsymbol{B}| = AB\sin\theta = (5.0)(4.0)\sin 60° = 20 \times 0.866 =$ **17.3** (a vector)

Direction: perpendicular to the plane of $\boldsymbol{A}$ and $\boldsymbol{B}$ (right-hand rule).

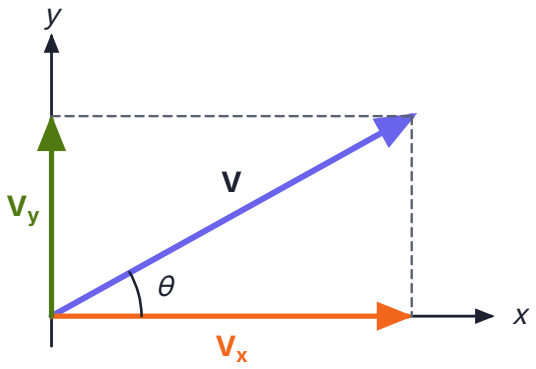
Scalar product	Vector product
$A \cdot B = AB \cos \theta$	$\lvert A \times B \rvert = AB \sin \theta$
Result is a scalar	Result is a vector
Used in work done	Used in torque
$A \cdot B = B \cdot A$	$A \times B \neq B \times A$
Zero when $\theta = 90°$	Zero when $\theta = 0°$
Maximum at $\theta = 0°$	Maximum at $\theta = 90°$

Component method

A vector V is resolved into x and y components, V_x and V_y. From simple trigonometry:

$$V_x = V \cos \theta$$

$$V_y = V \sin \theta$$



The **magnitude** of the resultant:

$$V = \sqrt{V_x^2 + V_y^2}$$

The **direction** of the resultant is the angle θ between V and the positive x-axis:

$$\tan \theta = \frac{V_y}{V_x}$$

Example 1.12

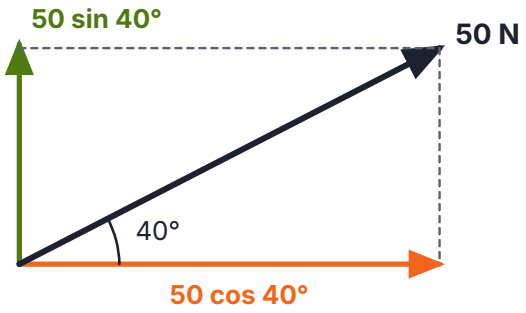
Calculate the horizontal and vertical components of a force of 50 N acting at 40° to the horizontal.

Solution

Step 1: sketch the force on the Cartesian axes. **Step 2**: find the x and y components.

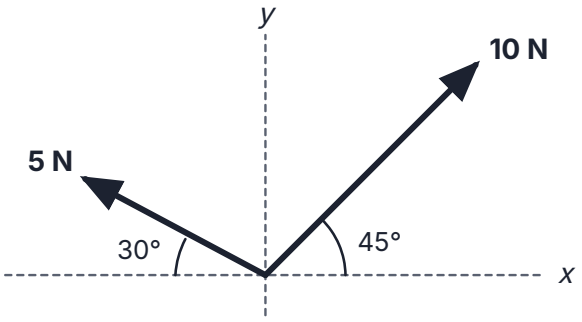
Horizontal component = 50 cos 40° = **38 N**

Vertical component = 50 sin 40° = **32 N**



Example 1.13

Find the resultant of the forces shown.



Solution

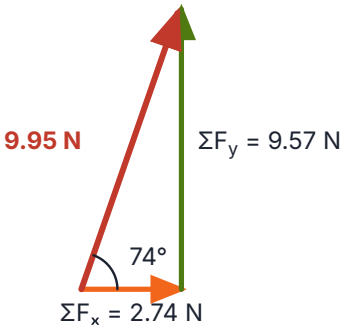
Step 1: find the x and y components of each force.

Force	x-component	y-component
10 N	10 cos 45° = +7.07 N	10 sin 45° = +7.07 N
5 N	−5 cos 30° = −4.33 N	5 sin 30° = +2.50 N

Step 2: ΣF_x = +7.07 − 4.33 = 2.74 N; ΣF_y = +7.07 + 2.50 = 9.57 N

Step 3: $F = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2} = \sqrt{(2.74)^2 + (9.57)^2}$ = **9.95 N**

Step 4: $\theta = \tan^{-1}\frac{9.57}{2.74}$ = 74°. **The resultant is 9.95 N at 74° to the positive x-axis.**



STPM Practice 1.2 · Question 1

Two forces of 8.0 N and 6.0 N act at a point. The angle between them is 60°. Determine the magnitude and direction of the resultant force. [5 marks]

Solution

Resolve along the 8.0 N force (x-axis): $F_x = 8.0 + 6.0\cos 60° = 11.0$ N; $F_y = 6.0 \sin 60° = 5.20$ N

$R = \sqrt{11.0^2 + 5.20^2} = \sqrt{148} = \mathbf{12.2\ N}$

$\theta = \tan^{-1}\dfrac{5.20}{11.0} = \mathbf{25.3°}$ from the 8.0 N force

STPM Practice 1.2 · Question 2

Two vectors are $\boldsymbol{A} = 3\boldsymbol{i} + 4\boldsymbol{j}$ and $\boldsymbol{B} = 5\boldsymbol{i} - 12\boldsymbol{j}$. Find (a) the scalar product $\boldsymbol{A} \cdot \boldsymbol{B}$, (b) the angle between $\boldsymbol{A}$ and $\boldsymbol{B}$. [5 marks]

Solution

(a) $\boldsymbol{A} \cdot \boldsymbol{B} = A_xB_x + A_yB_y = (3)(5) + (4)(-12) = 15 - 48 = \mathbf{-33}$

(b) $|\boldsymbol{A}| = \sqrt{3^2 + 4^2} = 5$; $|\boldsymbol{B}| = \sqrt{5^2 + 12^2} = 13$

$\cos\theta = \dfrac{-33}{(5)(13)} = -0.508 \Rightarrow \theta = \mathbf{120.5°}$

STPM Practice 1.2 · Question 3

A force $\boldsymbol{F} = (3\boldsymbol{i} + 4\boldsymbol{j})$ N acts at a point with position vector $\boldsymbol{r} = (2\boldsymbol{i} - \boldsymbol{j})$ m from a fixed axis. Using the vector product, calculate the torque of the force about the axis. [4 marks]

Solution

$\tau = \boldsymbol{r} \times \boldsymbol{F} = (r_x F_y - r_y F_x)\,\boldsymbol{k}$

$= [(2)(4) - (-1)(3)]\,\boldsymbol{k} = (8 + 3)\,\boldsymbol{k} =$ **11 $\boldsymbol{k}$ N m**

Magnitude 11 N m, directed out of the plane (anticlockwise rotation).

1.3 Uncertainties in measurements

Learning outcome (a): explain systematic errors and random errors in measurements.

The error in a measurement is decided by the instruments used and the physical constraints of the observer. The two main types are **systematic error** and **random error**.

Random error

(chance error) is a deviation between an observed value and the true value due to **unpredictable and fluctuating factors** during measurement.

- It makes readings **scatter** about the true value.
- Individual readings may be higher or lower than the true value.
- **Reduced** by taking repeated measurements and using the mean, and by increasing the sample size.

Common sources: parallax errors, the sensitivity of an instrument, the physical constraints of the observer, poorly controlled procedures.

Systematic error

is an error that **occurs consistently**, or changes in a predictable way, every time a measurement is made.

- All readings are shifted from the true value by the same amount in the same direction.
- **Eliminated** by improving the measurement method, e.g. using a different instrument.

Common sources: zero errors, instrumental error, incorrect assumptions.

Every measurement has an uncertainty, limited by the instrument and the observer. For example, the length of A4 paper measured with a metre ruler (error 0.1 cm) is written **29.7 ± 0.1 cm**.

(b) Uncertainty from repeated measurements: standard deviation

Standard deviation

expresses **by how much the members of a group differ from the mean value** of the group.

$$\sigma = \sqrt{\frac{\Sigma(x - \bar{x})^2}{N - 1}}$$

Standard deviation · Example

The diameter of a boiling tube is measured five times: 31.33, 31.15, 31.26, 31.02 and 31.20 mm. Find the mean, the average deviation and the standard deviation.

Solution

$$\text{Mean} = \frac{\text{sum of readings}}{\text{number of readings}} = \frac{155.96 \text{ mm}}{5} = \mathbf{31.19 \text{ mm}}$$

Reading	Diameter / mm	Deviation / mm
1	31.33	+0.14
2	31.15	−0.04
3	31.26	+0.07
4	31.02	−0.17
5	31.20	+0.01

Average deviation, d = **0.086 mm**

$$\sigma = \sqrt{\frac{(0.14)^2 + (-0.04)^2 + (0.07)^2 + (-0.17)^2 + (0.01)^2}{5 - 1}} = \mathbf{0.12 \text{ mm}}$$

(c) Uncertainty in a derived quantity

Trigonometric functions are not required. (d) Give a derived quantity to an appropriate number of significant figures.

$$Z = A \pm B \Rightarrow \Delta Z = \Delta A + \Delta B$$

$$Z = AB \text{ or } \frac{A}{B} \Rightarrow \frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

$$Z = A^n \Rightarrow \frac{\Delta Z}{Z} = n\frac{\Delta A}{A}$$

Adding or subtracting: add the absolute uncertainties. Multiplying or dividing: add the fractional uncertainties.

Addition and subtraction · Example 1

Mass of beaker with liquid, m_1 = (32.0 ± 0.1) g; mass of beaker, m_2 = (3.0 ± 0.1) g. Calculate the mass of the liquid.

Solution

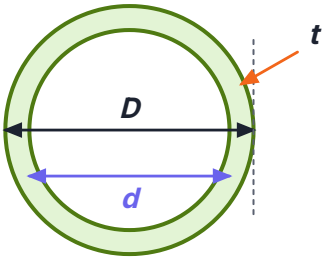
Mass of liquid = $m_1 - m_2$ = 32.0 − 3.0 = 29.0 g

Absolute uncertainty = 0.1 + 0.1 = 0.2 g

Mass of liquid = (29.0 ± 0.2) g

Addition and subtraction · Example 2

External diameter of a test tube, D = (1.23 ± 0.01) cm; internal diameter, d = (1.13 ± 0.01) cm. Calculate the thickness of the wall.



Solution

Thickness, $t = \frac{D - d}{2} = \frac{1.23 - 1.13}{2} = 0.05$ cm

$\Delta t = ½\Delta D + ½\Delta d = ½(0.01) + ½(0.01) = 0.01$ cm

Thickness = (0.05 ± 0.01) cm

Multiplication and division · Example

The dimensions of a piece of A4 paper are x = (297 ± 1) mm and y = (209 ± 1) mm. Calculate the area.

Solution

Area $A = xy = 297 \times 209 = 62\,073$ mm^2

$\frac{\Delta A}{A} = \frac{\Delta x}{x} + \frac{\Delta y}{y} \Rightarrow \Delta A = (\frac{1}{297} + \frac{1}{209})(62\,073) = 506$ mm^2 $= 5 \times 10^2$ mm^2 (1 s.f.)

Area, A = (6.21 ± 0.05) × 10^4 mm^2

STPM Practice 1.3 · Question 1

A student records five readings of the diameter of a wire (in mm): 0.52, 0.54, 0.53, 0.55, 0.51. Determine (a) the mean diameter, (b) the standard deviation, (c) the diameter with its uncertainty. [6 marks]

Solution

(a) $\bar{x} = \dfrac{0.52 + 0.54 + 0.53 + 0.55 + 0.51}{5} = \mathbf{0.530\ mm}$

(b) Deviations: −0.01, +0.01, 0.00, +0.02, −0.02; $\Sigma(x - \bar{x})^2 = 0.0010\ \text{mm}^2$

$\sigma = \sqrt{\dfrac{0.0010}{4}} = \mathbf{0.016\ mm}$

(c) **Diameter = (0.530 ± 0.016) mm**

STPM Practice 1.3 · Question 2

A small cube has mass m = (24.0 ± 0.1) g and side L = (2.00 ± 0.01) cm. Calculate its density ρ = m/L^3 and the uncertainty in the density. [6 marks]

Solution

$V = L^3 = 8.00\ \text{cm}^3$; $\rho = \dfrac{24.0}{8.00} = 3.00\ \text{g cm}^{-3}$

$\dfrac{\Delta\rho}{\rho} = \dfrac{\Delta m}{m} + 3\dfrac{\Delta L}{L} = \dfrac{0.1}{24.0} + 3\left(\dfrac{0.01}{2.00}\right) = 0.00417 + 0.0150 = 0.0192$

$\Delta\rho = 0.0192 \times 3.00 = 0.06\ \text{g cm}^{-3}$; **ρ = (3.00 ± 0.06) g cm^{-3}**

STPM Practice 1.3 · Question 3

In a cooling experiment the initial temperature is $\theta_1 = (75.5 \pm 0.5)$ °C and the final temperature is $\theta_2 = (28.0 \pm 0.5)$ °C. Calculate the temperature change and its uncertainty, to an appropriate number of significant figures. [4 marks]

Solution

$\Delta\theta = \theta_1 - \theta_2 = 75.5 - 28.0 = 47.5$ °C

For subtraction the absolute uncertainties add: $0.5 + 0.5 = 1.0$ °C

$\Delta\theta = (47.5 \pm 1.0)$ °C

STPM specimen question

In determining the density of a solid sphere, a student measures its diameter and its mass. The percentage errors in the diameter and the mass are 2.0% and 3.0% respectively. What is the percentage error in the density of the sphere?

A. 5.0%

B. 8.0%

C. 9.0%

D. 11%

Answer

Answer: C

$\rho = \frac{m}{V}$ and $V = \frac{4}{3}\pi r^3$. Since $r = d/2$, $V \propto d^3$, so $\rho \propto \frac{m}{d^3}$

$\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + 3\frac{\Delta d}{d} = 3.0\% + 3(2.0\%) =$ **9.0%**

STPM examination tips

- ✔ **Dimensions**: express dimensions using base quantities (M, L, T) only.
- ✔ **Homogeneity**: every term in an equation must have the same dimensions.
- ✔ **Constructing equations**: equate the indices of M, L and T separately.
- ✔ **Uncertainties**: for × and ÷, add the percentage uncertainties.
- ✔ **Significant figures**: the answer is no more accurate than the least accurate measurement.

Common mistakes

- ✖ mixing dimensions with units
- ✖ using T for the temperature dimension (it is θ)
- ✖ forgetting powers in uncertainty calculations
- ✖ assuming a homogeneous equation is automatically correct

Chapter 2 · Semester 1

KINEMATICS

Notes by Cikgu Suhaili

21 worked examples 3 simulations Exercises with answers

2.0 Introduction

Kinematics

is the study of **motion without regard to mass or force**. (From the Greek word for *motion*.)

In this chapter we study motion along

- **a straight line** (one-dimensional motion): linear motion, including **free fall**
- **a curved path** (two-dimensional motion): **projectile motion**

2.1 Linear motion

Distance and displacement

Distance, s

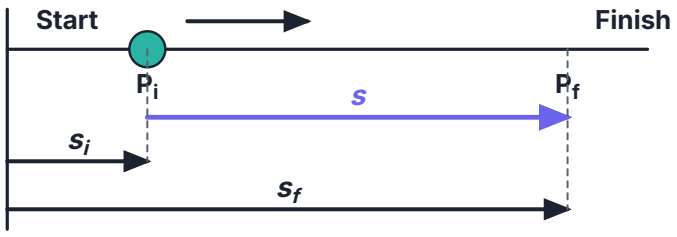
is the **length of the path** between two specified moments in time. It is a **scalar**.

Displacement, $\vec{s}$

is the **difference between the final position and the initial position** of a moving object. It is a **vector**.

$$s = s_f - s_i$$

where s_f = final position and s_i = initial position.



Speed and velocity

Speed, v

is the **rate of change of distance** (change of distance per unit time). Scalar.

$$v = \frac{\mathrm{d}s}{\mathrm{d}t}$$

Velocity, $\vec{v}$

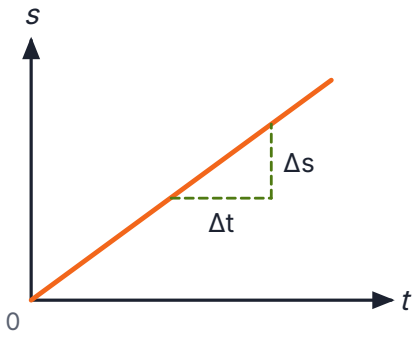
is the **rate of change of displacement** (change of displacement per unit time). Vector.

$$\vec{v} = \frac{\mathrm{d}s}{\mathrm{d}t}$$

Average speed, $\langle v \rangle$

is the **distance travelled divided by the elapsed time**.

$$\langle v \rangle = \frac{\Delta s}{\Delta t}$$



Average velocity, $\langle \vec{v} \rangle$

is the **displacement divided by the elapsed time**, in the same direction as the displacement.

$$\langle \vec{v} \rangle = \frac{\Delta s}{\Delta t} = \frac{s_\mathrm{f} - s_\mathrm{i}}{t_\mathrm{f} - t_\mathrm{i}}$$

Average velocity = gradient of the displacement–time graph:

$$\langle \vec{v} \rangle = \frac{\Delta s}{\Delta t}$$

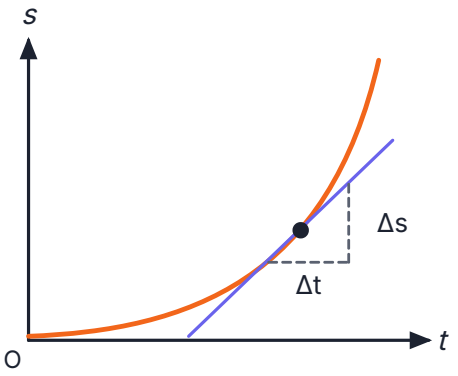
A **straight-line** s–t graph means the object moves with **uniform velocity**, so the velocity at every instant equals the average velocity.

Instantaneous velocity

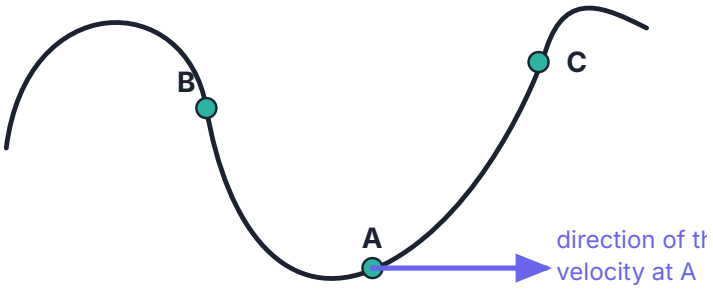
is the limit of the average velocity as the time interval approaches zero; it equals the instantaneous rate of change of displacement with time.

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{\mathrm{d}s}{\mathrm{d}t}$$

= **gradient of the tangent** to the s–t graph at that instant.



- Instantaneous velocity is a **vector**.
- Its **direction** at any instant is ***along the tangent*** to the path at that point (e.g. point A).
- Its **magnitude** at any instant is equal to the ***speed*** at that point.



Uniform velocity

Velocity is uniform (**constant**) if (a) its magnitude is constant and (b) its direction is unchanged. The average and instantaneous velocities are then equal.

Acceleration

Acceleration, $\vec{a}$

is the **rate of change of velocity**.

$$\vec{a} = \frac{\mathrm{d}v}{\mathrm{d}t}$$

- **Acceleration** is the rate of ***increase*** of velocity: acceleration and velocity point in the same direction (positive value).
- **Deceleration** is the rate of ***decrease*** of velocity: acceleration and velocity point in opposite directions (negative value).

Average acceleration, $\langle\vec{a}\rangle$

is the **change in velocity divided by the time taken**. It shows how the speed and its direction change with time.

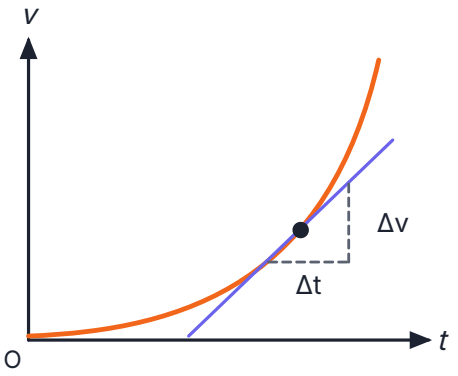
$$\langle\vec{a}\rangle = \frac{\Delta v}{\Delta t} = \frac{v_\mathrm{f} - v_\mathrm{i}}{t_\mathrm{f} - t_\mathrm{i}}$$

Instantaneous acceleration

is the limit of the average acceleration as the time interval approaches zero; it equals the instantaneous rate of change of velocity with time.

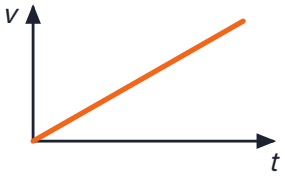
$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$

= **gradient of the tangent** to the v–t graph.



Uniform acceleration

Acceleration is uniform (**constant**) if (a) its magnitude is constant and (b) its direction is unchanged. The average and instantaneous accelerations are then equal.



Velocity increases uniformly: gradient constant, so a is uniform.

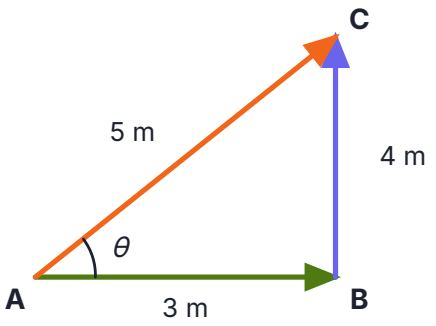


Constant velocity: gradient = 0, so $a = 0$.

Worked examples: distance, displacement, speed and velocity

Example 2.1.1

An object moves 3 m from A to B, then 4 m from B to C (at right angles). Find its distance and displacement.



Solution

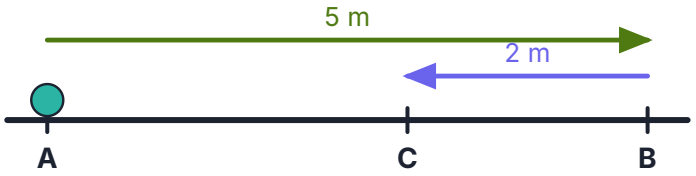
Distance = AB + BC = 3 + 4 = **7 m**

Displacement: magnitude AC = $\sqrt{3^2 + 4^2}$ = **5 m**

Direction: θ = $\cos^{-1}\frac{3}{5}$ = **53.1°** from AB

Example 2.1.2

An object moves from A to B (5 m) and then from B back to C (2 m). What is its distance? What is its displacement?



Solution

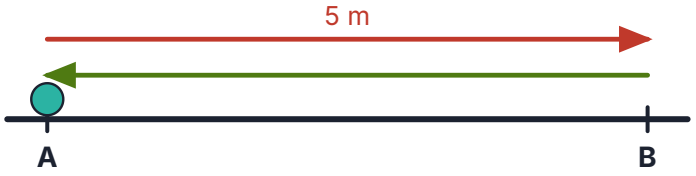
Distance = AB + BC = 5 + 2 = **7 m**

Displacement: AC = (+5) + (−2) = **+3 m**

Direction: **3 m to the right of A**

Example 2.1.3

An object moves from A to B (5 m) and then from B back to A. What are the distance and the displacement?



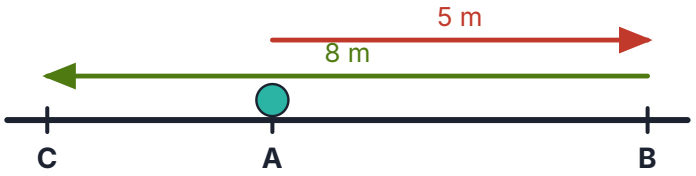
Solution

Distance = 5 + 5 = **10 m**

Displacement = (+5) + (−5) = **0 m**

Example 2.1.4

An object moves from A to B (5 m) and then from B to C (8 m). What is its distance? What is its displacement?



Solution

Distance = 5 + 8 = **13 m**

Displacement: AC = (+5) + (−8) = **−3 m**

Direction: **3 m to the left of A**

Speed, v

Scalar. Total distance moved, s, divided by the time, t.

$$v = \frac{s}{t}$$

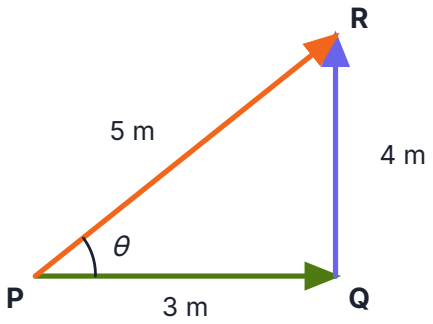
Velocity, $\vec{v}$

Vector. Displacement, s, divided by the time, t.

$$\vec{v} = \frac{s}{t} \text{ or } \vec{v} = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$$

Example 2.1.5

An object moves from P to Q (3 m) in 3 s and from Q to R (4 m, at right angles) in 2 s. What are its speed and velocity?



Solution

Distance = PQ + QR = 3 + 4 = 7 m; time = 3 + 2 = 5 s

Speed = $\frac{7}{5}$ = **1.4 m s^{-1}**

Displacement = PR = 5 m; time = 5 s

Velocity: magnitude = $\frac{5}{5}$ = **1 m s^{-1}**

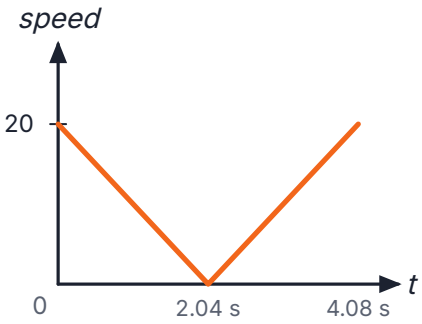
Direction: tan θ = $\frac{4}{3}$, θ = **53.1°** from PQ

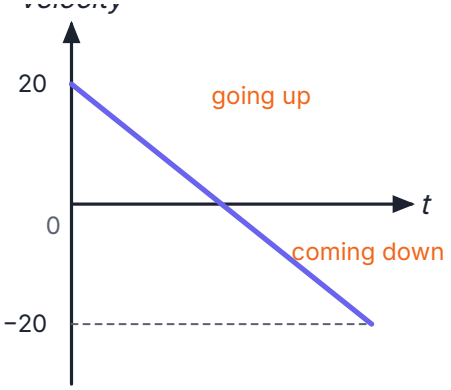
Example 2.1.6

A stone is thrown vertically upward with velocity u = 20 m s^{-1} and returns to its original position. Sketch the graphs of speed against time and velocity against time.

Solution

Going up, the speed falls from 20 m s^{-1} to 0 at the top; coming down it rises back to 20 m s^{-1}. The velocity changes uniformly from +20 m s^{-1} to −20 m s^{-1} (negative = downward); the gradient is $-g$ throughout.

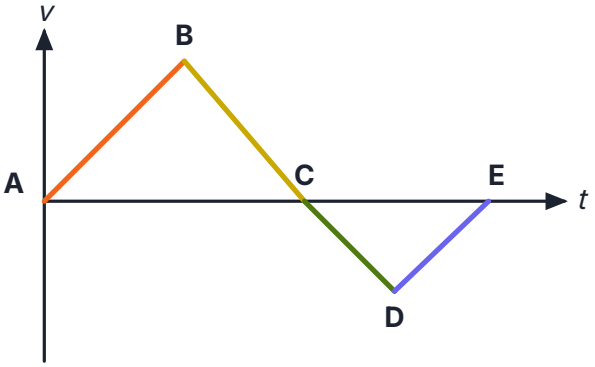




Acceleration and deceleration from a graph

Example 2.1.7

Based on the v–t graph below, describe the motion of the particle.

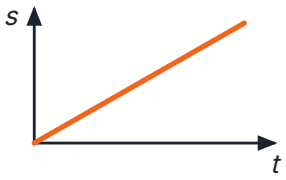
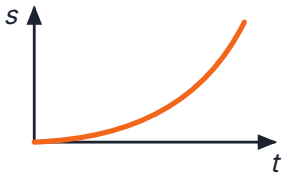
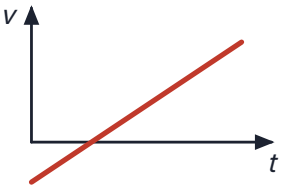
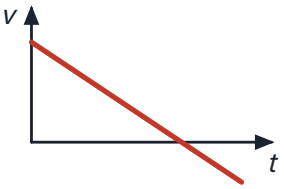
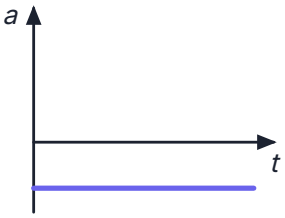


Solution

- **AB**: v increases ⇒ **acceleration** (constant)
- **BC**: v decreases ⇒ **deceleration** (constant)
- **CD**: v increases in the opposite direction ($-v$) ⇒ **acceleration**
- **DE**: v decreases, still in the opposite direction ($-v$) ⇒ **deceleration**

Graphs for linear motion

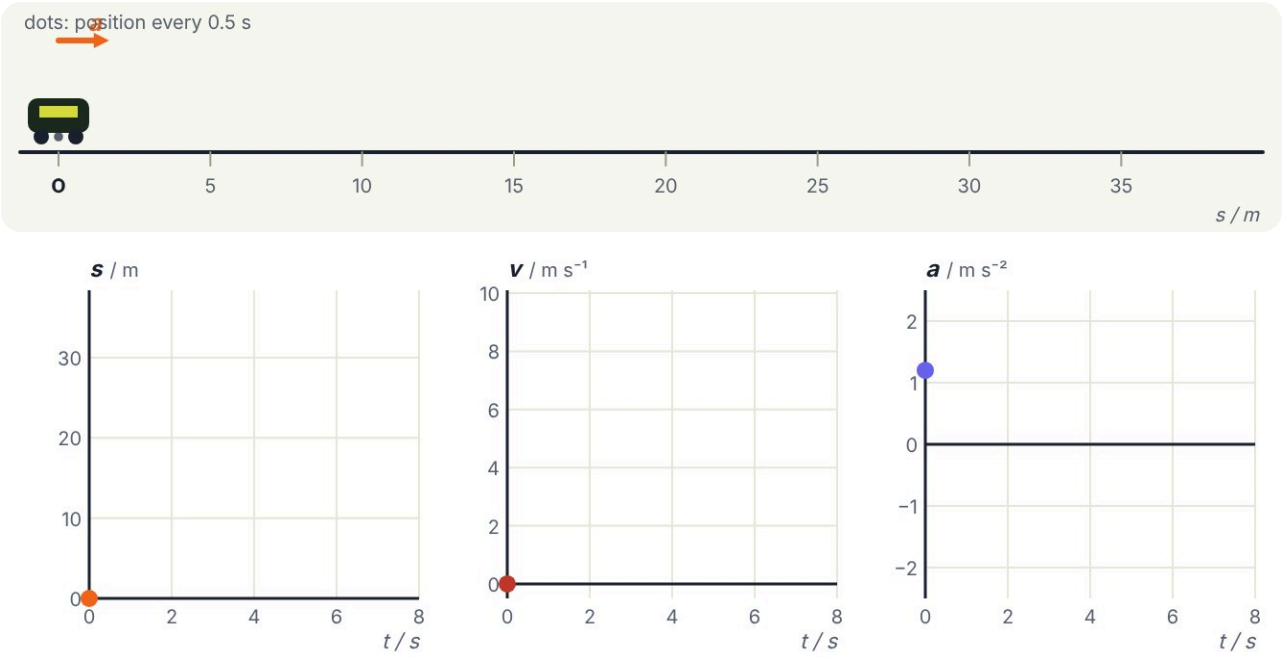
Gradient of s–t = velocity. Gradient of v–t = acceleration. Area under v–t = displacement. Area under a–t = change in velocity.

	s–t graph	v–t graph	a–t graph
Uniform velocity	 gradient = velocity (constant)	 gradient = 0, area = displacement	 a = 0
Uniform acceleration	 gradient increases: velocity increases	 velocity increases uniformly	 a positive, area = change in velocity
Uniform deceleration	 gradient decreases: velocity decreases	 velocity decreases uniformly	 a negative, area = change in velocity

Simulation

Motion graphs: watch s–t, v–t and a–t being drawn

Pick a motion, or set the initial velocity and acceleration yourself, then press Play. Turn on "Area under v–t" to see that the shaded area equals the displacement.

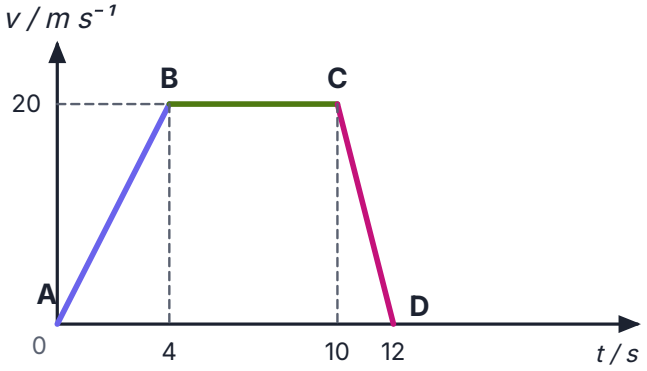


t	displacement s	velocity v	acceleration a
0.0 s	**0.0 m**	**0.0 m/s**	**1.2 m/s²**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 2 to use the simulation.

Example 2.1.8

From the v–t graph, describe the motion of the object and determine (a) the acceleration, (b) the deceleration, (c) the total distance moved.



Solution

- At A: $t = 0$, $v = 0$
- **AB**: v increases uniformly, constant acceleration
- **BC**: constant v, $a = 0$
- **CD**: v decreases uniformly, constant deceleration
- At D: $v = 0$

(a) Acceleration, $a = \dfrac{20}{4} = \mathbf{5\ m\,s^{-2}}$

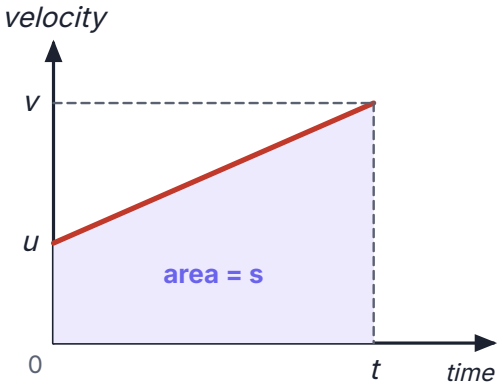
(b) Deceleration, $a = \dfrac{0 - 20}{12 - 10} = \mathbf{-10\ m\,s^{-2}}$

(c) Total distance = area under graph = $\dfrac{1}{2}[(10 - 4) + 12] \times 20 = \mathbf{180\ m}$

2.1 Equations of motion (uniform acceleration)

From the v–t graph:

- v = final velocity
- u = initial velocity
- a = acceleration
- t = time
- s = area under the graph = displacement



From $a = \frac{dv}{dt} = \frac{v-u}{t}$, rearranging: (i) $\boldsymbol{v = u + at}$

Area under the graph (trapezium): (ii) $\boldsymbol{s = \frac{1}{2}(u+v)t}$

Substitute (i) into (ii): $s = \frac{1}{2}[u + (u + at)]t$, so (iii) $\boldsymbol{s = ut + \frac{1}{2}at^2}$

From (i), $t = \frac{v-u}{a}$. Substitute into (iii): $s = u(\frac{v-u}{a}) + \frac{1}{2}a(\frac{v-u}{a})^2$

$2as = 2uv - 2u^2 + (v^2 - 2uv + u^2)$, so (iv) $\boldsymbol{v^2 = u^2 + 2as}$

$$s = \frac{1}{2}(u+v)t$$

$$s = ut + \frac{1}{2}at^2$$

$$v = u + at$$

$$v^2 = u^2 + 2as$$

Equations of motion with uniform acceleration

Sign convention

Quantity	To the left	To the right
Displacement, s	−ve	+ve
Velocity, v	−ve	+ve

- Acceleration: +ve; deceleration: −ve.
- Special case, uniform velocity: $u = v \Rightarrow a = 0$ and $s = ut$.

Example 2.2.1

An object is moving along the x-axis. Its velocity at the starting point O is 5 m s^{-1}. If it has a uniform acceleration of 6 m s^{-2}, calculate (a) its position after 2 s, (b) its velocity after 3 s.

Solution

Given: $u = 5$ m s^{-1}, $a = 6$ m s^{-2}

(a) $s = ut + \frac{1}{2}at^2 = (5)(2) + \frac{1}{2}(6)(2)^2 =$ **22 m** (to the right)

(b) $v = u + at = 5 + (6)(3) =$ **23 m s^{-1}** (to the right)

Example 2.2.2

An airplane lands with an initial velocity of 70.0 m/s and then decelerates at 1.50 m/s² for 40.0 s. What is its final velocity?

Solution

Given: $u = 70.0$ m/s, $a = -1.50$ m/s², $t = 40.0$ s

$v = u + at = 70.0 + (-1.50)(40.0) =$ **10.0 m/s**

Example 2.2.3

A vehicle accelerates uniformly along a straight road, increasing its speed from 4.0 m/s to 30.0 m/s in 13 s. Calculate (a) its acceleration, (b) the distance it moves in this time.

Solution

(a) $a = \frac{v - u}{t} = \frac{30.0 - 4.0}{13} =$ **2.0 m/s²**

(b) $s = \frac{1}{2}(u + v)t = \frac{1}{2}(4.0 + 30.0)(13) = 221 \approx$ **220 m**

Example 2.2.4

A body with initial velocity 8.0 m/s moves along a straight line with constant acceleration and travels 640 m in 40 s. For the 40 s interval, find (a) the average velocity, (b) the final velocity, (c) the acceleration.

Solution

Given: u = 8.0 m/s, s = 640 m, t = 40 s

(a) $\langle \vec{v} \rangle = \frac{s}{t} = \frac{640}{40} =$ **16 m/s**

(b) $\langle \vec{v} \rangle = \frac{u + v}{2} \Rightarrow v = 2(16) - 8.0 =$ **24 m/s**

(c) $a = \frac{v - u}{t} = \frac{24 - 8.0}{40} =$ **0.40 m/s²**

Exercises: equations of motion

1. A vehicle on a straight downhill road accelerated uniformly from 4.0 m/s to 29 m/s over a distance of 850 m, when the driver braked and stopped the vehicle in 28 s.
   (a) Calculate (i) the time taken to reach 29 m/s from 4 m/s, (ii) its acceleration during this time.
   (b) Calculate (i) the distance it travelled during deceleration, (ii) its average deceleration as it slowed down.

   Answer

   (a)(i) 52 s (ii) 0.49 m/s² (b)(i) 406 m (ii) −1.04 m/s²

2. A rail wagon moving at 2.0 m/s on a level track reached a steady incline, which slowed it down in 15.0 s and caused it to reverse. Calculate (a) the distance it moved up the incline, (b) its acceleration on the incline, (c) its velocity and position on the incline after 20.0 s.

   Answer

   (a) 15 m (b) −0.133 m/s² (c) 0.67 m/s downwards, 13.3 m from the start

2.1 Linear motion: free fall

Free fall

Free-fall motion is the motion of an object under the influence of the ***Earth's gravitational attraction*** only.

- An object in free fall has only gravity acting on it (***no air resistance or friction***).
- On Earth all free-falling objects have the same ***acceleration due to gravity, g*** = 9.81 m s^{-2}.
- The direction of g is **downward**.

$$s = ut - \frac{1}{2}gt^2$$

$$v = u - gt$$

$$v^2 = u^2 - 2gs$$

Upward is positive, so $a = -g$. Put g = 9.81 m s^{-2} into these equations: the minus sign is already in them.

Sign convention

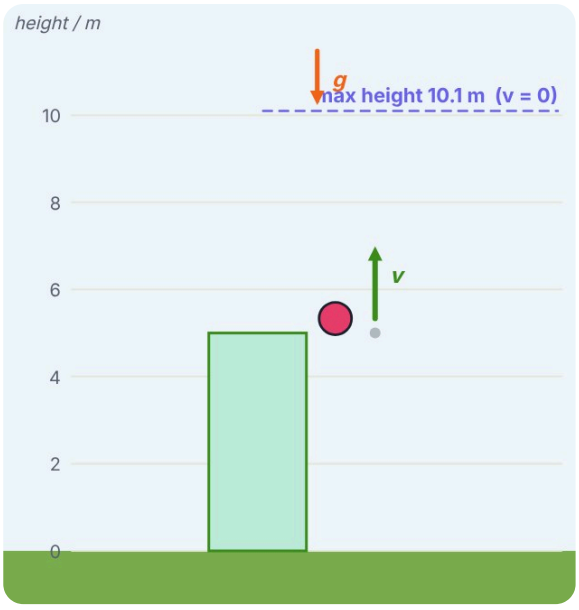
Quantity	Upward motion	Downward motion
Displacement, s	+ve	−ve
Velocity, v	+ve	−ve
Acceleration due to gravity	−ve	−ve

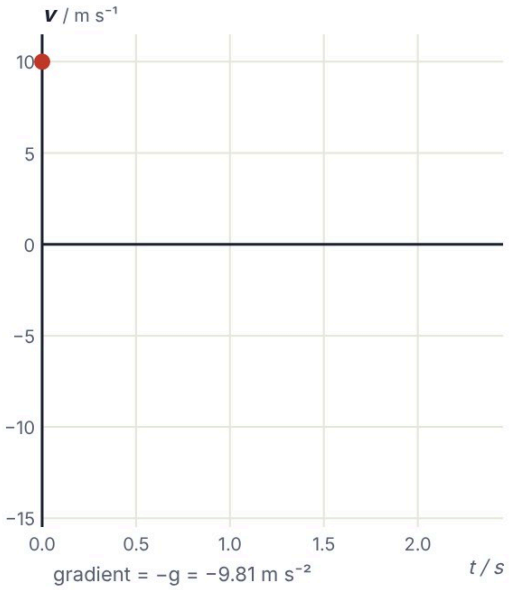
An object released from rest ($u = 0$) at height h has displacement $-s$ when it reaches the ground.

Simulation

Vertical motion: drop it, throw it up, launch it from a height

Set the launch velocity (upward is positive) and the starting height, or tap one of the examples. The ball, its velocity arrow and the v–t graph run together; the results appear when it lands.





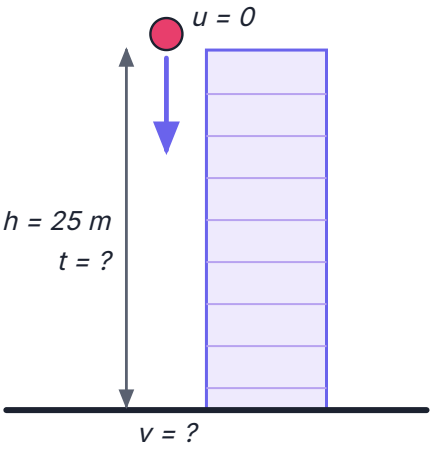
t	displacement s	velocity v	max height (above ground)	time to top
0.00 s	0.00 m	10.00 m/s	10.10 m	1.02 s

time of flight	impact velocity
…	…

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 2 to use the simulation.

Example 2.3.1

An object is released from rest at a height of 25 m from a building. Determine the time and velocity when it reaches the ground.



Solution

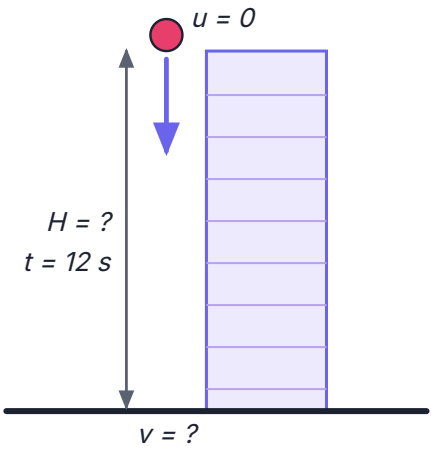
$u = 0$, $s = -25$ m, $g = 9.81\ \text{m s}^{-2}$

$s = ut - \frac{1}{2}gt^2$: $-25 = 0 - \frac{1}{2}(9.81)t^2$, $t = \mathbf{2.26\ s}$

$v = u - gt = 0 - (9.81)(2.26) = \mathbf{-22.2\ m s^{-1}}$ (−ve: downward)

Example 2.3.2

An object is released from rest at height H from a building; it takes 12 s to reach the ground. Determine the height and the velocity when it reaches the ground.



Solution

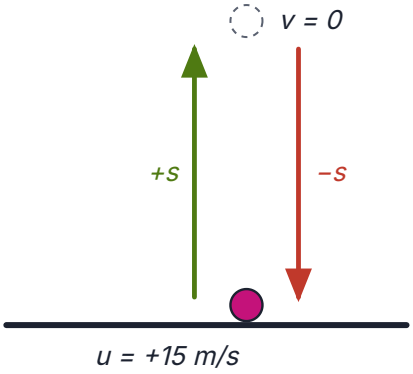
$u = 0$, $t = 12$ s, $g = 9.81\ \text{m s}^{-2}$

$s = ut - \frac{1}{2}gt^2 = 0 - \frac{1}{2}(9.81)(12)^2 = -706$ m, so the height is **706 m**

$v = u - gt = 0 - (9.81)(12) =$ **$-118\ \text{m s}^{-1}$** (−ve: downward)

Example 2.3.3

An object is thrown upward with a velocity of 15 $\mathrm{m\,s^{-1}}$. What is its **final velocity** when it reaches the ground? What is the **flight time**?



Solution

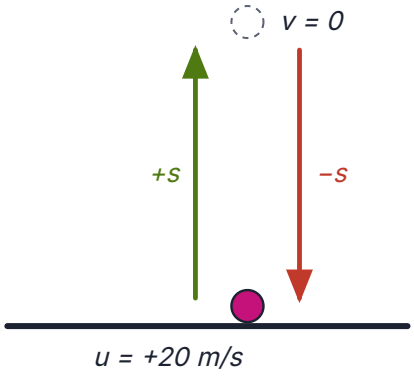
Back on the ground: displacement $s = 0$; $u = +15\ \mathrm{m\,s^{-1}}$

$v^2 = u^2 - 2gs = (15)^2 - 2(9.81)(0) \Rightarrow v =$ **$-15\ \mathrm{m\,s^{-1}}$** (moving down)

$s = ut - \frac{1}{2}gt^2$: $0 = 15t - \frac{1}{2}(9.81)t^2 \Rightarrow t =$ **3.06 s**

Example 2.3.4

A ball is thrown vertically upward with $u = 20\ \text{m s}^{-1}$. Determine the **maximum height** and the **time taken to reach the maximum height**.



Solution

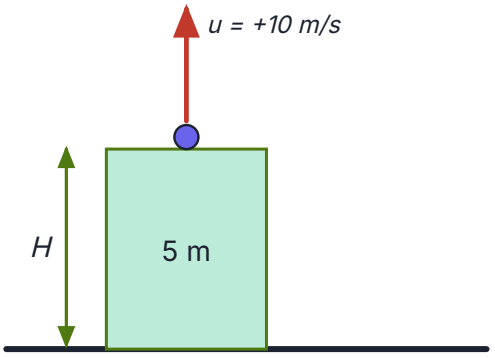
At maximum height, $v = 0$. $u = 20\ \text{m s}^{-1}$, $g = 9.81\ \text{m s}^{-2}$

$v^2 = u^2 - 2gs$: $0 = 20^2 - 2(9.81)s \Rightarrow h =$ **20.4 m**

$v = u - gt$: $0 = 20 - 9.81t \Rightarrow t =$ **2.04 s**

Example 2.3.5

An object is launched upward from a table of height 5 m with velocity 10 $\mathrm{m\,s^{-1}}$. Determine (a) the **maximum height**, (b) the **time of flight**, (c) its **velocity** when it hits the ground.



Solution

(a) At maximum height $v = 0$: $0 = 10^2 - 2(9.81)s \Rightarrow s = 5.10$ m above the table.

Maximum height above the ground, $h = s + H = 5.10 + 5 =$ **10.10 m**

(b) Ground is 5 m below the launch point: $s = -5$ m.

$-5 = 10T - \frac{1}{2}(9.81)T^2 \Rightarrow T =$ **2.45 s**

(c) $v = u - gt = 10 - (9.81)(2.45) = -14.03\ \mathrm{m\,s^{-1}}$

or $v^2 = u^2 - 2gs = 10^2 - 2(9.81)(-5) \Rightarrow v = \pm 14.07$, take **$-14.07\ \mathrm{m\,s^{-1}}$** (moving downward)

The two answers differ slightly only because T was rounded to 2.45 s. The v^2 method does not need T, so it is the more accurate one.

Exercises: free fall

1. A pebble, released at rest from a bridge, took 0.9 s to hit the water. Calculate (a) the distance it fell before hitting the water, (b) its speed just before hitting the water.

   Answer

   (a) 4.0 m (b) 8.8 m/s

2. A spanner was dropped from a hot-air balloon when the balloon was at rest 50 m above the ground. Calculate (a) the time taken for the spanner to hit the ground, (b) the speed of impact of the spanner on hitting the ground.

   Answer

   (a) 3.2 s (b) 31 m/s

3. A stone is thrown vertically upward with a speed of 12.0 m/s from the edge of a cliff 75.0 m high. (a) How much later does it reach the bottom of the cliff? (b) What is its speed just before hitting? (c) What total distance did it travel?

   Answer

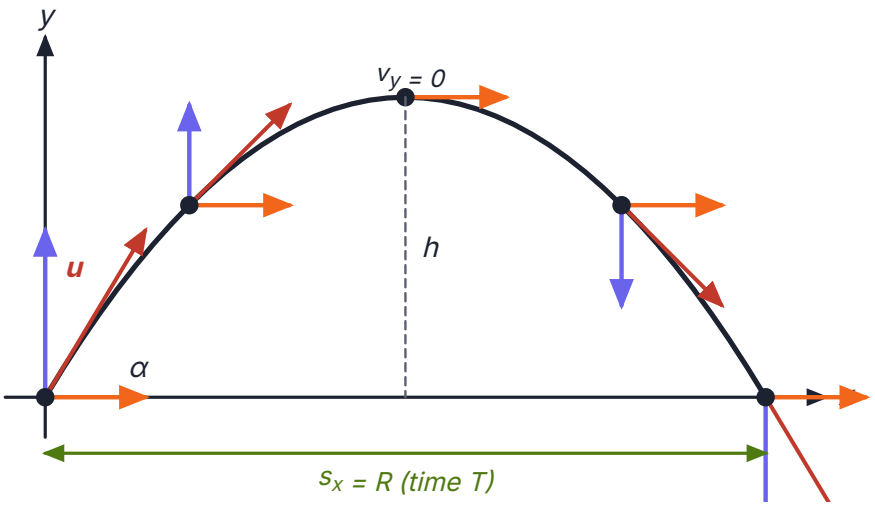
   (a) 5.33 s (b) 40.2 m/s (c) 89.7 m

2.2 Projectile motion

Projectile motion

is a combination of ***horizontal motion with constant velocity*** and ***vertical motion with constant acceleration***.

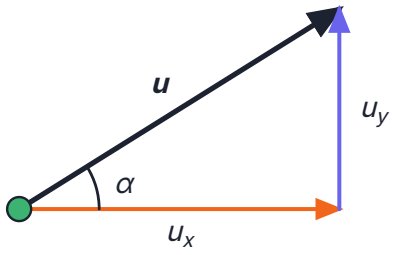
- **Horizontal**: the velocity in the **x-direction is constant** (v_x = constant, $a_x = 0$).
- **Vertical**: the velocity in the **y-direction has a constant acceleration due to gravity**, air resistance ignored ($a_y = -g = -9.81\ \text{m s}^{-2}$).



Parabola: the path is symmetrical.

u = initial velocity; v = velocity at time t; u_x, u_y = initial x- and y-components; v_y = y-component at time t; s_x = R = horizontal range; α (or θ) = launch angle; h = maximum height; T = time of flight.

Equations of projectile motion



$$u_x = u\cos\alpha$$

$$u_y = u\sin\alpha$$

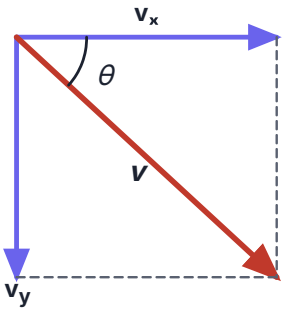
$$\alpha = \tan^{-1}\frac{u_y}{u_x}$$

Quantity	Horizontal (x-component)	Vertical (y-component)
Displacement, s	$s_x = u_x t$	$s_y = u_y t - \frac{1}{2}gt^2$
Velocity, v	$v_x = u_x$ (constant)	$v_y = u_y - gt$ $v_y^2 = u_y^2 - 2gs_y$
Acceleration, a	$a_x = 0$	$a_y = -g = -9.81\ \text{m s}^{-2}$ (constant)

Velocity at any point

Magnitude: $|v| = \sqrt{v_x^2 + v_y^2}$

Direction: $\theta = \tan^{-1}\frac{v_y}{v_x}$



Maximum height, H

Vertically: $u_y = u\sin\theta$. At maximum height $v_y = 0$.

$$v_y^2 = u_y^2 - 2gs_y \Rightarrow 0 = (u\sin\theta)^2 - 2gH$$

$$H = \frac{u^2\sin^2\theta}{2g}$$

Time of flight, T

Vertically: at the highest point $v_y = 0$. Using $v_y = u_y - gt$: $0 = u \sin\theta - gt$, so the time to reach the maximum height is

$$t = \frac{u \sin\theta}{g}$$

The path is symmetrical: **time to go up = time to come down**.

$$T = 2t = \frac{2u \sin\theta}{g}$$

Horizontal range, R

Horizontally: $a_x = 0$, $u_x = u \cos\theta$, $s_x = R = u_x T$

$$R = (u \cos\theta)\left(\frac{2u \sin\theta}{g}\right) \quad \text{where } \sin 2\theta = 2 \sin\theta \cos\theta$$

$$R = \frac{u^2 \sin 2\theta}{g}$$

The horizontal range is the horizontal displacement of the projectile (back to the launch level).

Maximum range and maximum height

$$H = \frac{u^2 \sin^2\theta}{2g}$$

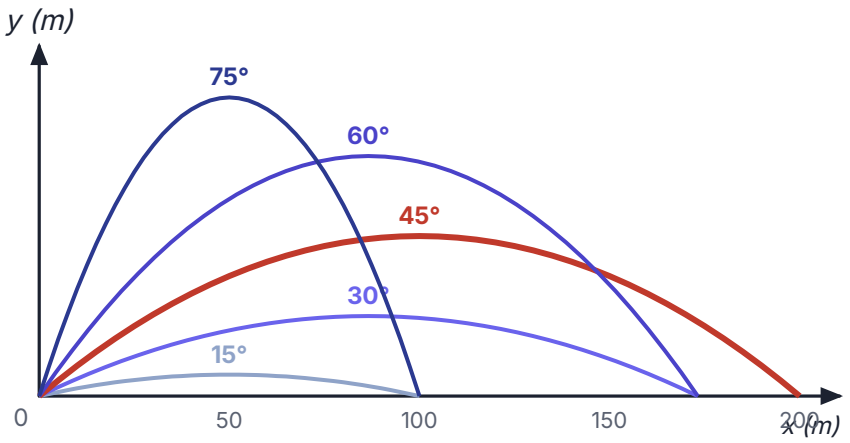
maximum when **$\theta = 90°$**:

$$H_{max} = \frac{u^2}{2g}$$

$$R = \frac{u^2 \sin 2\theta}{g}$$

maximum when $2\theta = 90°$, i.e. **$\theta = 45°$**:

$$R_{max} = \frac{u^2}{g}$$

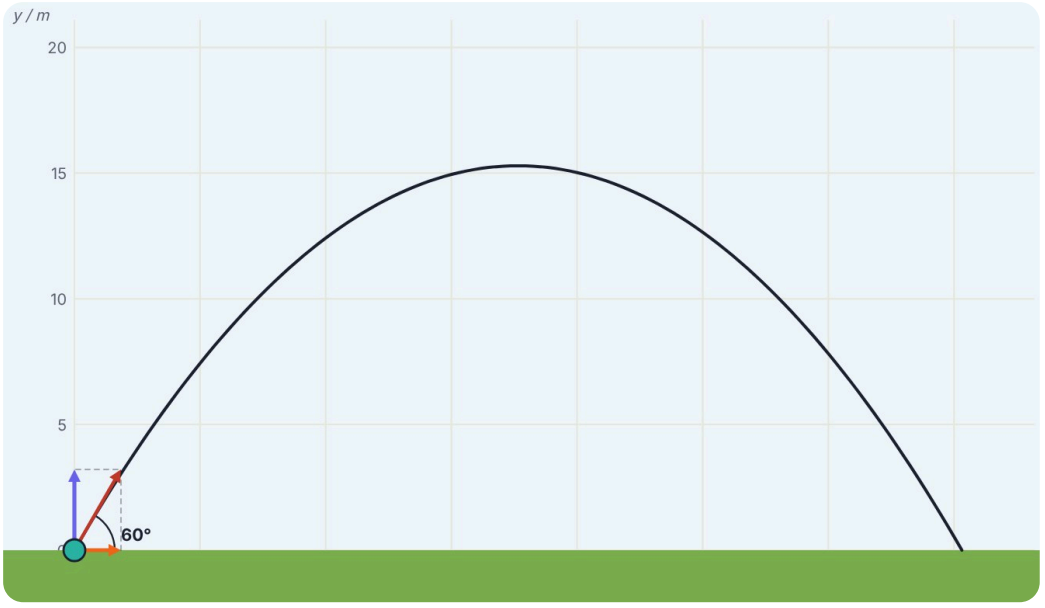


Same launch speed, different angles: 45° goes furthest; 30° and 60° (and 15° and 75°) land at the same place.

Simulation

Projectile launcher

Change the launch speed, angle and height. Watch the horizontal component stay the same while the vertical component shrinks, reverses and grows. "Compare angles" draws 15° to 75° together; "Air resistance" shows the real, lopsided path.



t	v_x (horizontal)	v_y (vertical)	speed $\lvert v \rvert$	max height H
0.00 s	**10.00 m/s**	**17.32 m/s**	**20.00 m/s**	**15.29 m**

time of flight T	range R	impact
3.53 s	**35.31 m**	**20.0 m/s, 60.0° below**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 2 to use the simulation.

Example 2.4.1

A ball is launched with a velocity of 20 m s^{-1} at an angle of 60° with the horizontal at point A. Calculate (a) the **maximum height**, H, (b) the **time of flight**, (c) the **horizontal range**, R, (d) its **velocity** when it strikes the ground.

Solution

(a) Vertically: u_y = 20 sin 60°, v_y = 0 at maximum height.

$v_y^2 = u_y^2 - 2gs_y$: $0 = (20 \sin 60°)^2 - 2(9.81)H \Rightarrow H =$ **15.29 m**

(b) Vertically: s_y = 0 (back on the ground). $0 = u \sin\theta\, t - \frac{1}{2}gt^2 \Rightarrow T = \frac{2u \sin\theta}{g} = \frac{2(20 \sin 60°)}{9.81}$

= **3.53 s**

(c) Horizontally: u_x = 20 cos 60°, a_x = 0. $R = u_x T$ = (20 cos 60°)(3.53) = **35.3 m**

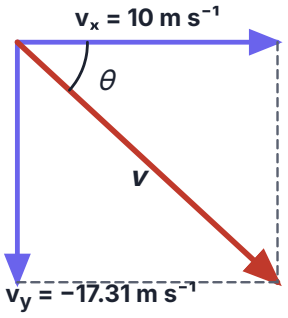
(d) $v_x = u_x$ = 20 cos 60° = 10 m s^{-1}

$v_y = u_y - gT$ = 20 sin 60° − (9.81)(3.53) = −17.31 m s^{-1} (downward)

Magnitude: $v = \sqrt{10^2 + (-17.31)^2}$ = **20 m s^{-1}**

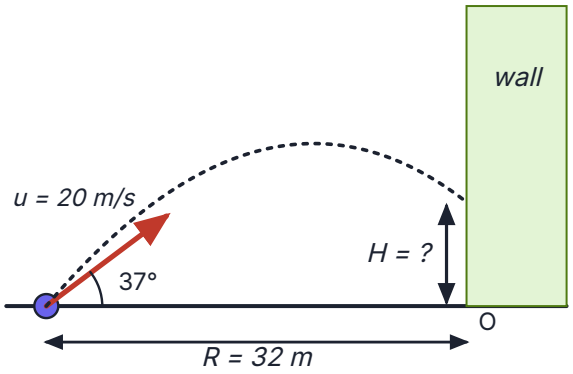
Direction: $\theta = \tan^{-1}\frac{-17.31}{10}$ = −60°

Its velocity is 20 m/s at 60° below the x-axis.



Example 2.4.2

Refer to the figure. Where will the ball hit the wall?



Solution

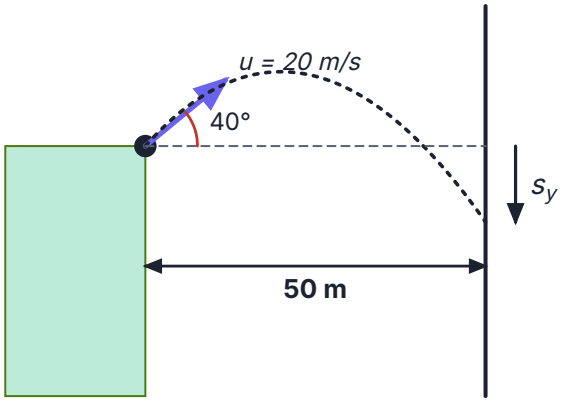
Horizontally: $u_x = 20 \cos 37°$, $s_x = 32$ m. $s_x = u_x t$: $32 = (20 \cos 37°)t \Rightarrow t = 2.0$ s

Vertically: $u_y = 20 \sin 37°$. $s_y = u_y t - \frac{1}{2}gt^2 = (20 \sin 37°)(2.0) - \frac{1}{2}(9.81)(2.0)^2 = 4.5$ m

The ball hits the wall **4.5 m above point O**.

Example 2.4.3

A ball is thrown from the top of one building toward a tall building 50 m away. The initial velocity of the ball is 20 m/s at 40° above the horizontal. How far above or below its original level will the ball strike the opposite wall?



Solution

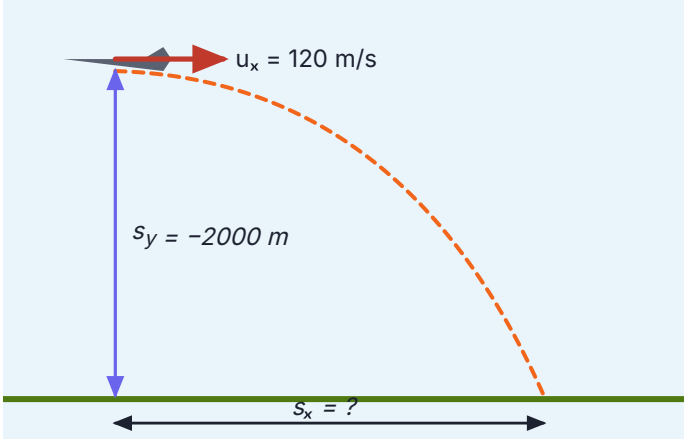
Horizontally: $u_x = 20 \cos 40°$, $s_x = 50$ m. $50 = (20 \cos 40°)t \Rightarrow t = 3.26$ s

Vertically: $u_y = 20 \sin 40°$. $s_y = (20 \sin 40°)(3.26) - \frac{1}{2}(9.81)(3.26)^2 = -10.22$ m

The ball hits **10.22 m below** its original level.

Example 2.4.4

A military airplane is flying horizontally at 120 m/s and accidentally drops a bomb at an elevation of 2000 m. (a) How much time is required for the bomb to reach the earth? (b) How far does it travel horizontally while falling? (c) Find the horizontal and vertical components of its velocity just before it strikes the earth. (d) Find the resultant velocity of the bomb just before it strikes the earth.



Solution

(a) Vertically: $u_y = 120 \sin 0° = 0$, $s_y = -2000$ m.

$-2000 = 0 - \frac{1}{2}(9.81)t^2 \Rightarrow t =$ **20.2 s**

(b) Horizontally: $u_x = 120 \cos 0° = 120$ m/s. $s_x = u_x t = (120)(20.2) =$ **2424 m**

(c) $v_x = 120\ \text{m s}^{-1}$; $v_y = u_y - gt = 0 - (9.81)(20.2) =$ **−198 m s^{-1}** (downward)

(d) Magnitude: $v = \sqrt{120^2 + (-198)^2} =$ **231.5 m s^{-1}**

Direction: $\theta = \tan^{-1}\frac{-198}{120} = -58.8°$

The bomb hits the ground at 231.5 m/s, 58.8° below the x-axis.

Exercises: projectile motion

1. A body is projected upward from level ground at an angle of 50° with the horizontal with an initial speed of 40 m/s. (a) How long will it take to hit the ground? (b) How far from the starting point will it strike? (c) At what angle with the horizontal will it strike?

Answer

(a) 6.3 s (b) 0.16 km (c) 50°

2. An object is projected horizontally at 16 m/s into the sea from a cliff top 45.0 m high. Calculate (a) how long it takes to reach the sea, (b) the horizontal range, (c) its impact velocity.

   Answer

   (a) 3.0 s (b) 49 m (c) 34 m/s at 62° to the horizontal

3. An object is thrown at 37° with the horizontal from a bridge 20.0 m above a river, with initial speed 30.0 m/s. Calculate (a) the time of flight, (b) the horizontal range, (c) the final velocity at impact.

   Answer

   (a) 4.57 s (b) 110 m (c) 35.9 m/s, 48.1° below the horizontal

4. An arrow is fired at 45 m/s at 30° above the horizontal. Calculate (a) its maximum height, (b) how long it takes to reach the maximum height, (c) how long it takes to return to the same horizontal level as it started, (d) the horizontal distance travelled to the point of return in (c).

   Answer

   (a) 25.8 m (b) 2.3 s (c) 4.6 s (d) 179 m

5. A ball is thrown upward at 30° to the horizontal and lands on the top edge of a building 20 m away. The top edge is 5.0 m above the throwing point. How fast was the ball thrown?

   Answer

   20 m/s

2.2 Air resistance effects on projectile motion

Without air resistance

- The trajectory is a perfect parabola.
- Horizontal velocity is constant; vertical acceleration = g.
- Maximum range is achieved at launch angle $\theta = 45°$.

With air resistance

- **Horizontal velocity decreases**: air drag opposes the horizontal motion.
- **Vertical deceleration increases** on the way up: net downward force = mg + drag.
- **The trajectory is no longer a symmetrical parabola.**
- **Maximum range occurs at an angle less than 45°.**
- **Horizontal range and maximum height are both reduced.**
- The time to descend is longer than the time to ascend.
- The impact speed is less than the launch speed.

Teacher's note

Open the projectile launcher above and switch on "Air resistance": the dashed path is the ideal parabola, the solid path is with drag. Try 45° and then 40°: with drag, 40° goes further.

Chapter 3 · Semester 1

DYNAMICS

Notes by Cikgu Suhaili

24 worked examples 3 simulations Exercises with answers

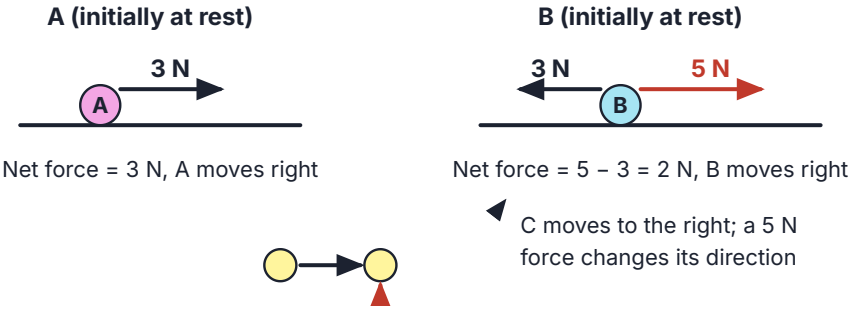
3.1 Newton's laws of motion

Newton's first law

Newton's first law

Every body **continues in its state of rest or uniform motion** in a straight line unless a **net force** acts on it. In equation form, $\Sigma F = 0$.

- A stationary particle starts to move because a net external force acts on it.
- A moving particle stops, or changes direction, because of a net external force.
- The first law tells us what a force does: it changes the state of motion.



Inertia

is **the tendency of a body to maintain its state of rest or of uniform motion in a straight line**. Newton's first law is also called the law of inertia.

- Inertia is a body's reluctance to start moving ("lazy") and its reluctance to stop once it is moving ("hardworking"), so a force is needed to move a body or to stop it.
- Inertia depends on **mass**: **mass is a measure of the inertia of a body**. A heavy body is harder to move, and harder to stop, than a light one.
- When a car stops suddenly the passenger continues to move forward because of his inertia, so a safety belt is needed during emergency braking.
- Classic inertia experiments: pulling a sheet from under a pile of books, and pushing two hanging tins, one empty and one full of sand.

Newton's second law

Newton's second law

The **rate of change of momentum** is **proportional to the resultant force** (net force).

$$\frac{\mathrm{d}p}{\mathrm{d}t} \propto F \Rightarrow \frac{mv - mu}{t} \propto F \Rightarrow ma \propto F \Rightarrow F = kma$$

- If m = 1 kg, a = 1 m s^{-2} and k = 1, then F = 1 N.
- The **direction of the acceleration** is the same as the direction of the applied net force.
- **1 newton** is the force that produces an acceleration of 1 m s^{-2} when acting on a 1 kg mass: **1 N = 1 kg m s^{-2}**.

$$F = \frac{\mathrm{d}(mv)}{\mathrm{d}t} = v\frac{\mathrm{d}m}{\mathrm{d}t} + m\frac{\mathrm{d}v}{\mathrm{d}t}$$

If the mass does not change, $\frac{\mathrm{d}m}{\mathrm{d}t} = 0$ and the acceleration is $a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{v - u}{t}$, so

$$F = ma \quad \text{or} \quad F = \frac{m(v - u)}{t}$$

Example 3.1.1

The velocity of a particle of mass **200 g** moving in a straight line increases from **10 to 50 cm s^{-1} in 2 s**. Find the **force** acting on the particle.

Solution

m = 200 × 10^{-3} kg, v = 50 × 10^{-2} m s^{-1}, u = 10 × 10^{-2} m s^{-1}

$$F = \frac{m(v - u)}{t} = \frac{200 \times 10^{-3}[(50 - 10) \times 10^{-2}]}{2} = \mathbf{4.0 \times 10^{-2}\ N}$$

Example 3.1.2

A sprinter accelerates constantly from **rest** and covers **50 m** in **6 s**. Her mass is **54 kg**. Ignoring all resistance, what **horizontal force** is she producing?

Solution

$m = 54$ kg, $u = 0$, $s = 50$ m, $t = 6$ s

$s = ut + \frac{1}{2}at^2 \Rightarrow a = \frac{s - ut}{\frac{1}{2}t^2} = \frac{50 - 0}{\frac{1}{2}(6)^2} = 2.8 \text{ m s}^{-2}$

$F = ma = (54)(2.8) = \mathbf{151\ N}$

Example 3.1.3

At time t seconds, the acceleration of a body of mass 5 kg is $(10 + 70t - 20t^2)$ m s^{-2}. Find the **force** on the body at $t = 0$, and the **maximum force** on the body.

Solution

At $t = 0$: $a = 10 + 0 - 0 = 10$ m s^{-2}, so $F = ma = (5)(10) = \mathbf{50\ N}$

Maximum force: $F_{\text{max}} = ma_{\text{max}}$. For a_{max}, $\frac{da}{dt} = 0$: $70 - 2(20)t = 0 \Rightarrow t = 1.75$ s

$a = 10 + 70(1.75) - 20(1.75)^2 = 71.25$ m s^{-2}

$F_{\text{max}} = 5(71.25) = \mathbf{356.3\ N}$

Using $F = m\,dv/dt + v\,dm/dt$ when the velocity is constant

When the **velocity is constant**, $dv/dt = 0$ and the force is needed only to change the mass that is moving:

$$F = v\frac{dm}{dt}$$

dm/dt = rate at which mass is added to (or removed from) the moving stream

Additional example (syllabus 3.1(d)) · conveyor belt

Sand falls vertically onto a conveyor belt at a rate of 2.0 kg s^{-1}. The belt moves horizontally at a constant speed of 3.0 m s^{-1}. Find the extra horizontal force needed to keep the belt moving at this speed.

Solution

The sand lands with no horizontal velocity and must be brought up to 3.0 m s^{-1}. The belt's speed is constant, so $dv/dt = 0$.

$$F = m\frac{dv}{dt} + v\frac{dm}{dt} = 0 + (3.0)(2.0) = \mathbf{6.0\ N}$$

Additional example (syllabus 3.1(d)) · water jet

Water leaves a hose at 20 m s^{-1} at a rate of 0.50 kg s^{-1} and hits a wall horizontally without bouncing back. Find the force exerted by the water on the wall.

Solution

Each second, 0.50 kg of water changes its velocity from 20 m s^{-1} to 0. The jet speed is constant, so the force is $F = v\frac{dm}{dt}$.

$F = (20)(0.50) = 10$ N. The wall exerts 10 N on the water, so by Newton's third law **the water exerts 10 N on the wall**, in the direction of the jet.

Exercises: F = ma

1. A lorry of mass 5000 kg can accelerate at 1 m s^{-2} when it is unloaded. What will its acceleration be when it carries a load of 3000 kg?

   Answer

   0.625 m s^{-2}

2. A car of mass 1000 kg experiences a net horizontal force of 4000 N, which accelerates it from rest to 10 m s^{-1}. How far does it travel during the acceleration?

   Answer

   12.5 m

3. A car of mass 800 kg experiences a force of 1200 N. How long does it take to reach 30 m s^{-1} from rest?

Answer

20 s

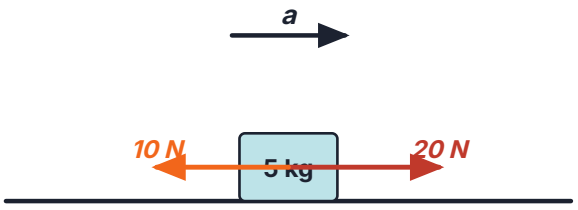
4. At time t seconds, the force on a body of mass 10 kg is $(4t - t^2)$ N. Find the acceleration at t = 1 s and the maximum acceleration.

Answer

0.3 m s^{-2}; 0.4 m s^{-2} (at t = 2 s)

Example 3.1.4

Two forces act on an object P of mass 5 kg: 20 N to the right and 10 N to the left. Find the **acceleration**.



Solution

Net force, F = 20 − 10 = 10 N to the right

$F = ma$: 10 = 5a ⇒ **a = 2 m s^{-2}** (to the right)

Exercises: net force and acceleration

1. Calculate the acceleration of a 10.0 kg box that moves horizontally when acted on by: (a) 3 N and 5 N, both to the right; (b) 3 N at 45° above the horizontal and 5 N, both towards the right; (c) 3 N vertically upward and 5 N to the right; (d) 3 N at 45° above the horizontal to the right and 5 N to the left.

   Answer

   (a) 0.8 m s^{-2} to the right (b) 0.71 m s^{-2} to the right (c) 0.5 m s^{-2} to the right (d) 0.29 m s^{-2} to the left

   Only horizontal components accelerate a box that moves horizontally: (b) 3 cos 45° + 5 = 7.12 N; (c) the vertical 3 N does not count, so 5 N; (d) 5 – 3 cos 45° = 2.88 N.

2. Find the acceleration of (a) a 6 kg body pulled by 12 N along a smooth floor; (b) a body of weight 49 N pulled up by two vertical forces of 12 N and 30 N.

   Answer

   (a) 2 m s^{-2} to the right (b) 1.4 m s^{-2} downward

3. Find the unknown forces: (a) force A gives a 6 kg body an acceleration of 3 m s^{-2} on a smooth floor; (b) force C lifts a 20 kg body (weight 196 N) with an upward acceleration of 1 m s^{-2}.

   Answer

   (a) A = 18 N (b) C = 216 N

Newton's third law

> **Newton's third law**
>
> For **every action (force)** there is a **reaction (opposing force)** of **equal magnitude but opposite direction**.

- If object A exerts a force on object B ($\mathbf{F}_{AB}$), then B exerts a force of the same size on A ($\mathbf{F}_{BA}$), in the opposite direction.
- The action and reaction act on **different bodies**.

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA} \quad \text{(action)} \quad \text{(reaction)}$$

- A hand pushes on a wall (action); the wall pushes on the hand (reaction).
- A hand presses on a desk (action); the desk pushes up on the hand (reaction).
- We can walk forward because, when one foot pushes backward on the ground, the ground pushes forward on the foot.

3.2 Conservation of linear momentum and impulse

Linear momentum, p

is the product of the **mass** of a particle, m, and its **velocity**, v: $\boldsymbol{p = mv}$.

- A **vector**, in the same direction as the velocity.
- Unit: kg m s^{-1} or N s.

Principle of conservation of linear momentum

When the **net external force** on a system is **zero**, the total **momentum** of the system is constant. Or: if there are **no external forces** acting on a system, the total momentum before a collision **equals** the total momentum after it.

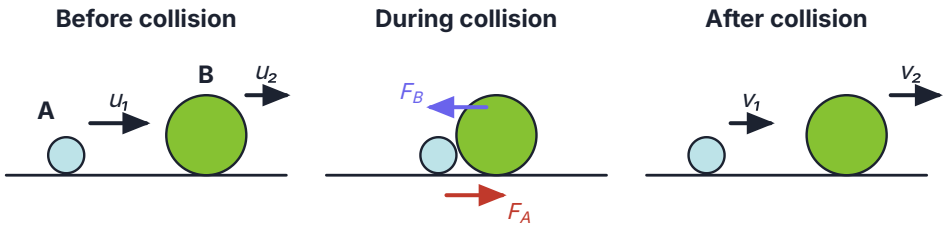
Newton's second law: $F = \dfrac{\Delta p}{\Delta t}$. If $\Sigma F = 0$ then $\Delta p = 0$, so p = constant

$$\Sigma p_i = \Sigma p_f$$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

For a collision between two bodies

Newton's third law proves the principle



During the collision: $F_A = m_1\dfrac{v_1 - u_1}{t}$ and $F_B = m_2\dfrac{v_2 - u_2}{t}$

By Newton's third law, $F_A = -F_B$ (the minus sign: F_B is opposite to F_A), so $m_1\dfrac{v_1 - u_1}{t} = -m_2\dfrac{v_2 - u_2}{t}$

Rearranging: $\boldsymbol{m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2}$. With no external force, Newton's third law proves the principle of conservation of linear momentum.

Elastic and inelastic collisions

Elastic collision

is one in which the **total kinetic energy is conserved**.

- Total momentum is conserved.
- Total kinetic energy is conserved.

Inelastic collision

is one in which the **total kinetic energy is not conserved**.

- Total momentum is conserved.
- Total kinetic energy is not conserved (some is changed to heat, sound and deformation).
- In a **perfectly inelastic** collision the bodies stick together after the collision.

Elastic collision in one dimension (head-on)

Momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

Kinetic energy: $\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$

If m_2 is at rest ($u_2 = 0$): from momentum, $m_1(u_1 - v_1) = m_2v_2$ **(1)**

From kinetic energy, $m_1(u_1^2 - v_1^2) = m_2v_2^2$ **(2)**

(2) ÷ (1): $u_1 + v_1 = v_2$ **(3)**

Solving (3) and (1):

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2}u_1 \qquad v_2 = \frac{2m_1}{m_1 + m_2}u_1$$

Elastic collision, m_2 initially at rest

Masses	After the collision
$m_1 = m_2$	$v_1 = 0$, $v_2 = u_1$ (they exchange velocities)
$m_1 > m_2$	$v_1 < u_1$; $v_2 > u_1$: m_1 slows down
$m_1 < m_2$	$v_1 < 0$; $v_2 < u_1$: m_1 recoils
$m_1 \gg m_2$	$v_1 \approx u_1$; $v_2 \approx 2u_1$
$m_1 \ll m_2$	$v_1 \approx -u_1$; $v_2 \approx 0$

Example 3.2.1

A 3000 kg truck moving at 10 m s^{-1} hits a 1000 kg parked car. The impact sets the car in motion at 15 m s^{-1}. Determine the **velocity** of the truck after the collision.

Solution

$(\Sigma p_{\text{before}})_x = (\Sigma p_{\text{after}})_x$: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$(3000)(10) + (1000)(0) = (3000)v_1 + (1000)(15) \Rightarrow 30\,000 - 15\,000 = 3000v_1$

$v_1 = 5.0$ **m** s^{-1} (to the right)

Check with the elastic-collision formula: $v_1 = \dfrac{3000 - 1000}{3000 + 1000}(10) = 5.0$ m s^{-1}

Perfectly inelastic collision in one dimension

After the collision the two objects stick together. By conservation of momentum:

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v \;\Rightarrow\; v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

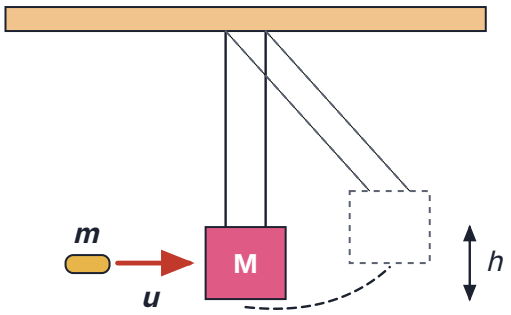
But kinetic energy is **not conserved**: $\tfrac{1}{2}m_1u_1^2 + \tfrac{1}{2}m_2u_2^2 \neq \tfrac{1}{2}(m_1 + m_2)v^2$. If m_2 is at rest:

$$\text{K.E.}_i = \tfrac{1}{2}m_1u_1^2, \quad \text{K.E.}_f = \tfrac{1}{2}(m_1 + m_2)v^2, \quad \frac{\text{K.E.}_f}{\text{K.E.}_i} = \frac{m_1}{m_1 + m_2} < 1$$

Ballistic pendulum

Ballistic pendulum

is a **device which measures the speed of a bullet**.



- M = mass of the block; m = mass of the bullet
- u = velocity of the bullet before the collision
- v = velocity of the block and bullet just after the collision
- h = maximum height reached by the block containing the bullet

The block is at rest. By conservation of momentum (perfectly inelastic): $\boldsymbol{mu = (M + m)v}$ (i)

After the collision, by conservation of energy (kinetic energy → gravitational potential energy): $\frac{1}{2}(M + m)v^2 = (M + m)gh \Rightarrow \boldsymbol{v = \sqrt{2gh}}$ (ii)

Substitute (ii) into (i) to get the speed of the bullet:

$$u = \frac{m + M}{m}\sqrt{2gh}$$

Momentum is conserved during the collision (it is too quick for the block to rise), but kinetic energy is not. Energy is conserved only afterwards, while the block swings up.

Example 3.2.7

In a ballistic pendulum experiment, h = 5.00 cm, m = 5.00 g and M = 1.00 kg. Find (a) the initial speed of the bullet, u, (b) the loss in kinetic energy due to the collision.

Solution

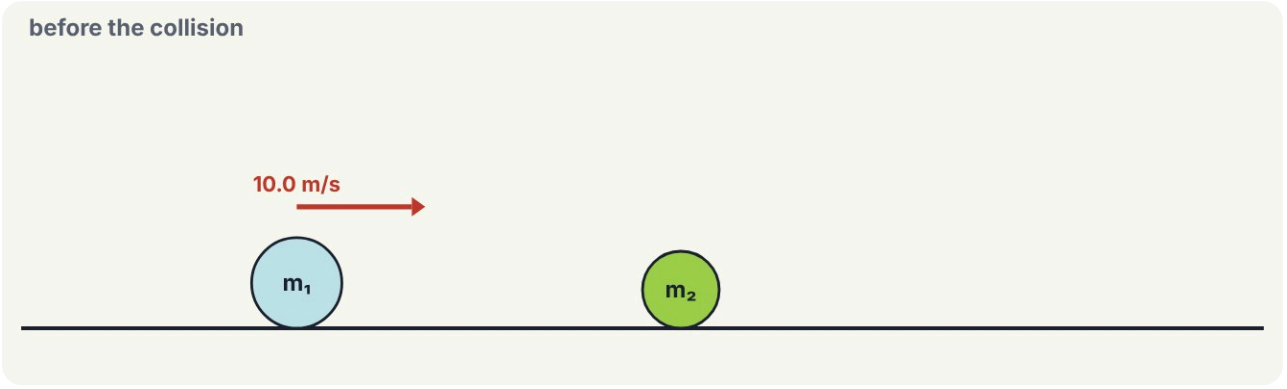
(a) $u = \dfrac{m + M}{m}\sqrt{2gh} = \dfrac{(5 \times 10^{-3}) + 1}{5 \times 10^{-3}}\sqrt{2(9.81)(5 \times 10^{-2})} = \mathbf{199\ m\ s^{-1}}$

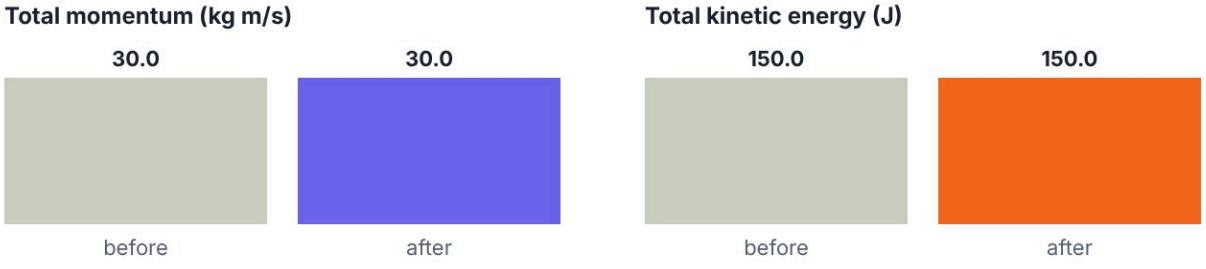
(b) $\Delta K = K_i - K_f = \tfrac{1}{2}mu^2 - \tfrac{1}{2}(m + M)v^2 = \tfrac{1}{2}mu^2 - \tfrac{1}{2}(m + M)(2gh)$

$= \tfrac{1}{2}(5 \times 10^{-3})(199)^2 - \tfrac{1}{2}(5 \times 10^{-3} + 1.00)(2)(9.81)(5 \times 10^{-2}) = 99.0 - 0.5 = \mathbf{98.5\ J}$

Simulation **Collisions: elastic, inelastic and perfectly inelastic**

Set the two masses and their velocities, choose the type of collision, then press Play. The bars show that total momentum is always the same before and after, while kinetic energy is conserved only in an elastic collision.





v_1 after	v_2 after	momentum before → after	KE lost
5.00 m/s	**15.00 m/s**	**30.0 → 30.0**	**0.0 J (0%)**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 3 to use the simulation.

Example 3.2.3

A 1800 kg truck stopped at a traffic light is struck from the rear by a 900 kg car and the two become entangled. If the car was moving at 20 m s^{-1} before the collision, what is the speed of the entangled mass after the collision?

Solution

m_1 = 900 kg, u_1 = 20 m s^{-1}; m_2 = 1800 kg, u_2 = 0

$\Sigma p_i = \Sigma p_f$: $m_1u_1 = (m_1 + m_2)v$

$(900)(20) = (1800 + 900)v \Rightarrow$ **v = 6.67 m s^{-1}**

Exercises: perfectly inelastic collisions

1. **Example 3.2.4** A truck of mass 3000 kg travelling at 3 m s^{-1} runs into a stationary truck of mass 5000 kg. The trucks couple together. Find their common speed.

   Answer

   1.13 m s^{-1}

2. **Example 3.2.5** A 16 g mass moves in the +x-direction at 30 cm s^{-1} while a 4.0 g mass moves in the −x-direction at 50 cm s^{-1}. They collide head on and stick together. Find their velocity after the collision.

   Answer

   0.14 m s^{-1} in the +x-direction

3. **Example 3.2.6** A railway engine of mass 20 000 kg moving at 3 m s^{-1} collides with and couples to a stationary train of wagons of total mass 10 000 kg. Calculate (a) the speed just after impact, (b) the loss of kinetic energy.

   Answer

   (a) 2 m s^{-1} (b) 30 000 J

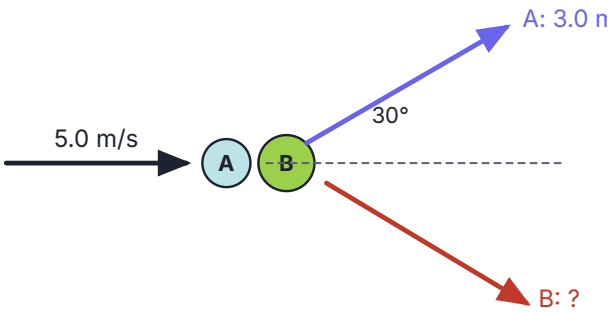
Collisions in two dimensions

Momentum is a vector, so in two dimensions it is conserved **separately in the x-direction and in the y-direction**:

$$\Sigma p_x \text{ before} = \Sigma p_x \text{ after} \quad \Sigma p_y \text{ before} = \Sigma p_y \text{ after}$$

Additional example (syllabus 3.3(c)) · collision in two dimensions

A ball A of mass 2.0 kg moving east at 5.0 m s^{-1} collides with a stationary ball B of mass 3.0 kg. After the collision A moves at 3.0 m s^{-1} at 30° north of east. Find (a) the velocity of B after the collision, (b) whether the collision is elastic.



Solution

Take east as +x and north as +y. Before: $p_x = (2.0)(5.0) = 10$ kg m s^{-1}, $p_y = 0$

x: $10 = (2.0)(3.0\cos 30°) + 3.0v_{Bx} \Rightarrow v_{Bx} = \dfrac{10 - 5.20}{3.0} = 1.60$ m s^{-1}

y: $0 = (2.0)(3.0\sin 30°) + 3.0v_{By} \Rightarrow v_{By} = -1.00$ m s^{-1}

(a) $v_B = \sqrt{1.60^2 + 1.00^2} =$ **1.89 m s^{-1}** at $\theta = \tan^{-1}\dfrac{1.00}{1.60} =$ **32° south of east**

(b) K.E. before = $\frac{1}{2}(2.0)(5.0)^2 = 25$ J; K.E. after = $\frac{1}{2}(2.0)(3.0)^2 + \frac{1}{2}(3.0)(1.89)^2 = 9.0 + 5.3 = 14.3$ J. Kinetic energy is not conserved, so the collision is **inelastic**.

Impulse of a force

Impulse

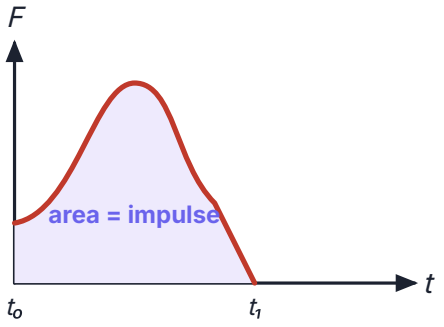
is the product of the applied force, F, and the time interval Δt for which it acts (for a constant force): $I = F\Delta t$.

From Newton's second law, $F = \frac{\Delta(mv)}{\Delta t} = \frac{\Delta p}{\Delta t}$

$$F\Delta t = \Delta p = m(v - u)$$

Impulse = change of momentum

- F is the impulsive force.
- A vector, in the direction of the force.
- Unit: kg m s^{-1} or N s.



Impulse = area under the graph of F against t = $\int_{t_0}^{t_1} F\,dt$

Applications

From $F\Delta t = m(v - u)$: $F = \frac{m(v - u)}{\Delta t}$

- To **increase** the impulsive force, the change of momentum must happen in a **very short time**: a karate chop, smashing a shuttlecock, kicking a ball, squash and golf.
- In a car accident, to **reduce** the impulsive force, the collision time is increased by using softer bodies (crumple zones, airbags).

Example 3.2.15

A car of mass 800 kg is travelling at 25 m s^{-1}. Find the **constant force** needed to stop it in 4 s.

Solution

u = 25 m s^{-1}, v = 0, t = 4 s, m = 800 kg

$F = \frac{m(v - u)}{t} = \frac{800(0 - 25)}{4}$ = −5000 N, so a force of **5000 N** opposite to the motion

Example 3.2.16

A ball of mass 0.2 kg is dropped from a height of 20 m. It rebounds to a height of 10 m. Find the **impulse** exerted by the ground on the ball.

Solution

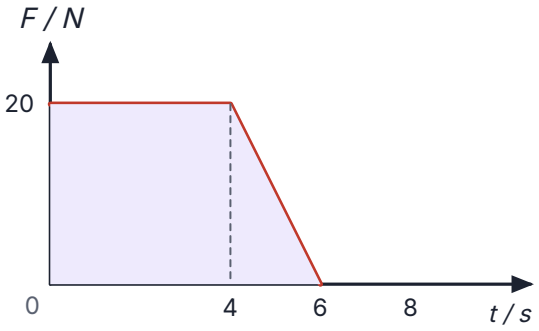
Speed just before impact: $u = \sqrt{2gh} = \sqrt{2(9.81)(20)} = 19.81$ m s^{-1} (downward)

Speed just after: $v = \sqrt{2(9.81)(10)} = 14.01$ m s^{-1} (upward)

Taking upward as positive: impulse = $m(v - u)$ = 0.2[14.01 – (−19.81)] = **6.76 N s** upward

Example 3.2.17

The force acting on an object is shown below. Find the impulse at (a) t = 6 s, (b) t = 8 s.



Solution

(a) Impulse = area = ½(4 + 6)(20) = **100 N s**

(b) No force acts after 6 s, so the total area is still 100 + 0 = **100 N s**

Seat belts, airbags and crumple zones

In a crash the passenger's momentum must fall from mv to zero: the **impulse** $F\Delta t = \Delta p$ is fixed by the speed. What safety features change is the **time** Δt over which it happens. A longer time means a **smaller force** on the body.

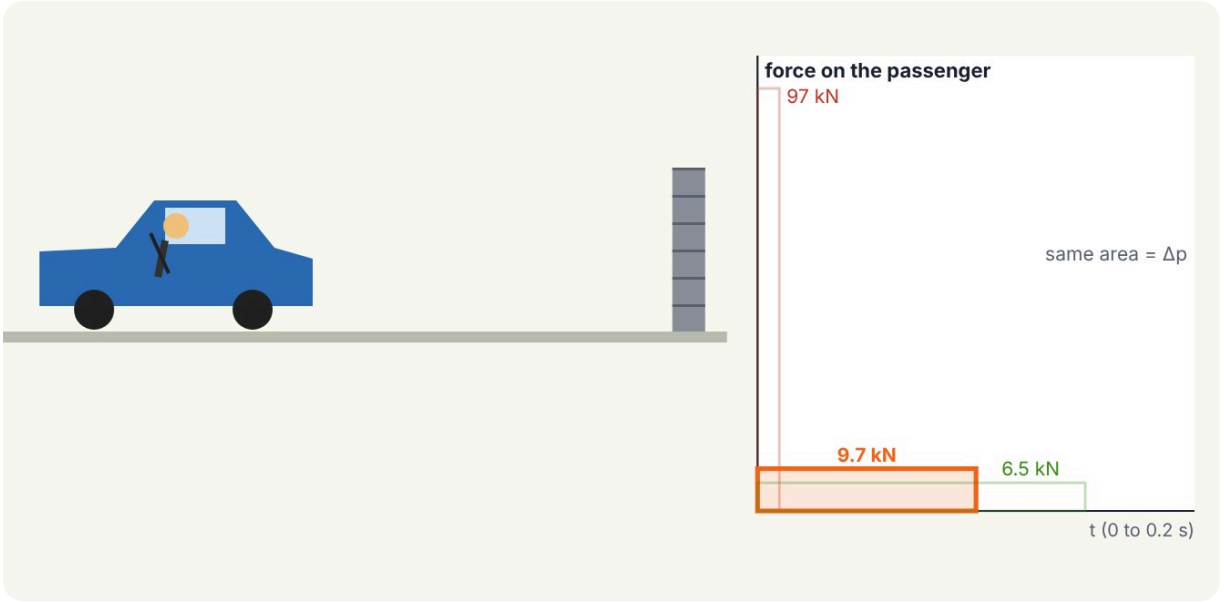
- **No seat belt:** the passenger keeps moving until they hit the dashboard or windscreen, and stop in about 0.01 s: an enormous force.
- **Seat belt:** it stretches slightly and stops the passenger together with the car over about 0.1 s.

- **Airbag:** a soft cushion that increases the stopping time further and spreads the force over a larger area.
- **Crumple zone:** the front of the car is designed to fold up, so the car itself takes longer to stop.

Simulation

Crash test: why seat belts and airbags save lives

Choose the speed and the safety system, then crash the car (it is shown slowed down). The change of momentum, and so the area under the force–time graph, is the same each time; a longer stopping time gives a much smaller force.



change of momentum Δp = mv	stopping time Δt	average force F = Δp/Δt	force ÷ body weight
972 N s	**100 ms**	**9.7 kN**	**14 ×**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 3 to use the simulation.

Example 3.2.18

A 70 kg passenger is in a car travelling at 50 km h^{-1} when it crashes. Find the average force on the passenger if they are stopped (a) by a seat belt in 0.10 s, (b) by hitting the dashboard in 0.010 s. Compare each with the passenger's weight.

Solution

v = 50 km h^{-1} = 13.9 m s^{-1}; $\Delta p = mv$ = (70)(13.9) = 972 N s

(a) $F = \dfrac{\Delta p}{\Delta t} = \dfrac{972}{0.10}$ = **9.7×10^3 N**, about 14 times the weight (687 N)

(b) $F = \dfrac{972}{0.010}$ = **9.7×10^4 N**, about 140 times the weight: enough to cause fatal injuries

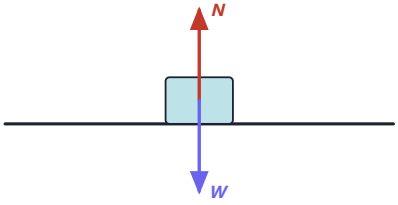
3.3 Reaction and frictional force

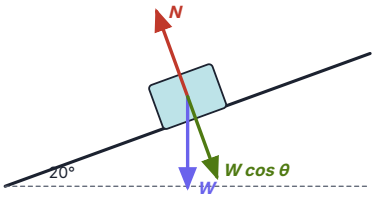
Normal (reaction) force, N or R

- Newton's third law example: the weight of an object presses on a table (action); the table pushes up on the object with a normal force (reaction).
- "Normal" means perpendicular.

Normal force

is the contact force exerted by the surface of one object on the surface of another, which prevents the objects from passing through each other. It acts **perpendicular to the surface**.



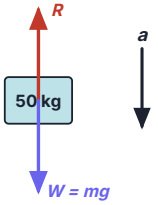


Free-body diagram

is a diagram of a body showing all the forces acting on it.

Example 3.3.1

A man of mass 50 kg stands on the floor of a lift which is descending with a downward acceleration of 0.8 m s^{-2}. Find the **reaction** of the floor of the lift on the man.



Solution

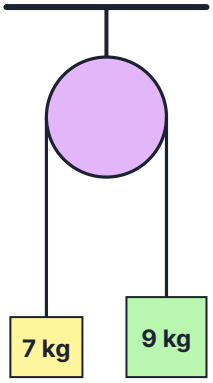
Free-body diagram: R upward, $W = mg$ downward, a downward

$\Sigma F_y = ma$ (downward positive): $ma = mg - R$

$R = mg - ma = 50(9.81 - 0.8) =$ **450.5 N**

Example 3.3.2

A cord passing over an easily turned pulley (massless and frictionless) has a 7.0 kg mass hanging from one end and a 9.0 kg mass from the other. Find the **acceleration** of the masses and the **tension** in the cord.



Solution

7.0 kg (accelerates up): $ma = T - mg \Rightarrow T = 7.0(a + 9.81)$ **(1)**

9.0 kg (accelerates down): $ma = mg - T \Rightarrow T = 9.0(9.81 - a)$ **(2)**

(1) = (2): a **= 1.23 m s^{-2}**, T **= 77.3 N**

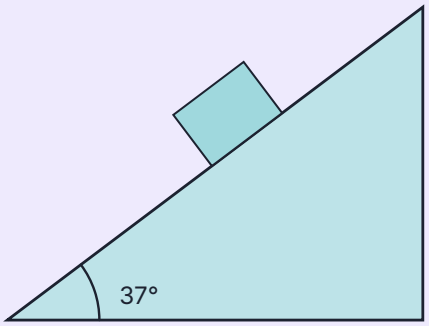
Exercises: normal force and connected bodies

1. A force of 15.0 N is applied at 30° below the horizontal (pushing down on the block) on a 0.750 kg block resting on a frictionless surface. Calculate (a) the magnitude of the resulting acceleration, (b) the magnitude of the normal force.

   Answer

   (a) 17.3 m s^{-2} (b) 14.85 N

2. A 10.0 kg mass is supported on a friction-free plane inclined at 37° and connected to a second mass M by an ideal string and pulley. M accelerates upward at 3.00 m s^{-2}. Calculate (a) the reaction force of the plane on the 10.0 kg mass, (b) the tension in the string, (c) the mass M.



Answer

(a) R = 78.3 N (b) T = 29.0 N (c) M = 2.27 kg

The slide answers 78.4 N, 28.8 N and 2.25 kg come from using g = 9.8 m s^{-2} with sin 37° ≈ 0.6 and cos 37° ≈ 0.8.

Frictional force, f

> **Frictional force**
> is the contact force exerted **between the surfaces** of two objects, **directed parallel to the surface**, that acts to **prevent (oppose) the objects** from slipping on each other. It results from forces between atoms and molecules when surfaces are in contact.

- The frictional force is **proportional to the normal force**: $f \propto N$.
- Friction occurs when an object moves on a rough surface or through a fluid.
- Three types: (a) static friction, (b) kinetic friction, (c) rolling friction.

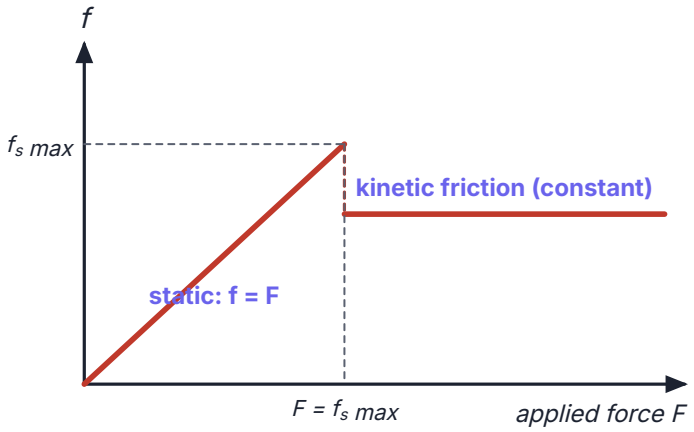
Static friction

- If the applied force F is increased gradually, at a certain value the book starts to move.
- The book just begins to move when F is slightly greater than the maximum static frictional force, f_s^{max}.
- The maximum frictional force produced by the rough surface is called the **limiting frictional force**.

Applied force	Static friction	Result
F = 3 N	f_s = 3 N	no movement
F = 5 N	f_s = 5 N	no movement
F = 7 N	f_s^{max} = 7 N	just about to move

$$f_s \le f_s^{max} \quad f_s^{max} = \mu_s N$$

μ_s = coefficient of static friction, N = normal force



Static friction grows with the applied force up to f_s^{max}; once the body slides, the kinetic friction is constant and smaller.

Simulation

Friction: push the block until it slides

Increase the applied force slowly. Static friction matches your push exactly until it reaches $\mu_s N$; then the block breaks away and the friction drops to $\mu_k N$. The graph of f against F builds up as you go.



normal force N = mg	limiting friction $\mu_s N$	kinetic friction $\mu_k N$	friction now	acceleration
19.6 N	**11.8 N**	**7.8 N**	**0.0 N**	**0**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 3 to use the simulation.

Coefficient of static friction, μ_s

is a **dimensionless constant, characteristic of the two surfaces**, which gives the relative strength of the force: $\mu_s = \dfrac{f_s^{max}}{N}$, where f_s^{max} is the limiting frictional force and N the normal force.

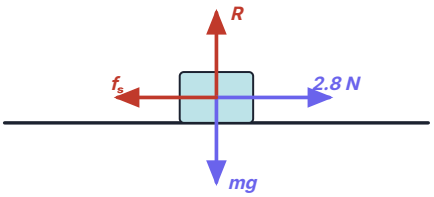
- μ_s depends on the **type** of material (wood on wood, rubber on concrete), the **condition** of the surfaces (polished, rough, lubricated) and other variables such as temperature.
- For smooth surfaces, $\mu_s = 0$.
- The maximum frictional force is **independent of the area of contact** if the normal reaction is the same.

Table 3.1 Coefficients of static and kinetic friction

Surfaces	Static, μ_s	Kinetic, μ_k
Rubber on dry concrete	1.0	0.7
Shoes on wood	0.9	0.7
Rubber on wet concrete	0.7	0.5
Steel on steel (dry)	0.6	0.3
Metal on wood	0.5	0.3
Wood on wood	0.4	0.2
Shoes on ice	0.1	0.05
Steel on steel (oiled)	0.05	0.03

Example 3.3.3

A body of mass 1.2 kg lies on a horizontal surface. A horizontal force of 2.8 N can just shift the body. Find the **coefficient of static friction**.



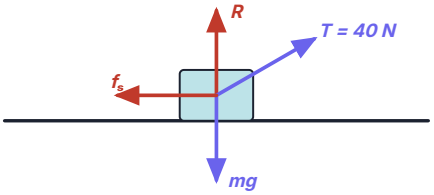
Solution

$\Sigma F_y = 0$: $R = mg$; $\Sigma F_x = 0$: $f_s = 2.8$ N

$f_s = \mu_s R = \mu_s mg \Rightarrow \mu_s = \dfrac{2.8}{(1.2)(9.81)} = \mathbf{0.24}$

Example 3.3.4

A body of mass 20 kg lies on a horizontal floor. A string attached to it at 30° to the horizontal has a tension of 40 N when the body begins to move. Find the coefficient of static friction.



Solution

$\Sigma F_y = 0$: $R + 40 \sin 30° = mg \Rightarrow R = mg - 40 \sin 30° = 176.2$ N

$\Sigma F_x = 0$: $40 \cos 30° = f_s = \mu_s R$

$$\mu_s = \frac{40 \cos 30°}{176.2} = \mathbf{0.197}$$

Kinetic friction

- Kinetic friction always **opposes the motion** of a moving object.
- It is less than the limiting static friction: the force needed to start motion is more than the force needed to keep it going, so $f_k < f_s^{max}$ and $\mu_k < \mu_s$.
- μ_k is independent of the velocity.

$$\mu_k = \frac{f_k}{N}, \quad f_k = \mu_k N$$

Example 3.3.5

A body is slid along a horizontal floor. It decelerates from 10 m s^{-1} to 5 m s^{-1} in 2 s. Find the **coefficient of friction** between the body and the floor.

Solution

$\Sigma F_y = 0$: $R = mg$; $\Sigma F_x = ma$: $ma = f_k = \mu_k mg$

$a = \dfrac{v - u}{t} = \dfrac{5 - 10}{2} = -2.5$ m s^{-2} (a deceleration of 2.5 m s^{-2})

$\mu_k = \dfrac{a}{g} = \dfrac{2.5}{9.81} = \mathbf{0.255}$

Example 3.3.6

A 70 kg box is slid along the floor by a 400 N force. The coefficient of kinetic friction is 0.50. Find the **acceleration** of the box.

Solution

$\Sigma F_y = 0$: $R = mg$; $\Sigma F_x = ma$: $ma = 400 - \mu_k mg$

$70a = 400 - (0.50)(70)(9.81) \Rightarrow \mathbf{a = 0.81\ m\ s^{-2}}$

Example 3.3.7

The 70 kg box is now pulled by a 400 N force at 30° to the horizontal. $\mu_k = 0.50$. Find the acceleration.

Solution

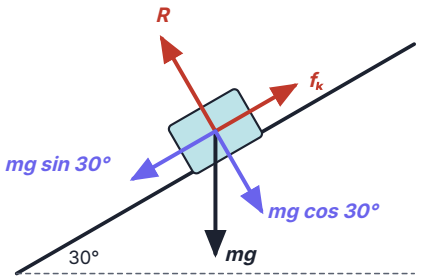
$\Sigma F_y = 0$: $R + 400 \sin 30° = mg \Rightarrow R = 486.7$ N

$\Sigma F_x = ma$: $ma = 400 \cos 30° - \mu_k R$

$70a = 346.4 - (0.50)(486.7) \Rightarrow \mathbf{a = 1.47\ m\ s^{-2}}$

Example 3.3.8

A 20 kg box sits on a 30° incline. μ_k between box and incline is 0.30. Find the acceleration of the box down the incline.



Solution

Perpendicular to the plane: $R = mg\cos 30° = 170$ N

Along the plane: $ma = mg\sin 30° - f_k = mg\sin 30° - \mu_k R$

$20a = (20)(9.81)(0.5) - (0.30)(170) \Rightarrow$ **$a = 2.36\ \text{m s}^{-2}$**

Exercises: friction on slopes and floors

1. A 10 kg block is placed on a plane inclined at 37°. μ = 0.4. When released, find (a) the normal force on the block, (b) its acceleration down the plane.

   Answer

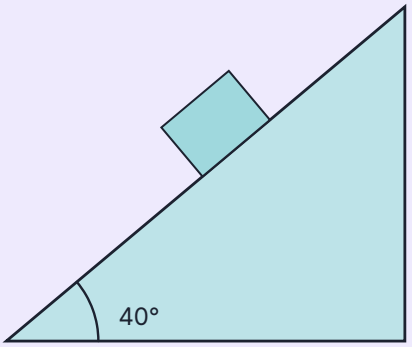
   (a) 78.3 N (b) 2.77 m s^{-2}

2. A 2 kg block is on a plane inclined at 30°; μ = 0.4. Find its acceleration (a) when pushed up the plane by a constant force of 20 N, (b) when pulled down the plane by the same force.

   Answer

   (a) 1.7 m s^{-2} (b) 11.51 m s^{-2}

3. A horizontal force of 500 N pushes a 25 kg box up a 40° incline with an acceleration of 0.75 m s^{-2}. Find the coefficient of kinetic friction. (g = 9.81 m s^{-2})



Answer

μ_k = 0.41

4. A body on a rough 30° slope slides down with an acceleration of 4 m s^{-2}. Find the coefficient of friction.

Answer

0.106

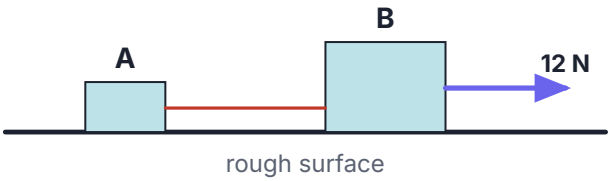
5. A horizontal force of 140 N pulls a 60 kg box across a horizontal floor at constant speed. What is the coefficient of friction?

Answer

0.24

Example 3.3.9

Blocks A (2 kg) and B (4 kg) are connected by a rope on a rough surface. The frictional forces on A and B are 2 N and 4 N. Find the acceleration and the tension in the rope if B is pulled by a force of 12 N.



Solution

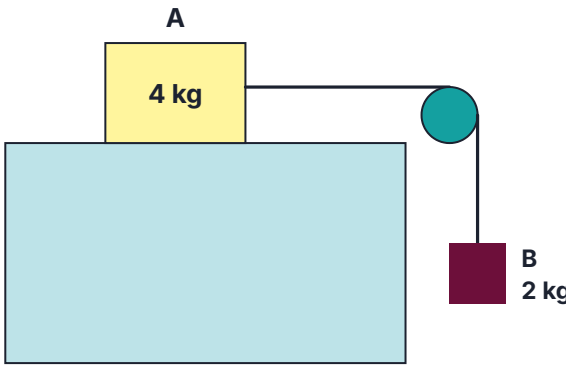
For A: $T - f_s = ma \Rightarrow T - 2 = 2a$ (1)

For B: $12 - T - f_s = ma \Rightarrow 8 - T = 4a$ (2)

(1) + (2): $6 = 6a \Rightarrow \boldsymbol{a = 1\ \text{m s}^{-2}}$; from (1): $\boldsymbol{T = 4\ \text{N}}$

Example 3.3.10

Object A (4 kg) on a rough table is connected to object B (2 kg) by a string over a pulley. The frictional force on A is 10 N. Find the acceleration, the tension in the string and μ_k.



Solution

For B: $m_B g - T = m_B a$; for A: $T - 10 = m_A a$

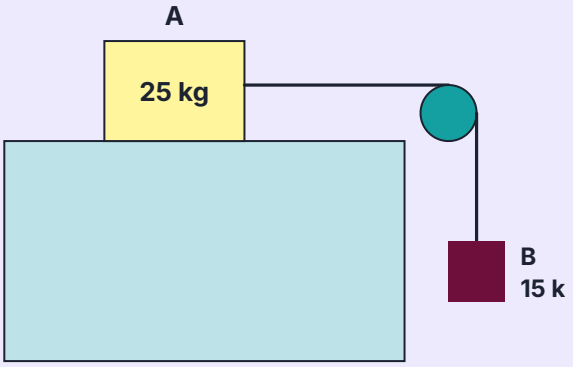
Adding: $2(9.81) - 10 = 6a \Rightarrow a = \mathbf{1.60\ m\ s^{-2}}$

$T = 2(9.81 - 1.60) = \mathbf{16.4\ N}$

$$\mu_k = \frac{f_k}{m_A g} = \frac{10}{4(9.81)} = \mathbf{0.25}$$

Exercises: connected bodies with friction

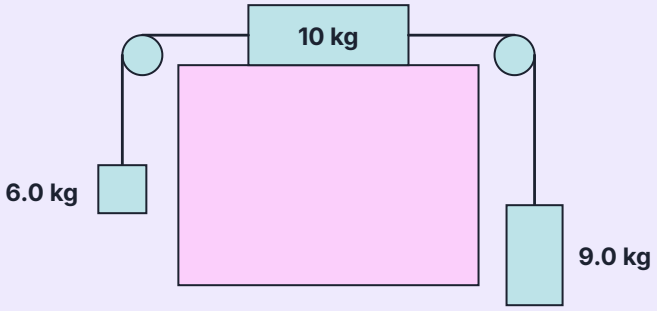
1. The coefficient of kinetic friction between block A (25 kg) and the table is 0.20. Block B (15 kg) hangs from a string over a pulley. How far will B drop in the first 3.0 s after the system is released?



Answer

11 m

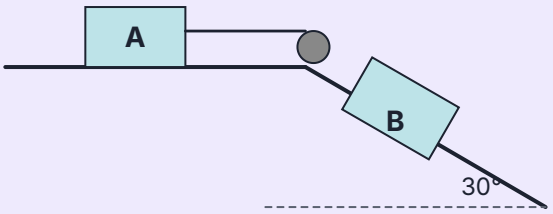
2. Three objects are connected by strings over massless, friction-free pulleys: 6.0 kg hanging on the left, 10 kg on the table, 9.0 kg hanging on the right. μ_k between the middle object and the table is 0.2. (a) What is the acceleration? (b) Calculate the tension in each string.



Answer

(a) 0.39 m s^{-2} (b) 61 N and 85 N

3. Two boxes of identical mass 40 kg are tied together: A on a horizontal surface, B on a 30° slope. Both experience sliding friction with $\mu_k = 0.15$. Find the acceleration and the tension in the tie cord.



Answer

1.1 m s^{-2}; 0.10 kN

Chapter 4 · Semester 1

WORK, ENERGY AND POWER

Notes by Cikgu Suhaili

17 worked examples 2 simulations Exercises with answers

4.1 Work and energy

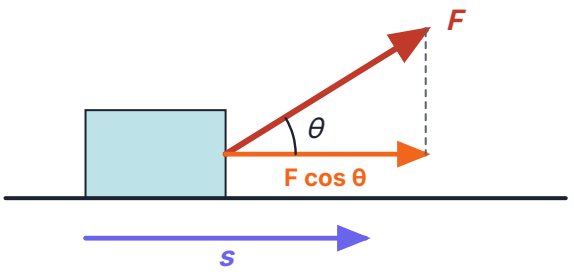
Work done by a constant force

Work done

is the **product of the component of the force parallel to the displacement and the magnitude of the displacement**.

$$W = \vec{F} \cdot \vec{s} = (F\cos\theta)s$$

F = magnitude of the constant force, s = magnitude of the displacement, θ = angle between **F** and **s**

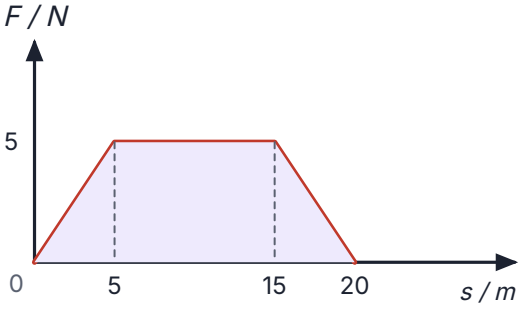


- Work is a **scalar**.
- Dimension: $[W] = [F][s] = \mathrm{M\,L^2\,T^{-2}}$.
- Unit: joule (J); 1 J = 1 N m = 1 kg m^2 s^{-2}.

Work done by a variable force

$$W = \int_{s_1}^{s_2} F\,\mathrm{d}s$$

Work done = area under the F–s graph.



Applications of W = F s cos θ for a constant force

Case	Angle	Work done
F in the same direction as **s**	θ = 0°	$W = Fs$
Two forces $\mathbf{F}_1$ and $\mathbf{F}_2$ both along **s**	θ = 0°	$W = (F_1 + F_2)s = F_{\text{net}}s = ma\,s$
F perpendicular to **s**	θ = 90°	$W = 0$
F opposite to **s**	θ = 180°	$W = -Fs$

W = 0 when $F = 0$, or $s = 0$, or θ = 90°.

Example 4.1.1

A physics reference book is pushed 1.20 m along a horizontal table by a horizontal force of 3.00 N. The frictional force is 0.60 N. Calculate the work done by (a) the 3.00 N force, (b) the frictional force, (c) the normal force, and (d) the total work done on the book.

Solution

(a) F = 3.00 N, θ = 0°: W = 3.00 cos 0° (1.20) = **3.60 J**

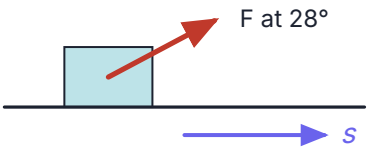
(b) f = 0.60 N, θ = 180°: W = 0.60 cos 180° (1.20) = **−0.72 J**

(c) R, θ = 90°: $W = R$ cos 90° (1.20) = **0 J**

(d) ΣW = 3.60 − 0.72 + 0 = **2.88 J**, or $\Sigma W = (F - f)$ cos 0° s = (3.00 − 0.60)(1.20) = 2.88 J

Example 4.1.2

An object is pulled along the ground by a 75 N force directed 28° above the horizontal. How much work does the force do in pulling the object 8.0 m?

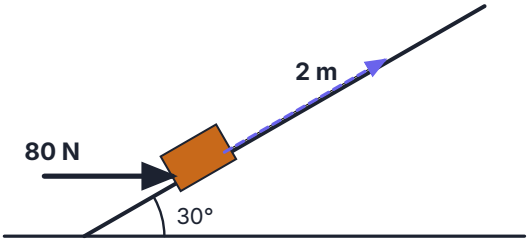


Solution

$$W = F \cos\theta \, s = 75 \cos 28° \, (8.0) = \mathbf{0.53\ kJ}$$

Example 4.1.3

A particle of mass 5 kg lies on a rough slope of angle 30°. A horizontal force of 80 N is applied to it. Find the work done by this force in pushing the particle 2 m up the slope.



Solution

The angle between the horizontal force and the displacement up the slope is 30°.

$$W_F = F \cos\theta \, s = 80 \cos 30° \, (2) = \mathbf{139\ J}$$

Exercises: work done by a constant force

1. A 300 g object slides 80 cm along a horizontal tabletop. How much work is done by friction if the coefficient of kinetic friction is 0.20?

   Answer

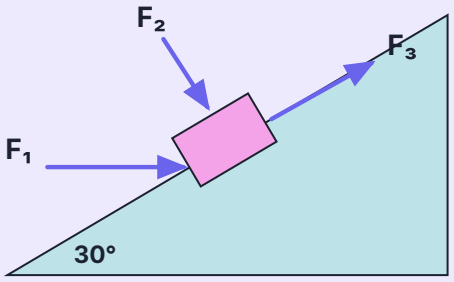
   −0.47 J

2. A force of (2**i** + 3**j**) N acts through a displacement of (4**i** + 5**j**) m. Find the work done.

Answer

23 J

3. A block moves 80 cm up a 30° incline under three forces: F_1 = 40 N (horizontal), F_2 = 20 N (perpendicular to the incline) and F_3 = 30 N (along the incline). Calculate the work done by each force.



Answer

W_1 = 28 J, W_2 = 0, W_3 = 24 J

4. A 5.00 kg package slides 1.50 m down a ramp inclined at 12.0° below the horizontal; μ_k = 0.310. Calculate the work done on the package by (a) friction, (b) gravity, (c) the normal force, (d) the total work. (e) If its speed at the top is 2.20 m s^{-1}, what is its speed after sliding 1.50 m? (g = 9.81 m s^{-2})

Answer

(a) −22.3 J (b) 15.3 J (c) 0 (d) −7.0 J (e) 1.4 m s^{-1}

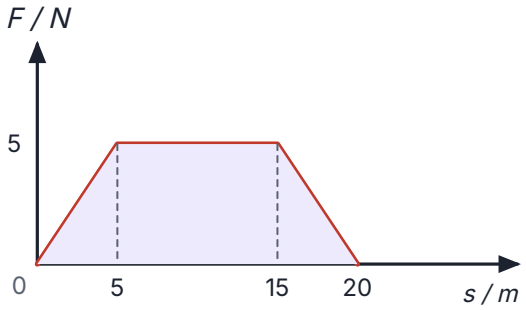
5. A 2.50 kg block is pushed 2.20 m along a frictionless horizontal table by a constant 16.0 N force directed 25.0° below the horizontal. Find the work done by (a) the applied force, (b) the normal force, (c) the gravitational force, (d) all the forces together.

Answer

(a) 31.9 J (b) 0 (c) 0 (both are perpendicular to the displacement) (d) 31.9 J

Example 4.1.4

Determine the work done on an object by the force F shown, if the object is displaced from s = 0 to s = 20 m.

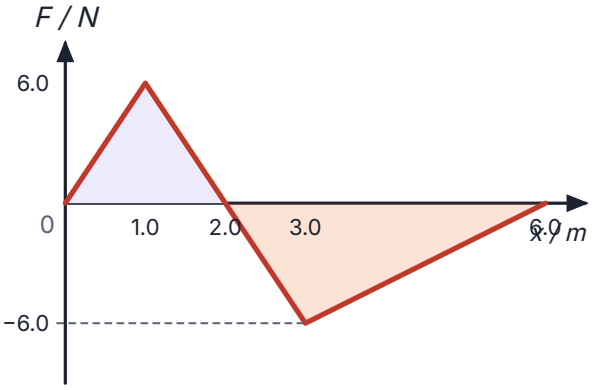


Solution

Work done = area under the graph = ½(5)(5) + (10)(5) + ½(5)(5) = **75 J**

Example 4.1.5

Calculate the work done by the variable force shown in the F–x graph.



Solution

Area above the axis (0 to 2.0 m) = ½(2.0)(6.0) = +6.0 J

Area below the axis (2.0 to 6.0 m) = ½(4.0)(−6.0) = −12 J

Total work = 6.0 − 12 = **−6 J**

4.2 Conservation of energy

Energy

is **the ability to do work**. Dimension $\mathrm{M\,L^2\,T^{-2}}$, unit joule (J), a scalar.

Two types of **mechanical energy**: potential energy, U, and kinetic energy, K. Two types of potential energy: gravitational, $U = mgh$, and elastic (spring), $U_s = \frac{1}{2}kx^2$.

Gravitational potential energy, U

Gravitational potential energy

is the **energy stored in a body or system because of its position**.

$$U = mgh$$

h = height of the body above the initial (reference) position

Elastic potential energy, U_s

Elastic potential energy

is the **energy stored in an elastic material as a result of stretching or compressing** it. Springs store elastic potential energy when compressed or extended.

Hooke's law

The restoring force, F_s, of a spring is **directly proportional to the extension or compression**, x, if the limit of proportionality is not exceeded.

$$F_s \propto -x \Rightarrow \boldsymbol{F_s = -kx}$$

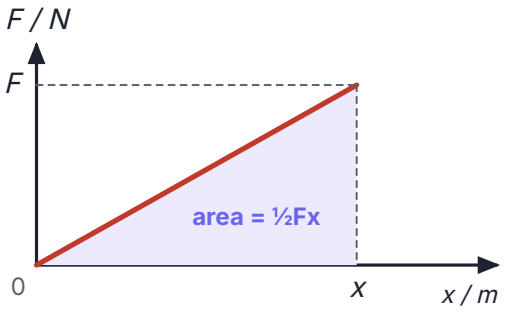
- k = spring constant (force constant); $x = x_f - x_i$.
- For calculation, if the object is stationary: $F_s = kx = F$ (the applied force).
- $[k] = \dfrac{[F_s]}{[x]} = \mathrm{M\,T^{-2}}$; unit kg $\mathrm{s^{-2}}$ or N $\mathrm{m^{-1}}$.

Work done by F to stretch the spring from its natural length by x: $W = \int_0^x F\,dx$, with $F = kx$

$W = \int_0^x kx\,dx = \tfrac{1}{2}kx^2$

$$U_s = \tfrac{1}{2}kx^2 = \tfrac{1}{2}F_s x$$

The work done equals the area under the straight-line F–x graph.



Example 4.2.1

A force of 600 N causes an extension of 10 cm in a spring. Find the elastic potential energy of the spring when (a) the extension is 12 cm, (b) a mass of 20 kg hangs vertically from it. (g = 9.81 m s^{-2})

Solution

$k = \dfrac{F}{x} = \dfrac{600}{0.1} = 6000$ N m^{-1}

(a) $U_s = \tfrac{1}{2}kx^2 = \tfrac{1}{2}(6000)(0.12)^2 =$ **43.2 J**

(b) $mg = kx \Rightarrow x = \dfrac{(20)(9.81)}{6000} = 3.27 \times 10^{-2}$ m; $U_s = \tfrac{1}{2}(6000)(3.27 \times 10^{-2})^2 =$ **3.21 J**

Exercises: springs

1. A 10 N force compresses a spring of spring constant 4.0×10^2 N m^{-1}. What is the compression?

   Answer

   2.5 cm

2. It takes 400 J of work to stretch a spring by 8.00 cm. What is the spring constant?

   Answer

   1.25×10^5 N m^{-1}

Kinetic energy, K

Kinetic energy

is the **energy of a body due to its motion**.

$$K = \tfrac{1}{2}mv^2$$

Work–energy theorem

Work–energy theorem

The **work done by the net force** on an object (the total work done on it) **equals the change in its kinetic energy**.

$$W_{\text{total}} = \Delta K = \tfrac{1}{2}mv^2 - \tfrac{1}{2}mu^2$$

A constant net force F acts on mass m through displacement s: $F = ma$

From $v^2 = u^2 + 2as$: $a = \dfrac{v^2 - u^2}{2s}$

$F = m\dfrac{v^2 - u^2}{2s} \Rightarrow Fs = \tfrac{1}{2}mv^2 - \tfrac{1}{2}mu^2 \Rightarrow \boldsymbol{W = K_f - K_i = \Delta K}$

Work done by a conservative force: W = −ΔU

The work done by a conservative force is equal to the **negative of the change in potential energy**.

- The work done by a **conservative** force (gravity, a spring) depends only on the initial and final positions; it is independent of the path.
- The work done against a **non-conservative** force (friction) depends on the path: the longer the path, the more work is done against friction.

When an object falls from h_1 to h_2, gravity does positive work and the potential energy decreases: $W_g = mg(h_1 - h_2) = U_i - U_f = -(U_f - U_i) = -\Delta U$

When an object is lifted, gravity does negative work and the potential energy increases: $W_g = -mg(h_2 - h_1) = -\Delta U$

For a spring: $W = -\Delta U_s = -(\tfrac{1}{2}kx_f^2 - \tfrac{1}{2}kx_i^2)$

Example 4.2.2

A 5.0 kg body is initially at rest on a horizontal frictionless surface. A 15 N force accelerates it to 12 m s^{-1}. Calculate (a) the distance travelled, (b) the work done by the force, (c) the final kinetic energy.

Solution

(a) $a = \dfrac{15}{5.0} = 3.0\ \text{m s}^{-2}$; $12^2 = 0 + 2(3.0)s \Rightarrow s =$ **24 m**

(b) $W = 15 \cos 0° \ (24) =$ **360 J**

(c) $K = \tfrac{1}{2}(5.0)(12)^2 =$ **360 J**, equal to the work done, as the work–energy theorem says

Example 4.2.3

How much work is needed for a 65 kg runner to accelerate from rest to 10.0 m s^{-1}?

Solution

$$W = \tfrac{1}{2}mv^2 - \tfrac{1}{2}mu^2 = \tfrac{1}{2}(65)(10.0)^2 - 0 = \textbf{3.25 kJ}$$

Example 4.2.4

A 4.1 kg box of books is lifted vertically from rest through 1.6 m by an upward force of 60.0 N. Find (a) the work done by the applied force, (b) the work done by gravity, (c) the final speed of the box.

Solution

(a) $W = 60.0 \cos 0° \, (1.6) =$ **96 J**

(b) $W = (4.1 \times 9.81) \cos 180° \, (1.6) =$ **−64 J**

(c) $W_{\text{total}} = 96 - 64 = 32 \text{ J} = \tfrac{1}{2}(4.1)v^2 - 0 \Rightarrow$ **$v = 3.9 \text{ m s}^{-1}$**

Exercises: work–energy theorem

1. Calculate the force required to stop a 1500 kg car travelling at 10 m s^{-1} within 500 m.

   Answer

   150 N

2. A 6.0 kg block initially at rest is pulled along a horizontal frictionless surface by a constant horizontal force of 12 N. Find its speed after it has moved 3.0 m.

   Answer

   3.5 m s^{-1}

3. A boy exerts a force of 11.0 N at 29.0° above the horizontal on a 6.40 kg box. Find the work done by the boy and the final speed of the box after it moves 2.00 m, if it starts at 0.500 m s^{-1} and slides without friction.

   Answer

   19.2 J; 2.50 m s^{-1}

4. An object is released from rest at P on a smooth plane inclined at 30°. At Q its speed is 3.2 m s^{-1}. Calculate the length PQ. (g = 9.81 m s^{-2})

   Answer

   1.0 m

5. A 2.00 kg block of ice slides 0.750 m from rest down a frictionless plane inclined at 36.9° below the horizontal. What is its final speed?

   Answer

   2.97 m s^{-1}

Principle of conservation of energy

Principle of conservation of energy

In an **isolated (closed) system**, the **total energy of the system is constant**. Energy may change form or be transferred from one system to another, but the total remains the same.

$$\Sigma E_i = \Sigma E_f \quad \text{(no external force)}$$

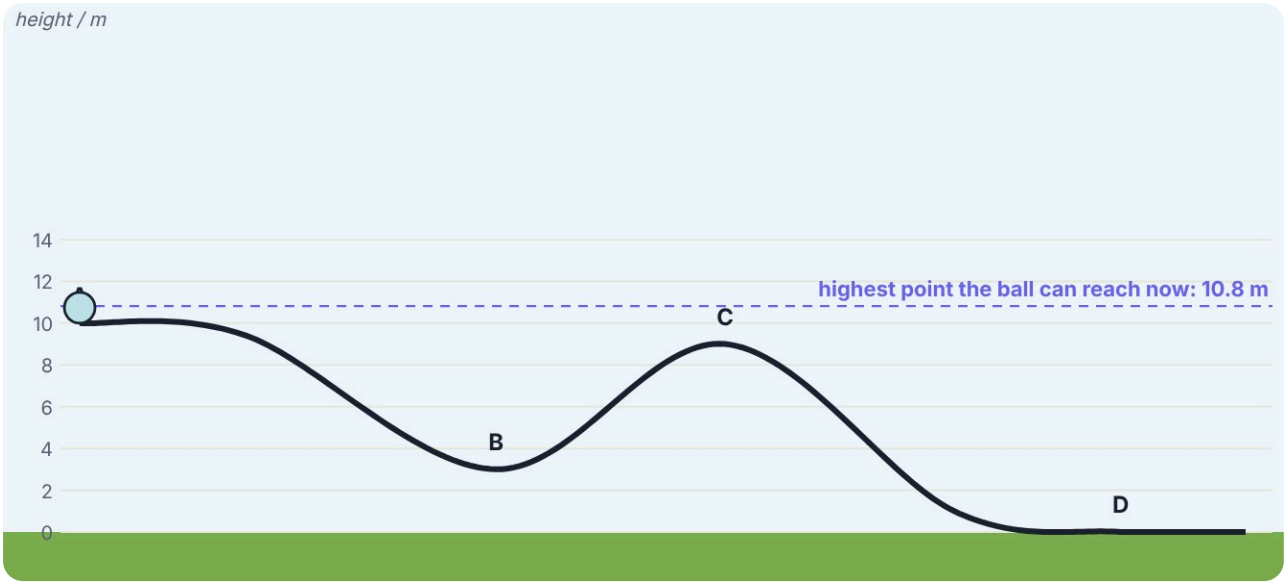
Without friction: $E = K + U$ = constant, so $K_i + U_i = K_f + U_f$

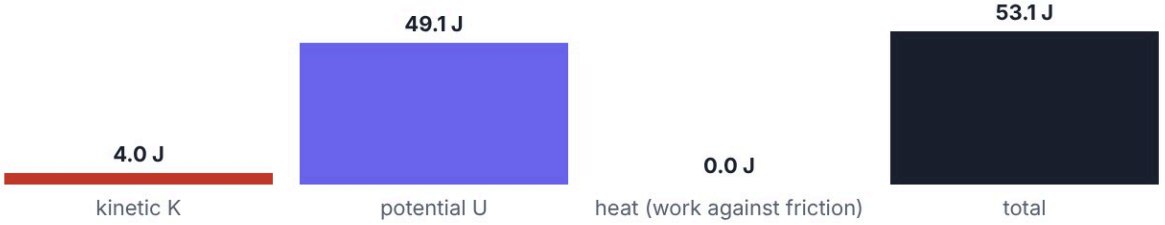
With friction, part of the mechanical energy becomes heat: $K_i + U_i = K_f + U_f +$ (work done against friction, fs).

Simulation

Energy on a track: kinetic, potential and heat

Release the ball from A and watch the energy bars. Without friction the total stays the same and the ball always reaches the same height. Turn on friction and the "heat" bar grows: that is the work done against friction.





speed	height	total energy at A	work done against friction
4.00 m/s	**10.00 m**	**53.1 J**	**0.0 J**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 4 to use the simulation.

Example 4.2.5

A cricket ball of mass 0.156 kg is thrown upward with a kinetic energy of 45 J. What is the maximum height it can reach?

Solution

$\Sigma E_i = \Sigma E_f$: $K_i + U_i = K_f + U_f \Rightarrow 45 + 0 = 0 + mgh$

$$h = \frac{45}{(0.156)(9.81)} = \mathbf{29.4\ m}$$

Example 4.2.6

A 1.20 kg piece of cheese is placed on a vertical spring of negligible mass and force constant 1800 N m^{-1} that is compressed 15.0 cm. When the spring is released, how high does the cheese rise from its initial position? (The cheese and spring are not attached; g = 9.81 m s^{-2}.)

Solution

$$K_i + U_i + U_{si} = K_f + U_f + U_{sf} \Rightarrow 0 + mgh_1 + \tfrac{1}{2}kx^2 = 0 + mgh_2 + 0$$

$$\tfrac{1}{2}kx^2 = mg(h_2 - h_1) = mg\Delta h$$

$$\Delta h = \frac{\tfrac{1}{2}(1800)(0.15)^2}{(1.20)(9.81)} = \mathbf{1.72\ m}$$

A real example: the bungee jump

- From the platform to the point where the cord becomes tight, the jumper falls freely: gravitational potential energy → kinetic energy.
- After that, the cord stretches like a spring: kinetic and gravitational potential energy → **elastic potential energy** $\tfrac{1}{2}kx^2$.
- The jumper is fastest where the cord's pull equals the weight ($kx = mg$), not where the cord first tightens. At the lowest point the speed is zero and all the energy lost from gravitational PE is stored in the cord.

Simulation

Bungee jump: where does the energy go?

Change the jumper's mass and the stiffness of the cord, then jump. Watch the energy bars: the total stays the same (apart from a little lost to air resistance) while it moves between gravitational, kinetic and elastic potential energy.



fallen so far	speed	cord stretch x	lowest point: stretch from $mg(L+x) = \frac{1}{2}kx^2$
0.0 m	**0.0 m s^{-1}**	**0.0 m**	**14.2 m**

fastest where kx = mg, at stretch
2.94 m

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 4 to use the simulation.

Example 4.2.7

A 60 kg jumper uses a bungee cord of unstretched length 20 m and force constant 200 N m^{-1}. Ignoring air resistance, find (a) how far the cord stretches at the lowest point, (b) the jumper's greatest speed.

Solution

(a) Lowest point (speed zero): loss of GPE = elastic PE: $mg(20 + x) = \tfrac{1}{2}kx^2 \Rightarrow (60)(9.81)(20 + x) = 100x^2$

$100x^2 - 589x - 11\,772 = 0 \Rightarrow x =$ **14.2 m** (total fall 34.2 m)

(b) Greatest speed where $kx = mg$: $x = \dfrac{(60)(9.81)}{200} = 2.94$ m

$\tfrac{1}{2}mv^2 = mg(20 + 2.94) - \tfrac{1}{2}kx^2 = 13\,503 - 864 = 12\,639$ J $\Rightarrow v =$ **20.5 m s^{-1}**

Exercises: conservation of energy

1. A body of mass 5 kg slides from rest down a smooth slope. Find its speed after it has fallen a vertical distance of 10 m.

   Answer

   14 m s^{-1}

2. A smooth wire is bent into a circle of radius a held in a vertical plane. A bead threaded on the wire is released from rest at the highest point. Find its speed at the lowest point.

   Answer

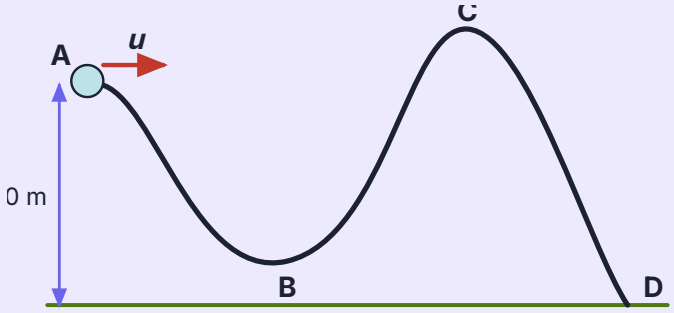
   $\sqrt{4ag}$

3. A 1.70 kg block slides on a horizontal frictionless surface into a spring of force constant 955 N m^{-1}. The block stops after compressing the spring 4.60 cm. Find its initial speed.

   Answer

   1.09 m s^{-1}

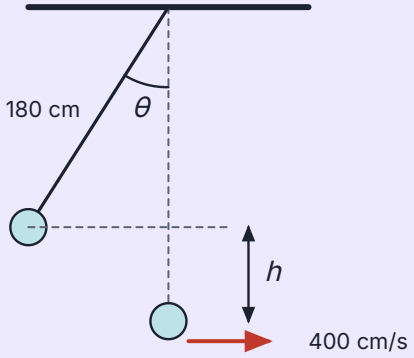
4. A 0.50 kg ball is at A, 10 m high, with initial speed 4 m s^{-1} (no friction). Calculate (a) the total energy at A, (b) the speed at B, 3 m high, (c) the speed at D on the ground, (d) the maximum height of C so that the ball can pass over it.



Answer

(a) 53.1 J (b) 12.4 m s^{-1} (c) 14.6 m s^{-1} (d) 10.8 m

5. A ball on the end of a 180 cm string swings as a pendulum. Its speed at the lowest position is 400 cm s^{-1}. (a) To what height h above this position does it rise before stopping? (b) What angle does the string then make with the vertical?



Answer

(a) 0.816 m (b) 56.9°

6. A 0.50 kg ball is thrown vertically upward with an initial kinetic energy of 80 J. Taking the launch point as the zero of potential energy, find (a) its kinetic and potential energies when it has travelled three-quarters of the way to its maximum height, (b) its speed there.

Answer

(a) K = 20 J, U = 60 J (b) 8.9 m s^{-1}

4.3 Power and mechanical efficiency

Power

is the **rate of doing work** or the **rate of energy transfer**.

$$P_{\text{av}} = \frac{\Delta W}{\Delta t} = \frac{\Delta E}{\Delta t}$$

Average power

$$P = \lim_{\Delta t \to 0} \frac{\Delta W}{\Delta t} = \frac{\mathrm{d}W}{\mathrm{d}t}$$

Instantaneous power

- Power is a **scalar**.
- $[P] = \dfrac{[\Delta W]}{[\Delta t]} = \dfrac{\mathrm{M\,L^2\,T^{-2}}}{\mathrm{T}} = \mathrm{M\,L^2\,T^{-3}}$; unit kg m^2 s^{-3} = J s^{-1} = watt (W).

$P = \dfrac{\mathrm{d}W}{\mathrm{d}t} = \dfrac{(F\cos\theta)\,\mathrm{d}s}{\mathrm{d}t}$ and $\dfrac{\mathrm{d}s}{\mathrm{d}t} = v$

$$P = Fv\cos\theta = \vec{F}\cdot\vec{v}$$

Example 4.3.1

A 1200 kg car can accelerate from rest to 25 m s^{-1} in 8.0 s. What average power must the motor produce? (Ignore friction.)

Solution

$$P_{\text{av}} = \frac{K_{\text{f}} - K_{\text{i}}}{\Delta t} = \frac{\tfrac{1}{2}(1200)(25)^2 - 0}{8.0} = \mathbf{47\ kW}$$

Example 4.3.2

An elevator of mass 1 Mg carries 10 passengers through a height of 15 m in 40 s. Calculate the average power required by the motor if no energy is lost. (g = 9.81 m s^{-2}; average mass per passenger 60 kg)

Solution

M = 1000 + (60 × 10) = 1600 kg

Work done = change in potential energy = $mgh_f - mgh_i$, with $h_i = 0$

$$P = \frac{Mgh}{\Delta t} = \frac{(1600)(9.81)(15)}{40} = \mathbf{5886\ W}$$

Example 4.3.3

A 100 kg block is pulled at a constant 5.0 m s^{-1} across a horizontal floor by a 122 N force at 37° above the horizontal. At what rate does the force do work on the block?

Solution

$$P = Fv\cos\theta = (122)(5.0)\cos 37° = \mathbf{487\ W}$$

Exercises: power

1. A 50 kg person runs 200 m up a straight road inclined at 20° in 50 s. Neglecting friction and air resistance, calculate (a) the work done, (b) the average power. (g = 9.81 m s^{-2})

   Answer

   (a) 3.36 × 10^4 J (b) 672 W

2. You push a 75 kg box across a floor where μ_k = 0.65 with a horizontal force. (a) How much power is needed to push it at 0.50 m s^{-1}? (b) How much work do you do in 35.0 s?

   Answer

   (a) 240 W (b) 8.4 kJ

Mechanical efficiency

Efficiency, η

is the **ratio of the useful work done (output) to the energy input**.

$$\eta = \frac{W_{out}}{E_{in}} \times 100\% = \frac{P_{out}}{P_{in}} \times 100\%$$

Example 4.3.4

An electric motor with a 2.0 horsepower input drives a machine with an efficiency of 45%. What is the energy output of the machine per second? (1 hp = 746 W)

Solution

$$0.45 = \frac{P_{out}}{2.0 \times 746} \Rightarrow P_{out} = 6.71 \times 10^2 \text{ W, so } \mathbf{6.71 \times 10^2\ J} \text{ per second}$$

Example 4.3.5

A machine lifts a 10 kg load through 2.5 m. The machine needs a force of 150 N to move 2 m. Determine its efficiency.

Solution

$W_{in} = F \cos\theta\ s = 150 \cos 0° (2) = 300$ J

$W_{out} = mgh = (10)(9.81)(2.5) = 245$ J

$\eta = \frac{245}{300} \times 100\% = \mathbf{81.7\%}$

Exercises: efficiency

1. The motor of an electric drill with an efficiency of 80% has a power input of 600 W. How much useful work is done by the drill in 30 s?

Answer

1.4×10^4 J

2. A hoist lifts a 3000 kg load through 8.00 m in 20.0 s. The power supplied to the engine is 18.0 hp. Calculate (a) the work output, (b) the power output and power input, (c) the efficiency of the engine and hoist. (1 hp = 746 W)

Answer

(a) 235 kJ (b) 11.8 kW; 13.4 kW (c) 87.7%

Chapter 5 · Semester 1

CIRCULAR MOTION

Notes by Cikgu Suhaili

13 worked examples 2 simulations Exercises with answers

Circular motion

Circular motion

is the motion of a body **along a circular path**.

Uniform circular motion (horizontal plane)	Non-uniform circular motion (vertical plane)
a ball swung in a horizontal circle	a bucket of water swung in a vertical circle
a car or motorcycle turning a corner (flat or banked)	a roller coaster car on a loop
a conical pendulum; a merry-go-round	

Uniform circular motion

is the motion of an object in a circle with a **constant speed**: the magnitude of its velocity stays constant but the **direction of its velocity changes continually**.

Linear motion	Circular motion
displacement, s (m)	angular displacement, θ (rad)
velocity, $v = \frac{ds}{dt}$ (m s^{-1})	angular velocity, $\omega = \frac{d\theta}{dt}$ (rad s^{-1})

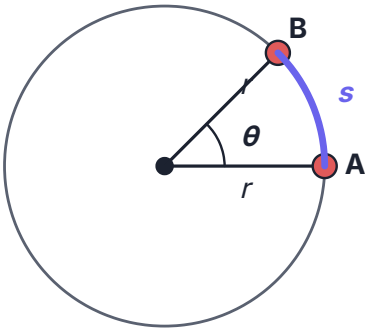
5.1 Angular displacement and angular velocity

Angular displacement, θ

is the **angle subtended at the centre by the arc length** travelled. Unit: radian (rad).

Linear distance, s

is the arc length between A and B.



$$s = r\theta$$

1 rad is the angle subtended at the centre by an arc equal in length to the radius; 2π rad = 360°.

Angular velocity, ω

is the **rate of change of angular displacement**: $\omega = \frac{d\theta}{dt}$.

- Unit: rad s^{-1}; other units: revolutions per second (rps) and revolutions per minute (rpm).
- It is a vector; its direction is perpendicular to the plane of motion (right-hand rule) and depends on whether the rotation is clockwise or anticlockwise.

Relationship between v and ω

$s = r\theta$. Divide both sides by t: $\frac{s}{t} = r\frac{\theta}{t}$

$$v = r\omega$$

The linear velocity at every point is along the tangent to the circle.

Period, T

is the time taken to make one complete revolution (unit: s).

Frequency, f

is the number of complete revolutions per second: $f = 1/T$ (unit: s^{-1} or Hz).

$$v = \frac{2\pi r}{T} = 2\pi r f \quad \omega = \frac{2\pi}{T} = 2\pi f$$

Example 5.1.1

An object moves in a circle with a uniform angular speed of 100 rpm. Calculate (a) the period, T, (b) the frequency, f.

Solution

$\omega = \frac{100 \text{ rev}}{1 \text{ min}} = \frac{100 \times 2\pi}{60} = 10.47 \text{ rad s}^{-1}$

(a) $T = \frac{2\pi}{\omega} = \frac{2\pi}{10.47} = \mathbf{0.60\ s}$

(b) $f = \frac{1}{T} = \frac{1}{0.60} = \mathbf{1.67\ Hz}$

Example 5.1.2

An object travels round a circle of radius 6 m at 30 rev min^{-1}. Calculate (a) its angular speed in rad s^{-1}, (b) its linear speed.

Solution

(a) $\omega = \frac{30 \times 2\pi}{60} = \mathbf{\pi\ rad\ s^{-1}}$ (3.14 rad s^{-1})

(b) $v = r\omega = 6\pi = \mathbf{18.85\ m\ s^{-1}}$

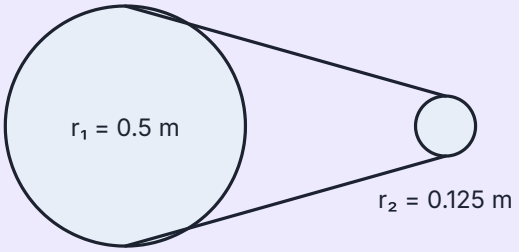
Exercises: angular velocity

1. A particle moves on a circular path of radius 0.5 m at a constant speed of 10 m s^{-1}. Calculate the time taken to complete 20 revolutions.

   Answer

   6.28 s

2. Two wheels of a machine are connected by a belt. The radius of the first wheel is 0.5 m and of the second 0.125 m. The bigger wheel turns at 3.5 Hz. What is the frequency of the smaller wheel?



Answer

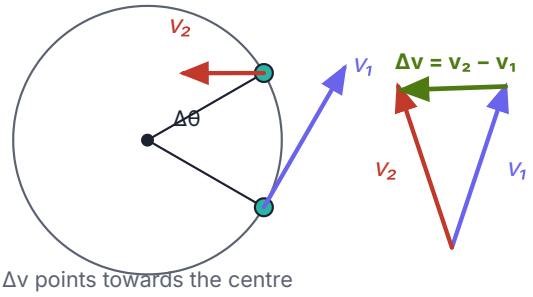
14 Hz (the belt gives both rims the same linear speed: $f_1 r_1 = f_2 r_2$)

5.2 Centripetal acceleration

An object moving in a circle of radius r at constant speed v keeps changing direction, so its velocity changes and it has an acceleration: the **centripetal acceleration**.

> **Centripetal acceleration, a_c**
>
> is the acceleration of an object moving in a circular path, **directed towards the centre of the circle**.

Direction, graphically: $a = \frac{\Delta \mathbf{v}}{\Delta t}$. As $\Delta t \to 0$, the change $\Delta \mathbf{v} = \mathbf{v}_2 - \mathbf{v}_1$ points towards the centre.



Δv points towards the centre

Additional derivation (syllabus 5.2(b)) · $a = v^2/r$

Derive the formula for centripetal acceleration.

Solution

In a short time Δt the object moves through an arc $\Delta s = v\Delta t$ and the radius turns through $\Delta\theta$.

The velocity vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ (both of magnitude v) also turn through $\Delta\theta$, so the velocity triangle and the radius triangle are similar: $\dfrac{\Delta v}{v} = \dfrac{\Delta s}{r}$

$$\Delta v = \frac{v\Delta s}{r} = \frac{v(v\Delta t)}{r} \Rightarrow a = \frac{\Delta v}{\Delta t} = \frac{v^2}{r}$$

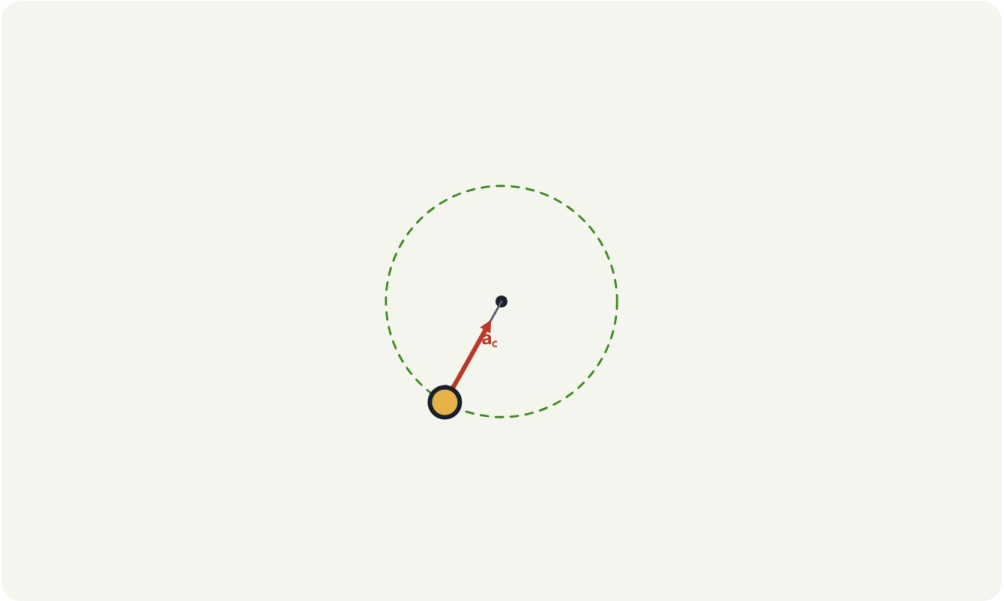
Since $v = r\omega$: $a = \dfrac{(r\omega)^2}{r} = r\omega^2$

$$a_c = \frac{v^2}{r} = r\omega^2 = v\omega = \frac{4\pi^2 r}{T^2}$$

Simulation

Uniform circular motion: velocity and acceleration

Change the radius and the angular velocity. The velocity arrow is always along the tangent and the acceleration arrow always points to the centre. Press "Cut the string" to see the object fly off along the tangent.



speed $v = r\omega$	period $T = 2\pi/\omega$	centripetal acceleration v^2/r	centripetal force mv^2/r
3.00 m/s	**4.19 s**	**4.50 m/s²**	**2.25 N**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 5 to use the simulation.

Example 5.2.1

Calculate the centripetal acceleration of a car travelling on a circular racetrack of radius 1000 m at 180 km h^{-1}.

Solution

$$v = \frac{180(1000)}{60(60)} = 50 \text{ m s}^{-1}$$

$$a_c = \frac{v^2}{r} = \frac{50^2}{1000} = \mathbf{2.5 \text{ m s}^{-2}}$$

5.3 Centripetal force

Centripetal force, F_c

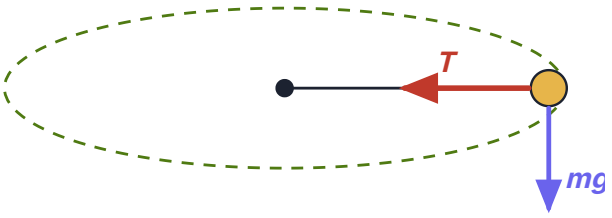
is the **net force required to keep an object of mass m moving at speed v on a circular path** of radius r.

$$F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$$

- Direction: towards the centre of the circle.
- F_c is perpendicular to **v**, so it does **no work** on the object.
- It is not a new kind of force: it is provided by another force. For the Moon orbiting the Earth, gravity provides it; for a car turning a corner, friction on the tyres provides it.

Uniform circular motion in a horizontal plane

Case 1: object whirled in a horizontal circle at steady speed. Two forces act: the weight mg, and the tension in the string. The tension is the only force in the radial direction, so it provides the centripetal force.



$$\Sigma F = ma_c \Rightarrow T = \frac{mv^2}{r}$$

Example 5.3.1

A 0.25 kg rock on a string 1.0 m long is whirled in a horizontal circle at a constant 10.0 m s^{-1}. Neglecting gravity, find the tension.

Solution

$$T = F_c = \frac{mv^2}{r} = \frac{0.25(10)^2}{1.0} = \mathbf{25\ N}$$

Case 2: a car going round a curve

(a) Flat curve. Static friction provides the centripetal force; $f_s = \mu R$.

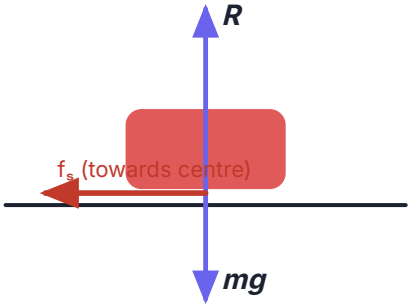
Vertical: $\Sigma F_y = 0 \Rightarrow R = mg$ (1)

Horizontal: $f_s = F_c \Rightarrow \mu R = \frac{mv^2}{r}$ (2)

(1) into (2): $\mu mg = \frac{mv^2}{r}$

$$v_{max} = \sqrt{\mu r g}$$

maximum speed without slipping



Example 5.3.2

A car travels round a flat curve of radius 50 m. μ_s between the tyres and the road is 0.75. Calculate the maximum speed without skidding.

Solution

$$f_s = \frac{mv^2}{r} \text{ and } f_s = \mu_s mg \Rightarrow v = \sqrt{\mu_s r g} = \sqrt{(0.75)(50)(9.81)} = \mathbf{19.18\ m\ s^{-1}}$$

Example 5.3.3

A 1200 kg car at 8.0 m s^{-1} goes round a flat curve of radius 9.0 m. (a) What horizontal force must the road exert on the tyres? (b) What coefficient of friction is needed for the car not to slip?

Solution

(a) $f_s = \frac{mv^2}{r} = \frac{1200(8.0)^2}{9.0} = \mathbf{8533\ N}$

(b) $\mu mg = \frac{mv^2}{r} \Rightarrow \mu = \frac{v^2}{rg} = \frac{(8.0)^2}{9.0(9.81)} = \mathbf{0.72}$

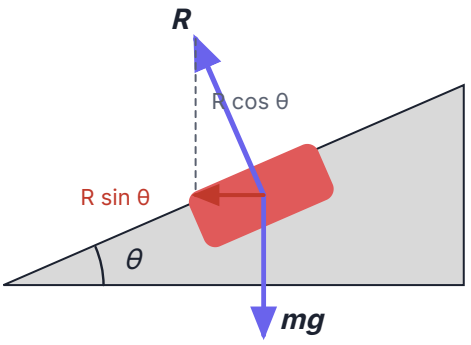
(b) Frictionless banked road. The horizontal component of the normal reaction provides the centripetal force.

x: $R\sin\theta = \frac{mv^2}{r}$ (1)

y: $R\cos\theta = mg$ (2)

(1) ÷ (2): $\tan\theta = \frac{v^2}{rg}$

$$v = \sqrt{rg\tan\theta}$$



Example 5.3.4

A curve of radius 30 m is to be banked so that a car can take it at 13 m s^{-1} without depending on friction. What must the banking angle be?

Solution

$$\tan\theta = \frac{v^2}{rg} \Rightarrow \theta = \tan^{-1}\frac{13^2}{30(9.81)} = \mathbf{30°}$$

Case 3: conical pendulum

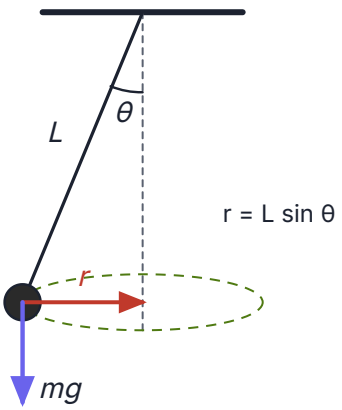
The bob moves in a horizontal circle of radius $r = L\sin\theta$. **$T\sin\theta$ provides the centripetal force.**

x: $T\sin\theta = \frac{mv^2}{r}$ (1)

y: $T\cos\theta = mg$ (2)

(1) ÷ (2): $\tan\theta = \frac{v^2}{rg}$

$$v = \sqrt{rg\tan\theta} = \sqrt{Lg\sin\theta\tan\theta}$$



Example 5.3.5

A 0.15 kg ball on a 1.2 m string moves in a horizontal circle; the string makes 30° with the vertical. Find the tension and the speed of the ball.

Solution

y: $T\cos 30° = mg \Rightarrow T = \dfrac{0.15(9.81)}{\cos 30°} = \mathbf{1.70\ N}$

x: $T\sin\theta = \dfrac{mv^2}{r}$ with $r = L\sin\theta = 0.60$ m $\Rightarrow v = \sqrt{\dfrac{rT\sin\theta}{m}} = \mathbf{1.84\ m\ s^{-1}}$

Exercise: conical pendulum

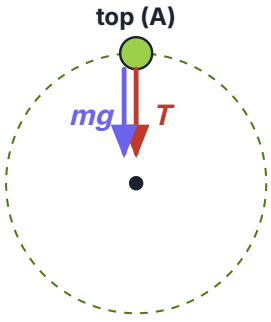
1. A pendulum bob of mass 1 kg on a string 1 m long revolves in a horizontal circle of radius 60 cm. Calculate the period of the motion and the tension in the string.

Answer

$T = 12.26$ N; period $= \dfrac{2\pi r}{v} = 1.8$ s

Circular motion in a vertical plane (non-uniform)

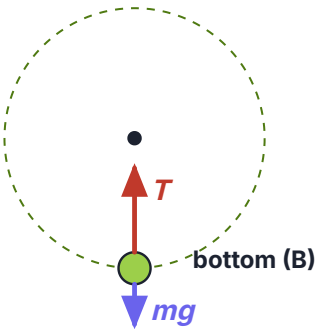
Case 1: a ball on a string moving in a vertical circle.



At the top (A): T and mg both point down, towards the centre.

$$T + mg = \frac{mv^2}{r} \Rightarrow T = \frac{mv^2}{r} - mg$$

T is minimum at the top.



At the bottom (B): T points up (to the centre), mg down.

$$T - mg = \frac{mv^2}{r} \Rightarrow T = \frac{mv^2}{r} + mg$$

T is maximum at the bottom.

Example 5.3.6

A 1.2 kg rock tied to a 90 cm string is whirled in a vertical circle at a constant 8 m s^{-1}. What are the tensions at the top and bottom of the circle?

Solution

m = 1.2 kg, r = 0.90 m, $\frac{mv^2}{r} = \frac{1.2(8)^2}{0.90}$ = 85.3 N

Top: T = 85.3 − (1.2)(9.81) = **73.6 N**

Bottom: T = 85.3 + (1.2)(9.81) = **97.1 N**

Minimum (critical) speed at the top

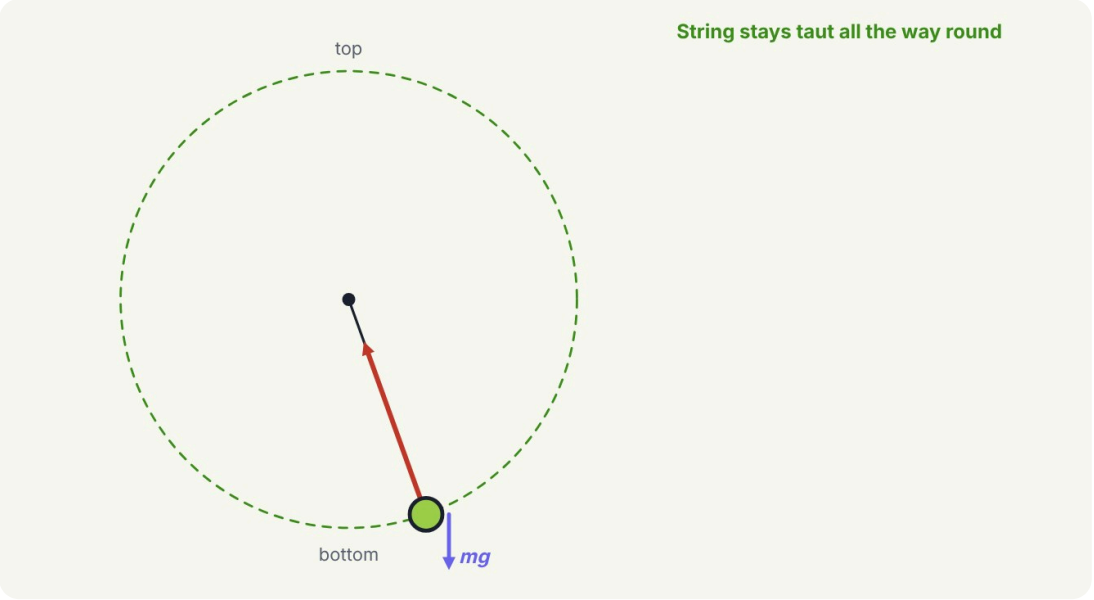
Circular motion is possible only while the string stays taut. If the string just becomes slack at the top, $T = 0$:

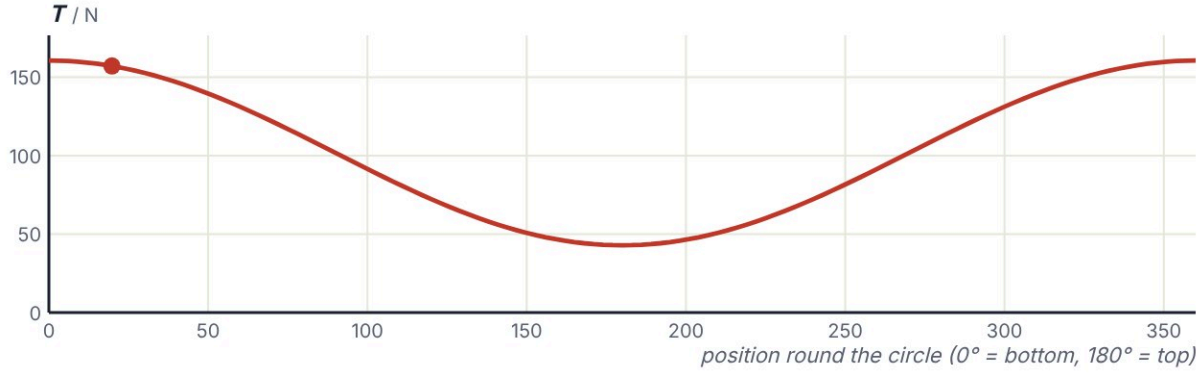
$$mg = \frac{mv^2}{r} \Rightarrow v_{\text{min}} = \sqrt{rg}$$

Simulation

Vertical circle: tension and the minimum speed

Set the speed at the top. The ball keeps its total energy, so it moves faster at the bottom. Watch the tension around the circle: below √(rg) at the top, the string goes slack.





speed now	tension now	T at top = $mv^2/r - mg$	T at bottom = $mv^2/r + mg$
7.45 m/s	**157.1 N**	**42.9 N**	**160.6 N**

minimum speed at top √(rg)
2.80 m/s

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 5 to use the simulation.

Example 5.3.7

A 2 kg ball on an 80 cm string is whirled in a vertical circle and has a speed of 5 m s^{-1} at the top. (a) What is the tension at that instant? (b) What is the minimum speed at the top needed to keep moving in the circle?

Solution

(a) $T = \frac{mv^2}{r} - mg = \frac{2(5)^2}{0.8} - 2(9.81) =$ **42.9 N**

(b) For v_{min}, $T = 0$: $v_{min} = \sqrt{rg} = \sqrt{(0.8)(9.81)} =$ **2.8 m s^{-1}**

Example 5.3.8

A bucket of water is swung in a vertical circle of radius 0.70 m. Calculate the minimum speed at the top so that the water does not spill.

Solution

The water does not spill if $T \geq 0$; at the limit $T = 0$: $v_{min} = \sqrt{rg} = \sqrt{(0.7)(9.81)} =$ **2.62 m s^{-1}**

Case 2: roller coaster on a circular track. The normal reaction R of the track replaces the tension.

Top: $R + mg = \frac{mv^2}{r} \Rightarrow R = \frac{mv^2}{r} - mg$ Bottom: $R - mg = \frac{mv^2}{r} \Rightarrow R = \frac{mv^2}{r} + mg$

A minimum speed (when $R = 0$) is needed to keep the car on the track: $v_{min} = \sqrt{rg}$.

Example 5.3.9

What minimum speed must a roller coaster have when upside down at the top of a loop of radius 8.0 m if the passengers are not to fall out?

Solution

At the top, for v_{min}, $R = 0$: $v_{min} = \sqrt{rg} = \sqrt{(8.0)(9.81)} =$ **8.9 m s^{-1}**

Chapter 6 · Semester 1

GRAVITATION

Notes by Cikgu Suhaili

29 worked examples 2 simulations Exercises with answers

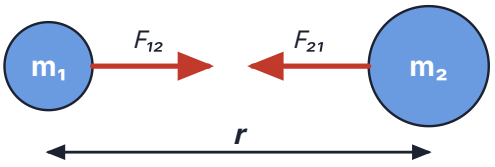
6.1 Newton's law of universal gravitation

Newton's law of universal gravitation

Every particle in the universe attracts every other particle with a force that is **directly proportional to the product of their masses** and **inversely proportional to the square of the distance between their centres**. The force acts along the line joining the two particles.

$$F = \frac{Gm_1m_2}{r^2}$$

G = universal gravitational constant = 6.67×10^{-11} N m^2 kg^{-2}



- F_{12} = force on object 1 by object 2; F_{21} = force on object 2 by object 1; $|F_{12}| = |F_{21}|$ (an action–reaction pair), even when the masses are different.
- The gravitational force is always **attractive** and is a **conservative** force.
- The weight of an object on Earth is the gravitational force between the object and the Earth.
- The gravitational force between the Earth and the Sun provides the centripetal force for the Earth's orbit.

Example 6.1.1

Calculate the gravitational force between two bodies of mass 3.0 kg and 2.0 kg whose centres are 50 cm apart.

Solution

$$F = \frac{(6.67 \times 10^{-11})(3.0)(2.0)}{(0.50)^2} = \mathbf{1.60 \times 10^{-9}\ N}$$

Example 6.1.2

Two objects attract each other with a force of 2.5×10^{-10} N when they are 0.25 m apart. Their total mass is 4.0 kg. Find their individual masses.

Solution

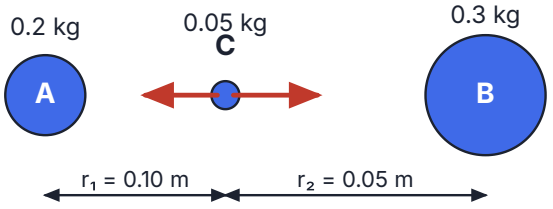
$m_1 m_2 = \dfrac{Fr^2}{G} = \dfrac{(2.5 \times 10^{-10})(0.25)^2}{6.67 \times 10^{-11}} = 0.234$; and $m_1 + m_2 = 4.0$

$m_1(4.0 - m_1) = 0.234 \Rightarrow m_1^2 - 4.0m_1 + 0.234 = 0$

$m_1 = 3.9$ kg, $m_2 = 0.1$ kg

Example 6.1.3

Particles A (0.2 kg) and B (0.3 kg) are 0.15 m apart. A third particle C (0.050 kg) is placed between them on the line joining them. Calculate (a) the gravitational force on C if it is 0.05 m from B, (b) the point on the line where there is no resultant gravitational force on C.



Solution

(a) $F_{CA} = G\dfrac{(0.05)(0.2)}{(0.10)^2} = 6.7 \times 10^{-11}$ N; $F_{CB} = G\dfrac{(0.05)(0.3)}{(0.05)^2} = 40.0 \times 10^{-11}$ N

Net force = $F_{CB} - F_{CA}$ = **33.3×10^{-11} N towards B**

(b) Let C be x from A: $G\dfrac{m_C m_A}{x^2} = G\dfrac{m_C m_B}{(0.15 - x)^2} \Rightarrow \dfrac{0.15 - x}{x} = \sqrt{\dfrac{0.3}{0.2}}$

$x = 0.067$ m from A

Exercises: Newton's law of gravitation

1. One small ball has mass 0.00150 kg and another 0.870 kg. Their centres are 0.100 m apart. Find the gravitational force each exerts on the other.

   Answer

   8.70×10^{-12} N

2. Three particles lie on a straight line, far from other objects: A (363 kg), B (517 kg) 0.500 m to the right of A, and C (154 kg) 0.250 m to the right of B. Find the net gravitational force on (a) A, (b) B, (c) C.

   Answer

   (a) 5.67×10^{-5} N to the right (b) 3.49×10^{-5} N to the right (c) 9.16×10^{-5} N to the left

3. The mass of the Earth is 6.0×10^{24} kg and of the Moon 7.4×10^{22} kg. Their centres are 3.8×10^{8} m apart. Find the point on the line between them where the net gravitational force is zero.

   Answer

   3.42×10^{8} m from the centre of the Earth

6.2 Gravitational field

Gravitational field

is a **region in which an object with mass experiences a force**.

Gravitational field strength, g

is the **gravitational force per unit mass** acting on a small mass (test mass) placed at that point in the field.

$$g = \frac{F}{m} = \frac{GM}{r^2}$$

- Unit: N kg^{-1} (equivalent to m s^{-2}).
- g is a vector, directed towards the centre of the mass M.
- g is also the **acceleration due to gravity**.

Gravitational force on mass m at distance r from a planet of mass M: $F = \frac{GMm}{r^2}$ (1)

If released, it falls towards the planet with acceleration a_g: $F = ma_g$ (2)

(1) = (2): $a_g = \frac{GM}{r^2}$. At the Earth's surface ($r = R_E$): $g = \frac{GM_E}{R_E^2}$

Equivalently, weight $mg = \frac{GM_E m}{R_E^2}$, so $GM_E = gR_E^2$

Example 6.2.1

Taking the distance from a baseball to the centre of the Earth as the Earth's radius, 6.38×10^6 m, find the mass of the Earth. ($g = 9.81\ \text{m s}^{-2}$)

Solution

$$M_E = \frac{gR_E^2}{G} = \frac{(9.81)(6.38 \times 10^6)^2}{6.67 \times 10^{-11}} = \mathbf{5.99 \times 10^{24}\ kg}$$

Example 6.2.2

On a distant planet the acceleration due to gravity is $5.00\ \text{m s}^{-2}$ and the radius is about 4560 km. Estimate the mass of the planet.

Solution

$$M = \frac{gr^2}{G} = \frac{(5.00)(4.56 \times 10^6)^2}{6.67 \times 10^{-11}} = \mathbf{1.56 \times 10^{24}\ kg}$$

Example 6.2.3

The Earth is a uniform sphere of radius 6.4×10^6 m and mean density $5500\ \text{kg m}^{-3}$. Calculate the acceleration due to gravity at its surface.

Solution

$M = \rho(\frac{4}{3}\pi R^3)$, so $g = \frac{GM}{R^2} = \frac{4}{3}\pi G\rho R$

$g = \frac{4}{3}\pi(6.67 \times 10^{-11})(5500)(6.4 \times 10^6) = \mathbf{9.83\ m\ s^{-2}}$

Example 6.2.4

An object weighs 100 N on the Earth's surface. What does it weigh on the Moon's surface? (r_m = $0.25r_E$, M_m = $0.01M_E$)

Solution

$$W_E = \frac{GM_E m}{r_E^2} = 100 \text{ N}$$

$$W_m = \frac{G(0.01M_E)m}{(0.25r_E)^2} = \frac{GM_E m}{r_E^2}\left(\frac{0.01}{(0.25)^2}\right) = 100 \times 0.16 = \mathbf{16\ N}$$

Exercises: field strength

1. The mass of the Earth is about 81 times the mass of the Moon and its radius is four times the Moon's. Find the acceleration due to gravity on the Moon. (g_E = 9.81 m s^{-2})

   Answer

   1.94 m s^{-2}

2. A planet has 1.5 times the radius of the Earth but the same mass. What is the acceleration due to gravity at its surface?

   Answer

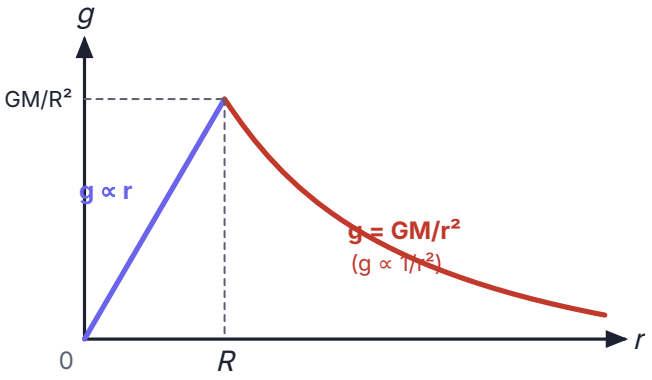
   4.36 m s^{-2}

Variation of g with distance from the centre of the Earth

On the surface ($r = R_E$): $g = \dfrac{GM_E}{R_E^2}$, constant if the Earth's radius is uniform.

Above the surface ($r > R_E$, $r = R_E + h$): $g = \dfrac{GM}{r^2}$, so $g \propto \dfrac{1}{r^2}$; $g \to 0$ as $r \to \infty$.

Below the surface ($r < R_E$), uniform density: only the mass inside radius r counts, $M = \rho(\frac{4}{3}\pi r^3)$, so $g = G\rho(\frac{4}{3}\pi r)$: $g \propto r$. g decreases linearly with depth.



Example 6.2.5

The Earth is a uniform sphere of radius 6.4×10^6 m and mass 6.0×10^{24} kg. Calculate g (i) at the surface, (ii) at a height of 0.5 times its radius above the surface.

Solution

(i) $g = \dfrac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{(6.4 \times 10^6)^2} = \mathbf{9.8\ N\ kg^{-1}}$

(ii) $r_1 = r + 0.5r = 1.5r$; $\dfrac{g_1}{g} = (\dfrac{r}{1.5r})^2 = 0.444 \Rightarrow g_1 = 0.444 \times 9.8 = \mathbf{4.35\ N\ kg^{-1}}$

Variation of g with latitude

- The Earth spins with angular velocity $\omega = \dfrac{2\pi}{T} = 7.27 \times 10^{-5}$ rad s^{-1} (T = 24 h).
- A body at latitude λ moves in a circle of radius $R\cos\lambda$, so part of the gravitational pull provides the centripetal force: $g_{\text{eff}} = \dfrac{GM}{R^2} - \omega^2 R\cos^2\lambda$.
- At the **equator** ($\lambda = 0°$) g is a minimum; at the **poles** ($\lambda = 90°$, $\cos\lambda = 0$) g is a maximum.
- The Earth is also slightly flattened (its radius is larger at the equator), which makes g larger at the poles as well.

STPM: calculations of the variation with latitude are limited to the equator and the poles.

Example · g at the equator and at the poles

Given $R = 6.37 \times 10^6$ m, $\omega = 7.27 \times 10^{-5}$ rad s^{-1} and $\frac{GM}{R^2} = 9.83$ N kg^{-1}, find g at the poles and at the equator.

Solution

$\omega^2 R = (7.27 \times 10^{-5})^2 (6.37 \times 10^6) = 0.0337$ m s^{-2}

Poles: $g = 9.83$ N kg^{-1}; equator: $g = 9.83 - 0.034 = 9.80$ N kg^{-1} (a difference of about 0.3%)

6.3 Gravitational potential

Gravitational potential energy, U

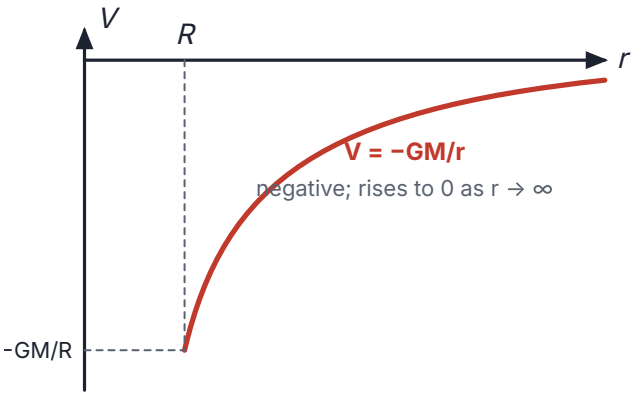
of a mass m at a point is the **work done by an external force in bringing the mass from infinity to that point**.

Bringing m from infinity to r, the external force balances gravity: $W = \int_\infty^r \frac{GM_E m}{r^2}\,dr$ (taken towards the Earth)

$$W = -\frac{GM_E m}{r} = U$$

$$U = -\frac{GMm}{r}$$

Unit: J



- At the Earth's surface, $U = -\frac{GM_E m}{R_E}$.
- The **minus sign** shows that the potential energy at the surface is less than at infinity (where $U = 0$). A body at infinity would "fall" towards the Earth; a body on the Earth does not "fall" to infinity.

Gravitational potential, V

at a point is the **gravitational potential energy per unit mass** of a small test mass placed at that point; equivalently, the work done per unit mass by an external force in bringing a small mass from infinity to that point.

$$V = \frac{U}{m} = -\frac{GM}{r}$$

Unit: J kg^{-1}

Variation of V with distance from the Earth

- For $r \geq R_E$, $V = -GM/r$: always negative, and it increases (becomes less negative) as r increases, with $V \to 0$ as $r \to \infty$.
- V is most negative at the surface: $V_{surface} = -\frac{GM}{R_E} = -gR_E \approx -6.3 \times 10^7$ J kg^{-1}.
- $V \propto 1/r$ while $g \propto 1/r^2$, so V falls off more slowly than g.

Example 6.3.1

Find the gravitational potential at the surface of the Earth, if the surface field strength is 9.8 N kg^{-1} and the mean radius is 6400 km.

Solution

$$g = \frac{GM_E}{R_E^2} \Rightarrow \frac{GM_E}{R_E} = gR_E; \quad V = -gR_E = -9.8(6.4 \times 10^6) = \mathbf{-6.3 \times 10^7\ J\ kg^{-1}}$$

Example 6.3.2

A 200 kg satellite orbits at an altitude of 2.00×10^6 m. Calculate its gravitational potential energy. ($M_E = 6.0 \times 10^{24}$ kg, $R_E = 6.4 \times 10^6$ m)

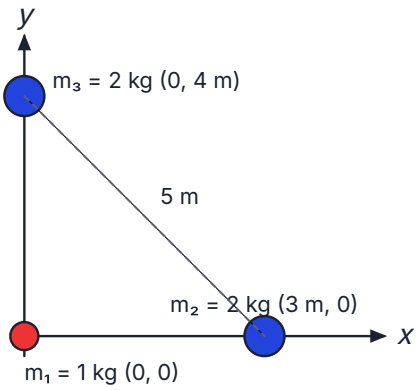
Solution

$r = R_E + h = 6.4 \times 10^6 + 2.00 \times 10^6 = 8.4 \times 10^6$ m

$$U = -\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})(200)}{8.4 \times 10^6} = \mathbf{-9.53 \times 10^9\ J}$$

Example 6.3.3

Calculate the total gravitational potential energy of the system of three masses shown. (G = 6.7 × 10^{-11} N m^2 kg^{-2})



Solution

$$U = U_{12} + U_{13} + U_{23} = -\frac{Gm_1m_2}{r_{12}} - \frac{Gm_1m_3}{r_{13}} - \frac{Gm_2m_3}{r_{23}}$$

$$= -6.7 \times 10^{-11}\left[\frac{(1)(2)}{3} + \frac{(1)(2)}{4} + \frac{(2)(2)}{5}\right] = \mathbf{-1.3 \times 10^{-10}\ J}$$

Work done and the change in potential energy

Work done (by an external force) to move m from r_1 to r_2 = change in gravitational potential energy: $W = U_2 - U_1 = \Delta U$

$$W = -\frac{GMm}{r_2} - \left(-\frac{GMm}{r_1}\right) = m(V_2 - V_1)$$

$$W = \Delta U = m\Delta V$$

Example 6.3.4

The potential energy of an object at A, a distance R from a planet, is −900 J. Calculate (a) the potential energy at B, a distance $3R$, (b) ΔU from A to B, (c) ΔU from B to A, (d) the work done to move the object from A to B.

Solution

(a) $U \propto \frac{1}{r}$: $\frac{U_B}{U_A} = \frac{R}{3R} \Rightarrow U_B = \mathbf{-300\ J}$

(b) $\Delta U_{AB} = U_B - U_A = (-300) - (-900) = \mathbf{600\ J}$

(c) $\Delta U_{BA} = U_A - U_B = \mathbf{-600\ J}$

(d) $W_{AB} = \Delta U_{AB} = \mathbf{600\ J}$

Example 6.3.5

How much energy is needed to move a 1000 kg object from the Earth's surface to an altitude of twice the Earth's radius? ($M_E = 6.0 \times 10^{24}$ kg, $R_E = 6.4 \times 10^6$ m)

Solution

Altitude $2R_E$ means $r = 3R_E$ from the centre.

$$W = m(V_f - V_i) = GMm\left(\frac{1}{R_E} - \frac{1}{3R_E}\right) = \frac{2GMm}{3R_E} = \mathbf{4.17 \times 10^{10}\ J}$$

ΔU over small distances: U = mgh is a special case

Move m from the surface ($r = R_E$) up a height h: $\Delta U = -\frac{GM_Em}{R_E + h} - \left(-\frac{GM_Em}{R_E}\right)$

$$= GM_Em\left(\frac{1}{R_E} - \frac{1}{R_E + h}\right) = GM_Em\,\frac{h}{R_E(R_E + h)} = GM_Em\,\frac{h}{R_E^2(1 + h/R_E)}$$

If $h \ll R_E$, $h/R_E \approx 0$: $\Delta U = \frac{GM_E}{R_E^2}mh = \boldsymbol{mgh}$, where $g = \frac{GM_E}{R_E^2}$

Relationship between g and V

Move a 1 kg test mass a small distance $\mathrm{d}r$ away from M. The gain in potential equals the work done per unit mass against the field: $\mathrm{d}V = -g\,\mathrm{d}r$

$$g = -\frac{\mathrm{d}V}{\mathrm{d}r}$$

g is the negative of the gradient of the V–r graph (the potential gradient). Units: J kg^{-1} m^{-1} = N kg^{-1}.

- The minus sign shows that g acts in the direction of **decreasing** potential, i.e. towards M.
- Check: $V = -\frac{GM}{r} \Rightarrow \frac{\mathrm{d}V}{\mathrm{d}r} = \frac{GM}{r^2} \Rightarrow g = -\frac{GM}{r^2}$: magnitude GM/r^2, directed towards M.
- Reading g from a V–r graph: $g = -\frac{\Delta V}{\Delta r} = -\frac{8.5 \times 10^6}{1.0 \times 10^6} = -8.5$ N kg^{-1}, i.e. 8.5 N kg^{-1} towards the centre.

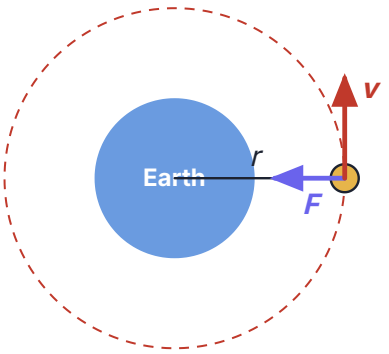
6.4 Satellite motion in a circular orbit

Satellite

is a body **orbiting a larger mass under the influence of its gravitational force**. Examples: the Moon (of the Earth), communication satellites, the Earth (of the Sun).

The gravitational attraction provides the centripetal force: $F_g = F_c$

$$\frac{GM_E m}{r^2} = \frac{mv^2}{r}$$



Orbital speed, $v = \sqrt{\frac{GM_E}{r}}$

- Orbital speed is the speed needed to stay in an orbit of fixed radius.
- It is **independent of the mass** of the satellite: a heavy satellite and a light one in the same orbit have the same speed.

Period: $T = \frac{2\pi r}{v}$. Substitute $v = \sqrt{\frac{GM_E}{r}}$:

$$T = \frac{2\pi r^{3/2}}{\sqrt{GM_E}} \Rightarrow T^2 = \frac{4\pi^2 r^3}{GM_E}$$

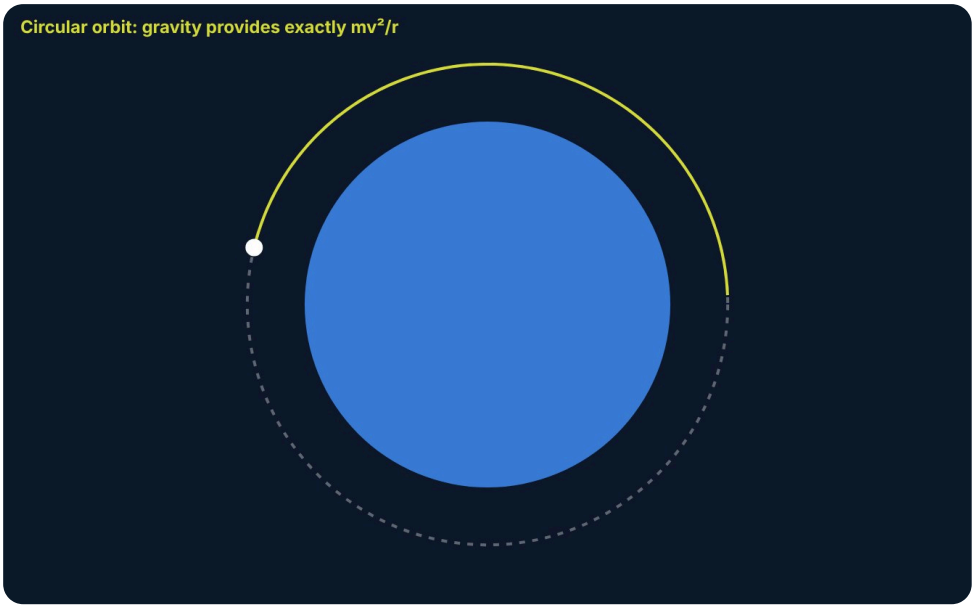
Since G and M_E are constant, $T^2 \propto r^3$ (Kepler's third law).

Geosynchronous satellites

- "Geosynchronous" means synchronised with the Earth's rotation.
- A **geosynchronous orbit** has the same rotational period (24 h) and direction as the rotation of the Earth.
- A geosynchronous satellite whose orbit lies over the equator is a **geostationary satellite**: it stays above the same point on the Earth, so it is used for communication.

Simulation **Satellite orbits: speed, period and escape**

Choose the orbit radius and launch speed. At exactly √(GM/r) the orbit is a circle; slower gives an ellipse that dips towards the Earth; at √2 times the circular speed the satellite escapes.



orbit radius r	circular speed √(GM/r)	period 2πr/v	escape speed √(2GM/r)
8.37 × 10⁶ m	**6.90 km/s**	**2.12 h**	**9.76 km/s**

launch speed	energy per kg now
6.90 km/s	**−23.8 MJ/kg**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 6 to use the simulation.

Example 6.4.1

A communication satellite orbits the Earth with a period of 24 hours. Find its height above the Earth's surface. (R_E = 6.4 × 10^6 m, M_E = 6.0 × 10^{24} kg, G = 6.67 × 10^{-11} N m^2 kg^{-2})

Solution

$T = 24 \times 60 \times 60 = 86\,400$ s

$r = \left(\frac{GM_E T^2}{4\pi^2}\right)^{1/3} = 4.23 \times 10^7$ m

$h = 4.23 \times 10^7 - 6.4 \times 10^6 =$ **3.59×10^7 m**

Example 6.4.2

The Moon takes 27.3 days to orbit the Earth. Calculate its average distance from the Earth. (g = 9.81 N kg^{-1}, R_E = 6400 km)

Solution

$GM_E = gR_E^2$; $T = 27.3 \times 24 \times 3600 = 2.36 \times 10^6$ s

$r^3 = \frac{gR_E^2 T^2}{4\pi^2} \Rightarrow$ **$r = 3.8 \times 10^8$ m**

Example 6.4.3

A satellite orbits 596 km above the Earth's surface. Find (a) its speed, (b) its period, (c) its speed and period if it moves to an orbit 400 km above the surface. (R_E = 6.38 × 10^6 m, M_E = 5.98 × 10^{24} kg)

Solution

(a) r = 6.38 × 10^6 + 596 × 10^3 = 6.98 × 10^6 m; $v = \sqrt{\frac{GM}{r}}$ = **7.56 × 10^3 m s^{-1}**

(b) $T = \frac{2\pi r}{v}$ = **5.80 × 10^3 s** (1.61 h)

(c) $v^2 r = GM$ = constant ⇒ $v_1^2 r_1 = v_2^2 r_2$, with r_2 = 6.78 × 10^6 m

(7.56 × 10^3)^2(6.98 × 10^6) = v_2^2(6.78 × 10^6) ⇒ **v_2 = 7.67 × 10^3 m s^{-1}**; $T_2 = \frac{2\pi r_2}{v_2}$ = **5554 s (1.54 h)**

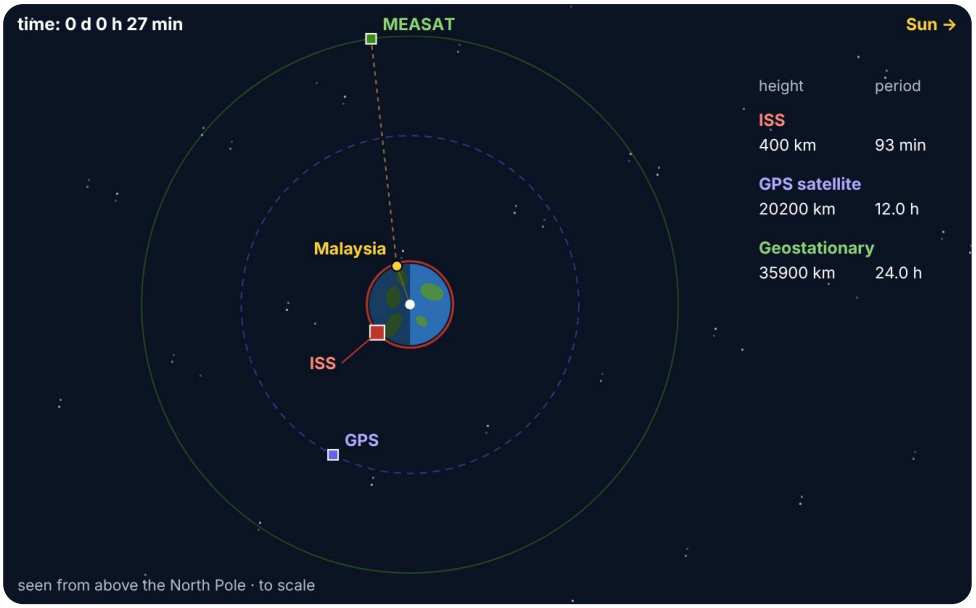
Real satellites above Malaysia

- **International Space Station (ISS):** a low orbit about 400 km up. It moves at almost 8 km s^{-1} and goes round the Earth in about 93 minutes, about 15 times a day.
- **GPS satellites:** about 20 200 km up, period 12 hours. The phone in your pocket uses signals from several of them to find your position.
- **Geostationary satellites** (e.g. the MEASAT satellites used for satellite TV in Malaysia): a geosynchronous orbit about 35 800 km above the **equator**, moving west to east. They stay above the same point on the Earth, so a satellite dish can be fixed pointing at one spot in the sky.
- The higher the orbit, the **slower** the satellite and the **longer** its period: $v = \sqrt{(GM/r)}$ and $T^2 \propto r^3$.

Simulation

The ISS, GPS and a geostationary satellite, drawn to scale

A view looking down on the North Pole, with the orbits drawn to the correct scale. Speed up time and watch: the ISS races round, a GPS satellite goes round twice a day, and the geostationary satellite stays above Malaysia as the Earth turns.



ISS: height	orbit radius r = R + h	speed v = √(GM/r)	period T = 2πr/v	orbits per day
400 km	**6800 km**	**7.67 km s^{-1}**	**92.8 min**	**15.5**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 6 to use the simulation.

Example 6.4.7

The International Space Station orbits 400 km above the Earth's surface. Find (a) its orbital speed, (b) its period in minutes, (c) the number of orbits it makes in one day. (R_E = 6.4 × 10^6 m, M_E = 6.0 × 10^{24} kg, G = 6.67 × 10^{-11} N m^2 kg^{-2})

Solution

$r = 6.4 \times 10^6 + 0.4 \times 10^6 = 6.8 \times 10^6$ m

(a) $v = \sqrt{\frac{GM_E}{r}} = \sqrt{\frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{6.8 \times 10^6}}$ = **7.67 × 10^3 m s^{-1}**

(b) $T = \frac{2\pi r}{v} = \frac{2\pi(6.8 \times 10^6)}{7.67 \times 10^3}$ = 5.57 × 10^3 s = **92.8 min**

(c) $\frac{24 \times 60}{92.8}$ = **15.5 orbits a day**

Exercises: orbits

1. The Earth orbits the Sun (mass 2 × 10^{30} kg) once a year. Find the distance of the Earth from the Sun.

   Answer

   1.50 × 10^{11} m

2. A satellite 36 000 km above the Earth's surface orbits once in 24 h. Calculate the mass of the Earth.

   Answer

   6.02 × 10^{24} kg

3. Satellites A and B are in different circular orbits about the Earth. A's orbital speed is twice B's. Find T_A/T_B.

   Answer

   0.125 ($r = GM/v^2$ and $T = 2\pi r/v = 2\pi GM/v^3$, so $T \propto 1/v^3$)

4. A satellite orbits an unknown planet at 1.70 × 10^4 m s^{-1} with radius 5.25 × 10^6 m. A second satellite orbits the same planet with radius 8.60 × 10^6 m. Find its orbital speed.

   Answer

   1.33 × 10^4 m s^{-1}

Energies of a satellite

Centripetal force = gravitational force: $\frac{mv^2}{r} = \frac{GM_E m}{r^2} \Rightarrow mv^2 = \frac{GM_E m}{r}$

K.E. = $\frac{1}{2}mv^2 = \frac{GM_E m}{2r}$; P.E. = $-\frac{GM_E m}{r}$

Total energy = K.E. + P.E. = $\frac{GM_E m}{2r} - \frac{GM_E m}{r}$

$$\text{K.E.} = \frac{GMm}{2r} \qquad \text{P.E.} = -\frac{GMm}{r} \qquad E = -\frac{GMm}{2r}$$

The total energy is negative: the satellite is bound to the Earth.

Example 6.4.4

An astronaut puts a bowling ball of mass 7.20 kg into a circular orbit 350 km above the Earth. Find (a) its K.E., (b) its P.E., (c) its total mechanical energy. (R_E = 6.4 × 10^6 m, M_E = 6 × 10^{24} kg)

Solution

$r = 6.4 \times 10^6 + 0.35 \times 10^6 = 6.75 \times 10^6$ m

(a) K.E. = $\frac{GM_E m}{2r}$ = **2.13 × 10^8 J**

(b) P.E. = $-\frac{GM_E m}{r}$ = **−4.27 × 10^8 J**

(c) E = K.E. + P.E. = **−2.13 × 10^8 J**

Weightlessness

- We judge our weight from the upward push of the floor (the reading on a scale): this is the **apparent weight**, R. The **actual weight** is the gravitational force, $W = mg$.
- If our feet are completely unsupported ($R = 0$) we experience **weightlessness**.

Lift	Newton's second law	Apparent weight
at rest or uniform velocity	$R = mg$	apparent weight = actual weight
accelerating downward	$ma = mg - R$	$R = m(g - a)$: less; $R = 0$ when $a = g$ (free fall)
accelerating upward	$ma = R - mg$	$R = m(g + a)$: more

- An astronaut in an orbiting spacecraft moves with constant speed round the orbit. The only force on him is gravity, which provides the centripetal force: $F_c = F$, so his acceleration $a = g'$, the acceleration due to gravity at that height.
- He and the spacecraft fall freely together, so the walls exert no force on him ($R = 0$). He experiences **apparent weightlessness** and floats about in the spacecraft. Gravity has not disappeared.

Example 6.4.5

A boy of mass 70 kg is in a lift moving upward with acceleration 2 m s^{-2}. What is his apparent weight?

Solution

$$ma = R - mg \Rightarrow R = m(g + a) = 70(9.81 + 2) = \mathbf{826.7\ N}\ \text{(he feels heavier)}$$

Example 6.4.6

A 70 kg boy weighs himself in a lift descending with acceleration a. His apparent weight is only 600 N. Find a.

Solution

$$ma = mg - R \Rightarrow a = \frac{mg - R}{m} = \frac{70(9.81) - 600}{70} = \mathbf{1.23\ m\ s^{-2}}$$

6.5 Escape velocity

Escape velocity

is the **minimum velocity** a body needs at the surface of a planet to **overcome its gravitational pull and escape completely** into space.

Launch from the surface (A, r) with speed v; at infinity (B) the body only just arrives, so $U = 0$ and K.E. $\approx$ 0.

Conservation of energy: $\Sigma E_A = \Sigma E_B \Rightarrow -\frac{GMm}{r} + \frac{1}{2}mv^2 = 0 + 0$

$$\frac{1}{2}mv^2 = \frac{GMm}{r}$$

$$v_e = \sqrt{\frac{2GM}{r}} = \sqrt{2gr}$$

using $g = GM/r^2$; independent of the mass of the body

- From the Earth an object needs more than about **11.2 km s^{-1}** (40 320 km h^{-1}) to escape into space.
- Gas molecules in air (average speed about 0.5 km s^{-1} at s.t.p.; hydrogen about three times faster) are far slower than this, so the Earth keeps its atmosphere.
- A planet with a small escape velocity may lose its gases into space and have no atmosphere.

Example 6.5.1

(a) Find the escape velocity for a 5000 kg spacecraft launched from the Earth. (b) Determine its minimum kinetic energy to escape the Earth's field. (M_E = 5.98 × 10^{24} kg, R_E = 6.37 × 10^6 m)

Solution

(a) $v_e = \sqrt{\frac{2(6.67 \times 10^{-11})(5.98 \times 10^{24})}{6.37 \times 10^6}}$ = **1.12 × 10^4 m s^{-1}**

(b) K.E. = $\frac{1}{2}mv_e^2 = \frac{1}{2}(5.00 \times 10^3)(1.12 \times 10^4)^2$ = **3.14 × 10^{11} J**

Example 6.5.2

The Moon's radius is 1600 km and its surface field strength is 1.6 N kg^{-1}. Calculate the escape velocity from the Moon.

Solution

$$v_e = \sqrt{2gr} = \sqrt{2(1.6)(1.6 \times 10^6)} = \mathbf{2.3 \times 10^3\ m\ s^{-1}}$$

Example 6.5.3

The escape velocity from the Earth is 11.2 km s^{-1}. Find the escape velocity from a planet with twice the Earth's mass and three times its radius.

Solution

$$\frac{v}{v_E} = \sqrt{\left(\frac{M}{M_E}\right)\left(\frac{R_E}{r}\right)} = \sqrt{(2)(⅓)}$$

v = 11.2 × 0.816 = **9.15 km s^{-1}**

Exercises: escape velocity

1. Calculate the escape speed from the Earth for a 5000 kg spacecraft, and the kinetic energy it must have at the surface to move infinitely far away. (M = 6.0 × 10^{24} kg, R = 6.4 × 10^{6} m)

   Answer

   11.2 km s^{-1}; 3.14 × 10^{11} J

2. A planet has radius 5.0 × 10^{5} m and mean density 3.0 × 10^{3} kg m^{-3}. Calculate the escape velocity from its surface.

   Answer

   $M = \rho\left(\frac{4}{3}\pi R^3\right)$ = 1.57 × 10^{21} kg; $v = \sqrt{\frac{2GM}{R}}$ = 6.5 × 10^{2} m s^{-1}

Past-year STPM questions

STPM 2021 · Section C Q18(a) and (b)

(a) State Newton's law of universal gravitation. (b) Explain the relation between the high-tide phenomenon and the position of the Earth when it lies in line between the Moon and the Sun.

Solution

(a) Every particle of matter in the universe attracts every other particle with a force that is directly proportional to the product of the masses of the particles and inversely proportional to the square of the distance between them.

(b) Tides are caused by the gravitational force of the Moon on the Earth's water. This force is inversely proportional to the square of the Earth–Moon distance, $F = Gm_E m_M / r_{EM}^2$. When the Sun, Earth and Moon lie in a straight line (new moon or full moon), the pulls of the Moon and the Sun act along the same line and add up, giving the highest high tides and the lowest low tides: **spring tides**.

Examiners' notes: (a) most candidates could state the law. Common mistakes: writing only "force" instead of gravitational (attractive) force; writing "radius between the two masses" instead of the distance between them. (b) Almost all candidates could not explain high tides and many left it blank. Not accepted: "closer" or "close" distance without relating force to distance; explaining with the Sun's gravity instead of the Moon's.

STPM 2021 · Section C Q18(c)

A satellite orbits the Earth in a circular path with a period of 6.2×10^3 s. Its mass is 5.5×10^3 kg. Determine (i) the radius of the orbit, (ii) the weight of the satellite, (iii) its potential energy.

Solution

(i) $\frac{GMm}{r^2} = \frac{4\pi^2 mr}{T^2} \Rightarrow r^3 = \frac{GMT^2}{4\pi^2} = 3.877 \times 10^{20} \Rightarrow \mathbf{r = 7.29 \times 10^6\ m}$

(ii) $W = \frac{GMm}{r^2} = \mathbf{4.12 \times 10^4\ N}$

(iii) $U = -\frac{GMm}{r} = \mathbf{-3.00 \times 10^{11}\ J}$

STPM 2021 · Section A Q7

Two planets A and B have masses M_A, M_B and radii R_A, R_B, with $M_A = ¼M_B$ and escape speeds $v_A = ⅓v_B$. What is R_A/R_B?

A. 0.19

B. 0.44

C. 2.25

D. 5.33

Answer

Answer: C

$$v = \sqrt{\frac{2GM}{R}} \Rightarrow R = \frac{2GM}{v^2}$$

$$\frac{R_A}{R_B} = \frac{M_A}{M_B}\left(\frac{v_B}{v_A}\right)^2 = ¼(3)^2 = \mathbf{2.25}$$

STPM 2023 · Section C Q19(a) and (b)

(a) State Newton's law of universal gravitation. [1] (b) Explain the change in gravitational acceleration when a meteoroid at a distance $3R$ from the surface of the Earth falls towards the surface, where R is the Earth's radius. [6]

Solution

(a) The gravitational force between two point masses is directly proportional to the product of their masses and inversely proportional to the square of the distance between them: $F = GM_1M_2/r^2$.

(b) Distance from the centre of the Earth: $r = R + 3R = 4R$, so $g = \frac{GM}{(4R)^2} = \frac{GM}{16R^2} = \frac{9.81}{16} = 0.61$ m s^{-2}.

As the meteoroid falls, r decreases and, since $g \propto 1/r^2$, g increases. At the surface $r = R$ and $g = 9.81$ m s^{-2}. The acceleration increases from 0.61 to 9.81 m s^{-2}, an increase of 9.20 m s^{-2}.

STPM 2023 · Section C Q19(c)

The Juno satellite is launched from the Earth to Jupiter and begins orbiting Jupiter at a radius of 7.05 × 10^4 km. The mass of Jupiter is 1.90 × 10^{27} kg. Determine (i) the escape speed of the satellite from the Earth's surface, [2] (ii) the distance travelled if it takes 1825 days to reach Jupiter, [2] (iii) the gravitational potential at its orbit, [2] (iv) the gravitational field strength at the satellite due to Jupiter. [2] (M_E = 5.97 × 10^{24} kg, R_E = 6.38 × 10^6 m, G = 6.67 × 10^{-11} N m^2 kg^{-2})

Solution

(i) $v = \sqrt{\dfrac{2GM_E}{R_E}}$ = **1.12 × 10^4 m s^{-1}**

(ii) t = 1825 × 24 × 3600 = 1.58 × 10^8 s; $s = vt$ = (1.12 × 10^4)(1.58 × 10^8) = **1.76 × 10^{12} m**

(iii) $V = -\dfrac{GM_J}{r}$ = **−1.80 × 10^9 J kg^{-1}**

(iv) $g = \dfrac{GM_J}{r^2}$ = **25.5 N kg^{-1}**

Chapter 7 · Semester 1

ROTATION OF RIGID BODIES

Notes by Cikgu Suhaili

21 worked examples 1 simulation Exercises with answers

7.1 Centre of mass and centre of gravity

Centre of mass (CM)

of a system is the **point at which the entire mass of the system may be considered to be concentrated**.

Centre of gravity (CG)

of a body is the **point at which the entire weight of the body may be considered to act**. The resultant of the gravitational forces on all the particles of the body passes through this point.

Centre of mass of a system of particles in a plane

The centre of mass is the mass-weighted average position of the particles:

$$\bar{x} = \frac{\Sigma m_i x_i}{\Sigma m_i} \qquad \bar{y} = \frac{\Sigma m_i y_i}{\Sigma m_i}$$

m_i = mass of the i-th particle at (x_i, y_i)

Additional example (syllabus 7.1(b)) · centre of mass of particles

Masses of 1 kg at (0, 0), 2 kg at (3 m, 0) and 2 kg at (0, 4 m) lie in a plane. Find the centre of mass.

Solution

$$\bar{x} = \frac{(1)(0) + (2)(3) + (2)(0)}{1 + 2 + 2} = \frac{6}{5} = \mathbf{1.2\ m}$$

$$\bar{y} = \frac{(1)(0) + (2)(0) + (2)(4)}{5} = \frac{8}{5} = \mathbf{1.6\ m}$$

Centre of mass at **(1.2 m, 1.6 m)**

Path of the centre of mass of a two-particle system

- Consider two particles m_1 and m_2 moving under their mutual interaction only (an explosion, or a collision), with no external force on the system.
- The internal forces are equal and opposite (Newton's third law), so they cancel and do not affect the motion of the centre of mass.
- The centre of mass therefore **keeps moving in a straight line at constant velocity** (or stays at rest), however complicated the paths of the individual particles are.
- If an external force acts, the centre of mass moves as if it were a single particle of mass ($m_1 + m_2$) acted on by that force. Example: after a shell explodes in flight, the centre of mass of the fragments continues along the original parabola.

When is the centre of mass also the centre of gravity?

- The centre of mass depends only on how the mass is distributed; the centre of gravity depends on how the weight is distributed.
- They **coincide when the gravitational field strength g is uniform** over the whole body (same magnitude and direction at every particle).
- This holds for everyday objects, which are tiny compared with the Earth. They would differ only for a very large body (a mountain, a very tall tower) over which g varies noticeably.

7.2 Moment of inertia

Inertia (linear motion) opposes a change in motion.
Moment of inertia (rotational motion) opposes a change in rotation.

> **Moment of inertia, I**
>
> of a body about an axis is the **sum of the products of the mass of each particle and the square of its perpendicular distance from the axis**. It measures the body's resistance to a change in its rotational motion.

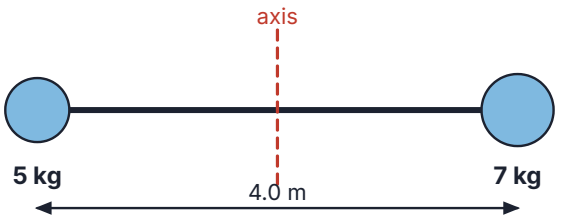
- It depends on (i) the mass, (ii) how the mass is distributed about the axis, and (iii) the position and orientation of the axis.
- It is a **scalar**; unit kg m^2.

Particle: $I = mr^2$ System of particles: $I = \Sigma m_i r_i^2 = m_1 r_1^2 + m_2 r_2^2 + \ldots + m_n r_n^2$

r = perpendicular distance of each particle from the axis of rotation

Example 7.2.1

Particles of 5 kg and 7 kg are mounted 4.0 m apart on a light rod. Calculate the moment of inertia of the system (a) about an axis halfway between the masses, (b) about an axis 0.50 m to the left of the 5 kg mass.



Solution

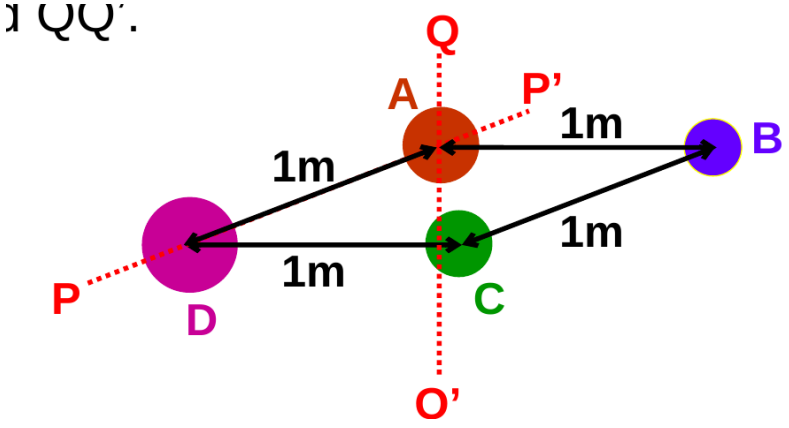
(a) $I = m_1r_1^2 + m_2r_2^2 = (5)(2)^2 + (7)(2)^2 =$ **48 kg m²**

(b) Axis 0.50 m to the left of the 5 kg mass: $I = (5)(0.50)^2 + (7)(4.50)^2 =$ **143 kg m²**

The slide gives 87 kg m² for (b). That is the answer for an axis 0.50 m to the **right** of the 5 kg mass (between the masses): $(5)(0.50)^2 + (7)(3.50)^2 = 87$ kg m².

Example 7.2.2

Calculate the moment of inertia of the system about the axis PP′ and about the axis QQ′. (m_A = 2 kg, m_B = 1 kg, m_C = 1.5 kg, m_D = 3 kg)



Solution

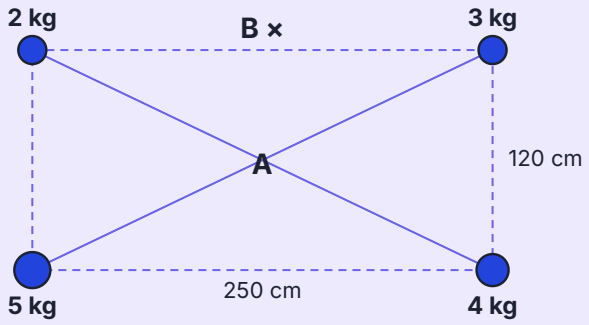
Masses on an axis contribute nothing (r = 0).

$I_{PP'} = m_B r_B^2 + m_C r_C^2 = (1)(1)^2 + (1.5)(1)^2 =$ **2.5 kg m²**

$I_{QQ'} = m_B r_B^2 + m_C r_C^2 + m_D r_D^2 =$ **7 kg m²**

Exercise: moment of inertia

1. Masses of 2 kg, 3 kg, 4 kg and 5 kg are at the corners of a 250 cm × 120 cm rectangle. Calculate their moment of inertia about an axis perpendicular to the page through (a) the centre A, (b) the midpoint B of the top edge.



Answer

(a) $d = \sqrt{1.25^2 + 0.60^2}$ = 1.39 m for all four masses: $I_A = 14 \times 1.92 =$ 27 kg m^2
(b) top masses d = 1.25 m; bottom masses $d = \sqrt{1.25^2 + 1.2^2}$ = 1.73 m: $I_B = (2 + 3)(1.25)^2 + (5 + 4)(1.73)^2 =$ 34.8 kg m^2 (the printed answer is 33 kg m^2)

Moments of inertia of some uniform bodies

STPM: the moment of inertia of these shapes is given in the question; you do not need to derive them.

Body	Axis	Moment of inertia
particle	distance r from the axis	mr^2
thin ring, hoop or cylindrical shell	through the centre, perpendicular to the plane	mr^2
thin rod, length L	through the centre, perpendicular to the rod	$\frac{1}{12}mL^2$
solid cylinder or disc	along the cylinder axis	$\frac{1}{2}mr^2$
solid sphere	through the centre	$\frac{2}{5}mr^2$
thin spherical shell	through the centre	$\frac{2}{3}mr^2$

Example 7.2.3

Calculate the moment of inertia of a 10.8 kg solid sphere of radius 0.648 m about an axis through its centre.

Solution

$$I = \frac{2}{5}mr^2 = \frac{2}{5}(10.8)(0.648)^2 = \mathbf{1.81\ kg\ m^2}$$

7.3 Torque and angular acceleration

Torque, τ

is the **tendency of a force to rotate a body about an axis**: the quantitative measure of the ability of a force to cause or change rotational motion.

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} \quad \tau = rF\sin\theta$$

- A **vector**; unit N m.
- Sign convention: clockwise negative, anticlockwise positive.
- $r\sin\theta$ is the perpendicular distance from the axis to the line of action of **F**.

Angular acceleration, α

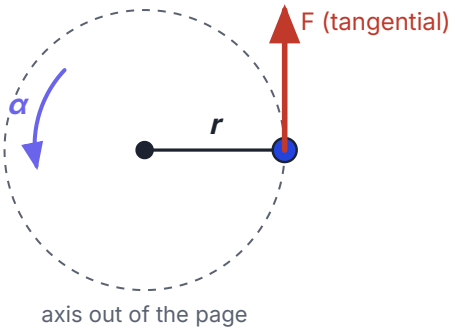
is the **rate of change of angular velocity**: $\alpha = \frac{d\omega}{dt}$ (rad s^{-2}). For a point at radius r, the tangential acceleration is $a_t = r\alpha$.

A net torque produces an angular acceleration: $\tau = I\alpha$

A constant tangential force F acts on a particle of mass m at radius r: $F = ma_t$ and $a_t = r\alpha$, so $F = mr\alpha$

Torque of this force ($\theta = 90°$): $\tau = rF = mr^2\alpha = I\alpha$ ($I = mr^2$ for a particle)

A rigid body is made of n particles: $\tau = m_1r_1^2\alpha + m_2r_2^2\alpha + \ldots = (\Sigma m_i r_i^2)\alpha$



$$\tau_{net} = I\alpha$$

the rotational form of Newton's second law, $F = ma$; τ_{net} is the net torque on the body

Example 7.3.1

A force of 200 N acts tangentially on the rim of a wheel of radius 25 cm. (a) Calculate the torque. (b) Repeat if the force makes an angle of 40° with the spoke of the wheel.

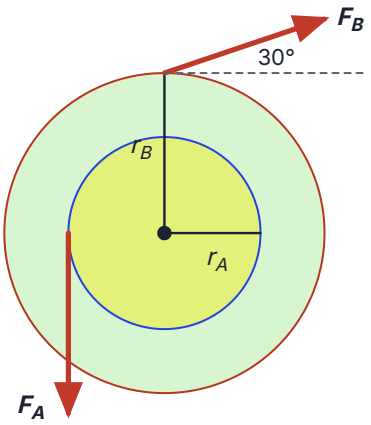
Solution

(a) $\tau = rF \sin\theta = (0.25)(200 \sin 90°) =$ **50 N m**

(b) $\tau = (0.25)(200 \sin 40°) =$ **32 N m**

Example 7.3.2

Two thin discs of radius r_A = 30 cm and r_B = 50 cm are fixed together on an axle through their centres. Calculate the net torque due to the two forces shown, each of 50 N.



Solution

$\tau_{net} = r_A F_A \sin 90° - r_B F_B \sin 60°$

$= (0.30)(50) - (0.50)(50) \sin 60° =$ **−6.7 N m** (6.7 N m clockwise)

Example 7.3.3

An 8.0 kg wheel has a radius of 30 cm. (a) What is its moment of inertia? (b) What torque gives it an angular acceleration of 3.0 rad s^{-2}? (Treat it as a uniform disc.)

Solution

(a) $I = \frac{1}{2}mr^2 = \frac{1}{2}(8.0)(0.30)^2 =$ **0.36 kg m²**

(b) $\tau = I\alpha = (0.36)(3.0) =$ **1.08 N m**

Example 7.3.4

A grinding disc of radius 0.6 m and mass 90 kg rotates at 460 rev min^{-1}. What frictional force, applied tangentially at the edge, stops it in 20 s?

Solution

$I = \tfrac{1}{2}mr^2 = \tfrac{1}{2}(90)(0.6)^2 = 16.2$ kg m^2

$\omega_0 = \dfrac{460 \times 2\pi}{60} = 48.2$ rad s^{-1}; $\alpha = \dfrac{0 - 48.2}{20} = -2.41$ rad s^{-2}

$rF = I\alpha \Rightarrow F = \dfrac{I\alpha}{r} = \dfrac{16.2(-2.41)}{0.6} = -65.1$ N: a force of **65.1 N** opposing the rotation

Example 7.3.5

What tangential force applied to the rim of a 2 kg disc-shaped wheel of radius 0.5 m gives it an angular acceleration of 4.8 rad s^{-2}? (Neglect friction.)

Solution

$I = \tfrac{1}{2}mr^2 = \tfrac{1}{2}(2)(0.5)^2 = 0.25$ kg m^2

$rF = I\alpha \Rightarrow F = \dfrac{0.25(4.8)}{0.5} = \mathbf{2.4\ N}$

Example 7.3.6

A 15.0 N force is applied to a cord wrapped round a pulley of mass 4.00 kg and radius 33.0 cm. There is a friction torque of 1.10 N m at the axle. Calculate the angular acceleration of the pulley.

Solution

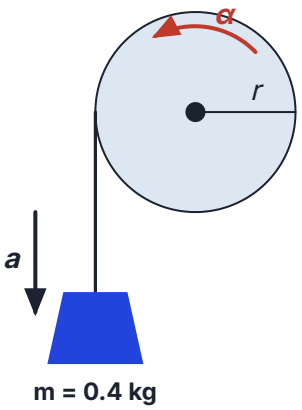
$I = \tfrac{1}{2}mR_0^2 = \tfrac{1}{2}(4.00)(0.33)^2 = 0.22$ kg m^2

$\tau_{net} = I\alpha$: $rF - \tau_{fr} = I\alpha$

$\alpha = \dfrac{(0.33)(15.0) - 1.10}{0.22} = \mathbf{17.5\ rad\ s^{-2}}$

Example 7.3.7

A 400 g mass hangs from a cord wound on a wheel of radius 15 cm. When released from rest it falls 2.0 m in 2 s. Calculate the moment of inertia of the wheel.



Solution

Mass (linear): $s = ut + \frac{1}{2}at^2 \Rightarrow 2.0 = \frac{1}{2}a(2)^2 \Rightarrow a = 1\ \text{m s}^{-2}$

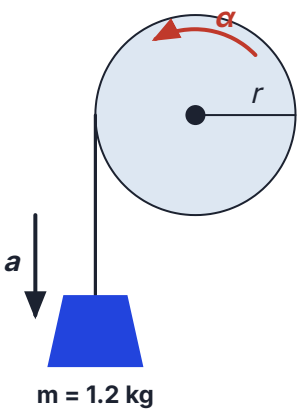
$mg - T = ma \Rightarrow T = 0.4(9.81 - 1) = 3.52\ \text{N}$

Wheel (rotation): $Tr = I\alpha$ with $\alpha = \frac{a}{r} = \frac{1}{0.15} = 6.67\ \text{rad s}^{-2}$

$I = \frac{Tr}{\alpha} = \frac{(3.52)(0.15)}{6.67} = \mathbf{0.079\ kg\ m^2}$

Example 7.3.8

A uniform disc of mass 2.5 kg and radius 20 cm is mounted on a fixed horizontal axle. A 1.2 kg block hangs from a light cord wrapped round its rim. Calculate (a) the acceleration of the block, (b) the angular acceleration of the disc, (c) the tension in the cord.



Solution

Disc (rotation): $TR = I\alpha = (\tfrac{1}{2}MR^2)(a/R) \Rightarrow T = \tfrac{1}{2}Ma$ (1)

Block (translation): $mg - T = ma$ (2)

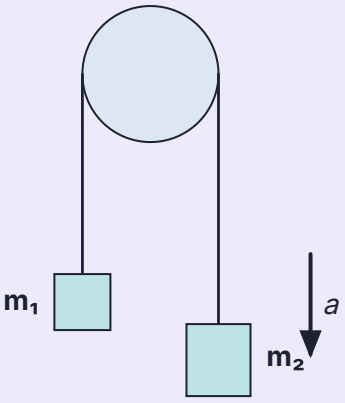
(1) into (2): $a = \dfrac{mg}{m + \tfrac{1}{2}M} = \dfrac{(1.2)(9.81)}{1.2 + 1.25} = \mathbf{4.80\ m\ s^{-2}}$

(b) $\alpha = \dfrac{a}{R} = \dfrac{4.80}{0.20} = \mathbf{24.0\ rad\ s^{-2}}$

(c) $T = \tfrac{1}{2}Ma = \tfrac{1}{2}(2.5)(4.80) = \mathbf{6.0\ N}$

Exercise: pulley with friction

1. Masses m_1 = 0.40 kg and m_2 = 0.80 kg hang from a pulley of mass 0.20 kg and radius 0.15 m. A constant friction torque of 0.35 N m acts at the axle. What is the acceleration of the masses?



Answer

$m_2g - T_2 = m_2a;\ T_1 - m_1g = m_1a;\ (T_2 - T_1)R - \tau_{fr} = \tfrac{1}{2}MR^2(a/R)$

$$a = \frac{(m_2 - m_1)g - \tau_{fr}/R}{m_1 + m_2 + \tfrac{1}{2}M} = \frac{3.92 - 2.33}{1.30} = 1.2\ \text{m s}^{-2}$$

Rotational motion with constant angular acceleration

If ω increases uniformly from ω_0 to ω in time t: $\omega = \omega_0 + \alpha t$ (1)

Angular displacement = average angular velocity × time: $\theta = \tfrac{1}{2}(\omega_0 + \omega)t$ (2)

(1) into (2): $\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2$ (3)

From (1), $t = (\omega - \omega_0)/\alpha$; substitute into (2): $\omega^2 = \omega_0^2 + 2\alpha\theta$ (4)

Linear motion	Rotational motion
$v = u + at$	$\omega = \omega_0 + \alpha t$
$s = \tfrac{1}{2}(u + v)t$	$\theta = \tfrac{1}{2}(\omega_0 + \omega)t$
$s = ut + \tfrac{1}{2}at^2$	$\theta = \omega_0 t + \tfrac{1}{2}\alpha t^2$
$v^2 = u^2 + 2as$	$\omega^2 = \omega_0^2 + 2\alpha\theta$

Example 7.3.9

A wheel rotates with constant angular acceleration 2.5 rad s^{-2}. At t = 0 its angular velocity is 3.0 rad s^{-1}. Calculate (a) the angle turned in the first 2 s, (b) the angular velocity at t = 2 s.

Solution

(a) $\theta = \omega_0 t + \frac{1}{2}\alpha t^2 = (3.0)(2) + \frac{1}{2}(2.5)(2)^2 =$ **11 rad**

(b) $\omega = \omega_0 + \alpha t = 3.0 + (2.5)(2) =$ **8.0 rad s^{-1}**

Example 7.3.10

A car wheel speeds up from rest to 100 rad s^{-1} in 9 s. Calculate (a) the angular acceleration, (b) the angular speed at t = 10 s, (c) the angle turned in 20 s (the acceleration continues).

Solution

(a) $\alpha = \frac{100}{9} =$ **11.11 rad s^{-2}**

(b) $\omega = (11.11)(10) =$ **111.1 rad s^{-1}**

(c) $\theta = \frac{1}{2}(11.11)(20)^2 =$ **2222 rad**

Example 7.3.11

A wheel turns with constant angular acceleration 0.45 rad s^{-2}. (a) How long does it take to reach 8.00 rad s^{-1} from rest? (b) Through how many revolutions does it turn in this time?

Solution

(a) $t = \frac{\omega - \omega_0}{\alpha} = \frac{8.00}{0.45} =$ **17.8 s**

(b) $\theta = \frac{\omega^2}{2\alpha} = \frac{64}{0.90} = 71.1$ rad $= \frac{71.1}{2\pi} =$ **11.3 rev**

Exercises: uniform angular acceleration

1. The angular speed of a disc decreases uniformly from 12.00 to 4.00 rad s^{-1} in 16.0 s. Calculate the angular acceleration and the number of revolutions made in this time.

   Answer

   $\alpha = -0.500$ rad s^{-2}; $\theta = \frac{1}{2}(12 + 4)(16) = 128$ rad = 20.4 rev

2. A flywheel of radius 0.08 m speeds up from 5 rpm to 11 rpm in 100 s. Calculate (a) its angular acceleration, (b) the tangential acceleration of a point on the rim.

   Answer

   (a) 6.3×10^{-3} rad s^{-2} (b) $a_t = \alpha r = 5.04 \times 10^{-4}$ m s^{-2}

7.4 Angular momentum and its conservation

Angular momentum, L

of a particle about an axis is the **cross product of its position vector r and its linear momentum p**: **L = r × p**.

Magnitude: $L = rp \sin\theta$. For circular motion, **r** ⊥ **v** ($\theta = 90°$) and $p = mv$, so $L = mvr$

With $v = r\omega$: $L = mr^2\omega$

A rigid body is made of many particles: $L = (\Sigma m_i r_i^2)\omega = I\omega$

$$L = I\omega$$

vector (right-hand rule, along the axis); unit kg m^2 s^{-1}

Example 7.4.1

A 10 kg disc of radius 0.25 m has an angular momentum of 0.45 kg m^2 s^{-1}. What is its angular speed?

Solution

$$I = \tfrac{1}{2}(10)(0.25)^2 = 0.3125 \text{ kg m}^2; \quad \omega = \frac{L}{I} = \frac{0.45}{0.3125} = \mathbf{1.44 \text{ rad s}^{-1}}$$

Example 7.4.2

A 500 g steel rod 30 cm long is pivoted about its centre and rotates at 300 rev min^{-1}. What is its angular momentum?

Solution

$$I = \frac{1}{12}mL^2 = \frac{1}{12}(0.5)(0.3)^2 = 3.75 \times 10^{-3}\ \text{kg m}^2;\ \omega = \frac{300 \times 2\pi}{60} = 31.4\ \text{rad s}^{-1};\ L = I\omega = \mathbf{0.118\ kg\ m^2\ s^{-1}}$$

Principle of conservation of angular momentum

If **no external torque** acts on a system, its **total angular momentum is conserved** (constant).

$$I_1\omega_1 = I_2\omega_2$$

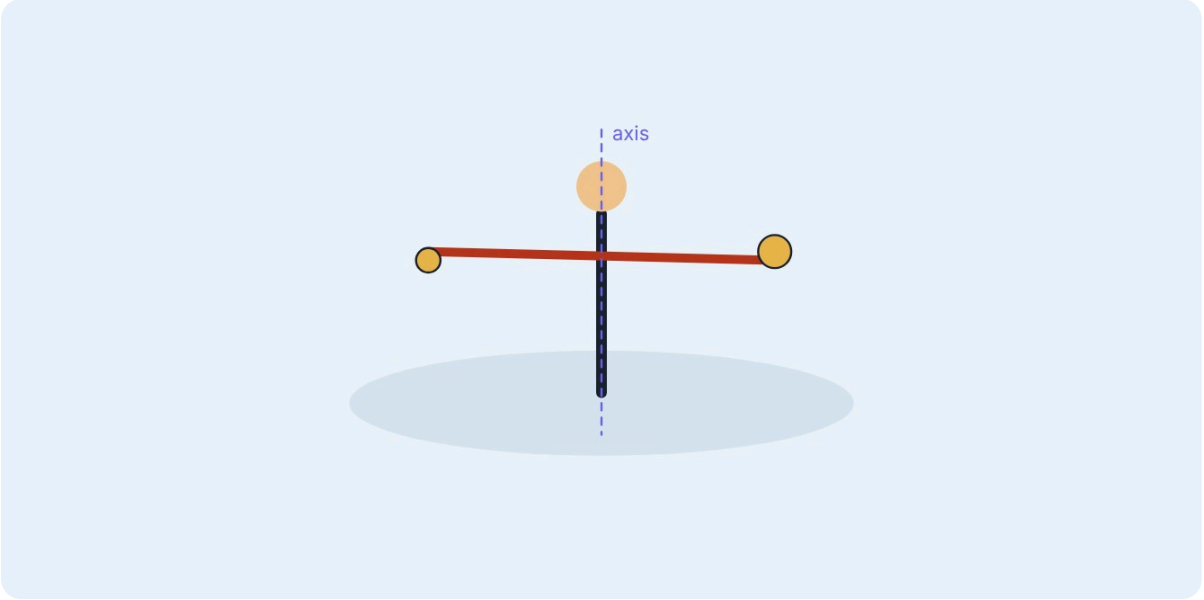
An ice skater starts spinning with her arms stretched out. Bringing her arms close to the axis reduces her moment of inertia I; since $I\omega$ is constant, her angular velocity ω increases. Divers, acrobats and gymnasts use the same idea.



Simulation

Conservation of angular momentum: the spinning skater

Pull the arms in and out. L = Iω stays the same, so when I gets smaller ω gets bigger. The kinetic energy ½Iω² is not conserved: the skater does work pulling her arms in.



moment of inertia I	angular velocity ω = L/I	L = Iω (constant)	rotational KE ½Iω²
5.86 kg m²	**1.02 rad/s**	**6.00 kg m²/s**	**3.1 J**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 7 to use the simulation.

Example 7.4.3

A horizontal disc rotates at 100 rpm about a vertical axis through its centre. A 10 g particle is dropped onto the disc 9.0 cm from the axis and sticks. The angular velocity falls to 90 rpm. Find the moment of inertia of the disc.

Solution

No external torque: $I\omega_0 = (I + mr^2)\omega$

$mr^2 = (0.010)(0.090)^2 = 8.1 \times 10^{-5}$ kg m^2

$I(100) = (I + 8.1 \times 10^{-5})(90) \Rightarrow$ **$I = 7.3 \times 10^{-4}$ kg m^2**

Example 7.4.4

A flywheel (moment of inertia 95 kg m^2) rotates at 60 rad s^{-1}. A coaxial gear wheel (5.0 kg m^2) is initially at rest. The two are joined together. Calculate the common angular velocity and the heat generated.

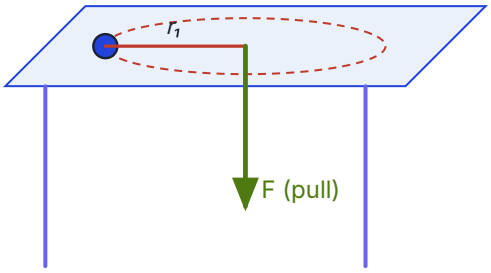
Solution

(a) $I_1\omega_{01} + I_2\omega_{02} = (I_1 + I_2)\omega \Rightarrow (95)(60) + 0 = (100)\omega \Rightarrow$ **$\omega = 57$ rad s^{-1}**

(b) Heat = loss in rotational kinetic energy = $\tfrac{1}{2}(95)(60)^2 - \tfrac{1}{2}(100)(57)^2 = 171\,000 - 162\,450 =$ **8550 J**

Example 7.4.5

A mass on a string moves in a circle on a frictionless table; the string passes through a hole in the table. Initially the speed is 2.4 m s^{-1} at radius 0.80 m. The string is pulled slowly until the radius is 0.48 m. What is the new speed?



Solution

The pull acts through the axis, so there is no torque: $L_{\text{initial}} = L_{\text{final}}$

$mr_1^2\omega_1 = mr_2^2\omega_2$ and $v = r\omega \Rightarrow v_1r_1 = v_2r_2$

$$v_2 = \frac{(2.4)(0.80)}{0.48} = \textbf{4.0 m s}^{-1}$$

Exercise: conservation of angular momentum

1. A merry-go-round of diameter 4.5 m rotates freely at 0.70 rad s^{-1}; its moment of inertia is 1750 kg m^2. Four people of 65 kg each step onto its edge. What is the new angular velocity?

Answer

$\Delta I = 4(65)(2.25)^2 = 1316$ kg m^2; $I_2 = 3066$ kg m^2; $\omega_2 = \dfrac{(1750)(0.70)}{3066} = \mathbf{0.40\ rad\ s^{-1}}$

7.5 Rotational kinetic energy

Rotational kinetic energy

is the **kinetic energy of a body due to its rotation**: the sum of the kinetic energies of all its particles, for rotation about a fixed axis (no translation).

$K_R = \tfrac{1}{2}m_1v_1^2 + \tfrac{1}{2}m_2v_2^2 + \dots + \tfrac{1}{2}m_nv_n^2$, with $v = r\omega$

$= \tfrac{1}{2}\omega^2(m_1r_1^2 + m_2r_2^2 + \dots) = \tfrac{1}{2}(\Sigma m_ir_i^2)\omega^2$

$$K = \tfrac{1}{2}I\omega^2$$

Example 7.5.1

A wheel (uniform disc) of mass 6.0 kg and radius 40 cm rotates at 300 rpm. Calculate its moment of inertia and its rotational kinetic energy.

Solution

$\omega = \dfrac{300 \times 2\pi}{60} = 31.4\ \text{rad s}^{-1}$

$I = \tfrac{1}{2}mr^2 = \tfrac{1}{2}(6.0)(0.40)^2 = \mathbf{0.48\ kg\ m^2}$

$K = \tfrac{1}{2}I\omega^2 = \tfrac{1}{2}(0.48)(31.4)^2 = \mathbf{2.37 \times 10^2\ J}$

Linear and rotational quantities compared

Linear motion	Rotational motion
displacement, s	angular displacement, θ
velocity, v	angular velocity, ω
acceleration, a	angular acceleration, α
mass, m	moment of inertia, I
force, F	torque, τ
$F = ma$	$\tau = I\alpha$
momentum, $p = mv$	angular momentum, $L = I\omega$
work, $W = Fs$	work, $W = \tau\theta$
$K = \frac{1}{2}mv^2$	$K = \frac{1}{2}I\omega^2$

Relationships: $s = r\theta$, $v = r\omega$, $a = r\alpha$; $\omega_{ave} = \theta/t = (\omega + \omega_0)/2$.

Chapter 8 · Semester 1

STATICS

Notes by Cikgu Suhaili

15 worked examples 1 simulation Exercises with answers

8.1 Equilibrium of a particle

- **Statics** is the study of forces in equilibrium. In equilibrium all the forces on a body are balanced, so there is no net change in its motion.
- An object is in **equilibrium** when there is no net force on it.

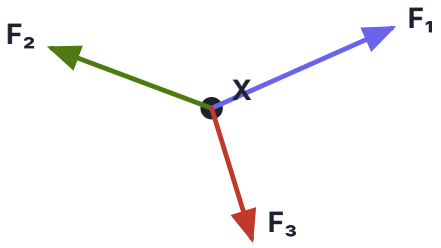
Type of equilibrium	Meaning
static equilibrium	at rest, $v = 0$
dynamic equilibrium	moving with constant velocity, $a = 0$
translational equilibrium	no linear acceleration
rotational equilibrium	no angular acceleration

Particle

is an object which has mass and is small enough to be regarded as a point.

Coplanar forces

are forces that lie in (act in) the same plane.



Concurrent forces

are forces whose lines of action intersect at a single point.

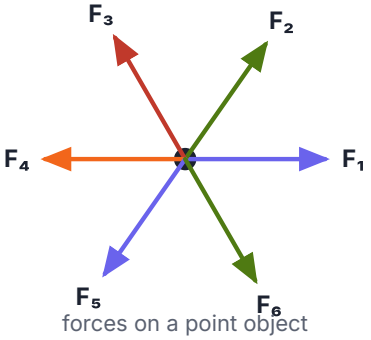
Condition for equilibrium of a particle

The **resultant (net) of the concurrent forces acting on the particle is zero**: $\Sigma\mathbf{F} = 0$, i.e. $\Sigma F_x = 0$ and $\Sigma F_y = 0$.

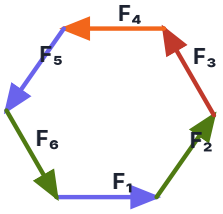
Two methods for a particle in equilibrium

- (i) **closed polygon of forces** (a scale drawing); (ii) **resolving forces into components**, using a free-body diagram (a diagram of a body showing all the forces acting on it).

> **Polygon of forces**
>
> A point object acted on by several forces is in equilibrium only if the force vectors, drawn head to tail, **form a closed polygon**.



forces on a point object



head to tail: a closed polygon

Draw arrows to show the magnitudes and directions of all the external forces.

Choose a scale, e.g. 1 cm : 2 N.

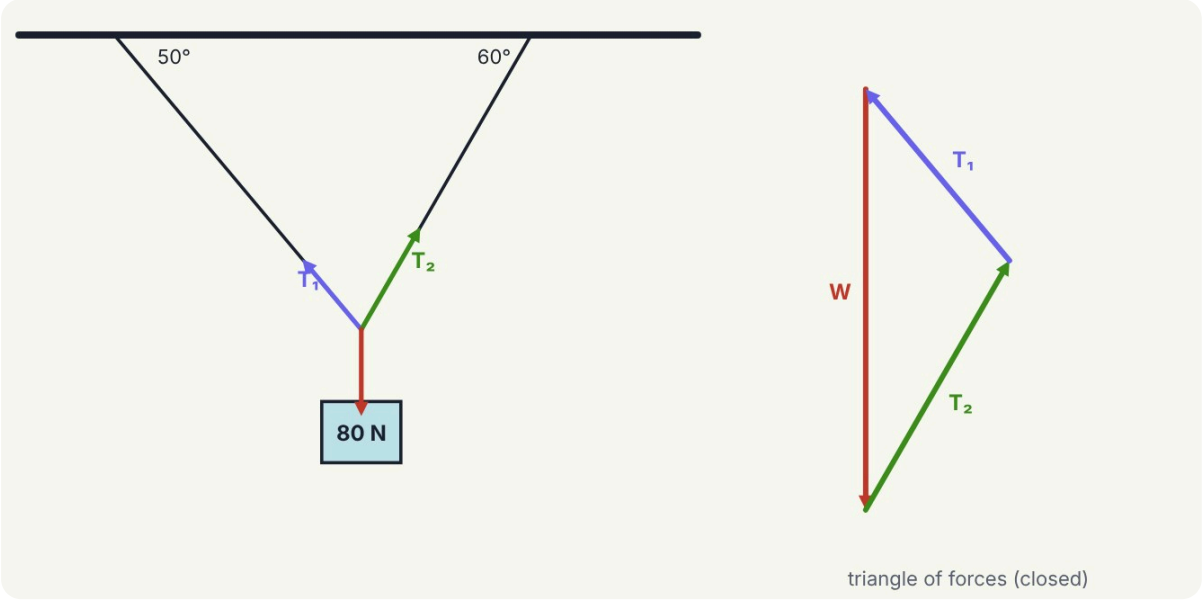
Add all the forces head to tail. The vector from the start to the end is the resultant (net) force; if the head of the last vector meets the tail of the first, the polygon is closed and the net force is zero.

Triangle of forces (STPM 8.1(e)): when a particle is in equilibrium under exactly three forces, the three force vectors drawn head to tail form a closed triangle.

Simulation

Three forces in equilibrium: the triangle of forces

A weight hangs from a knot held by two ropes. Change the angles and the weight: the tensions update, and the three forces drawn head to tail always close into a triangle.



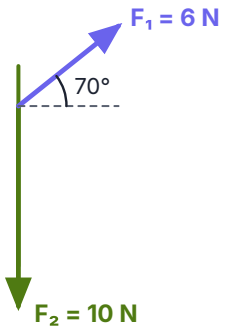
triangle of forces (closed)

T_1	T_2	$\Sigma Fx = T_2\cos - T_1\cos$	$\Sigma Fy = T_1\sin + T_2\sin - W$
42.6 N	**54.7 N**	**0.00 N**	**0.00 N**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 8 to use the simulation.

Example 8.1.1

A system of concurrent forces acts on a particle: F_1 = 6 N at 70° above the horizontal and F_2 = 10 N vertically downward. Find the force F_3 that keeps the particle in equilibrium.



Solution

Closed polygon (1 cm : 2 N): the closing side measures 2.4 cm, so F_3 ≈ 4.8 N at about 117°.

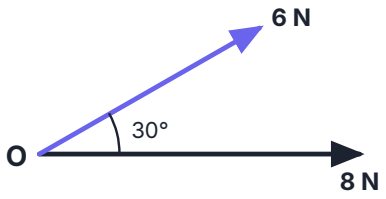
Components: $\Sigma F_y = 6 \sin 70° - 10 = -4.4$ N; $\Sigma F_x = 6 \cos 70° = 2.1$ N

Resultant of F_1 and F_2: $F = \sqrt{2.1^2 + 4.4^2} = 4.9$ N at $\theta = \tan^{-1}(-4.4/2.1) = -64.5°$

F_3 must cancel it: **F_3 = 4.9 N at 115.5°** (from the positive x-axis)

Example 8.1.2

A point object O is acted on by a 6 N force and an 8 N force at 30° to one another. What single third force will balance them?



Solution

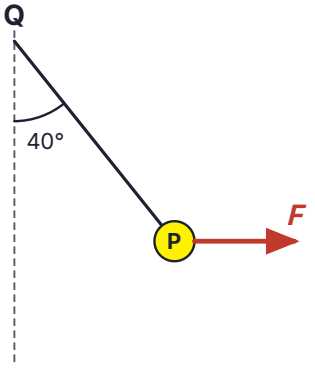
Take the 8 N force along +x: $\Sigma F_x = 8 + 6\cos 30° = 13.20$ N; $\Sigma F_y = 6 \sin 30° = 3.00$ N

Resultant = $\sqrt{13.20^2 + 3.00^2}$ = 13.5 N at 12.8°

The balancing force is equal and opposite: **13.5 N at 192.8°**

Example 8.1.3

A 5 kg body P hangs from a light string fixed at Q. A horizontal force is applied to P, and in equilibrium PQ makes 40° with the vertical. Find the horizontal force and the tension in the string.



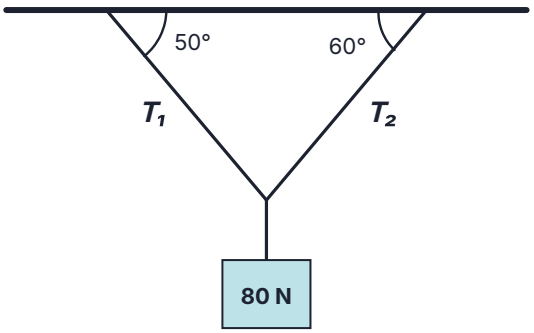
Solution

Resolve vertically: $T\cos 40° - mg = 0 \Rightarrow T = \dfrac{(5)(9.81)}{\cos 40°}$ = **64 N**

Resolve horizontally: $F - T\sin 40° = 0 \Rightarrow F = 64 \sin 40°$ = **41 N**

Example 8.1.4

The object shown weighs 80 N and is in equilibrium. Find T_1 and T_2.



Solution

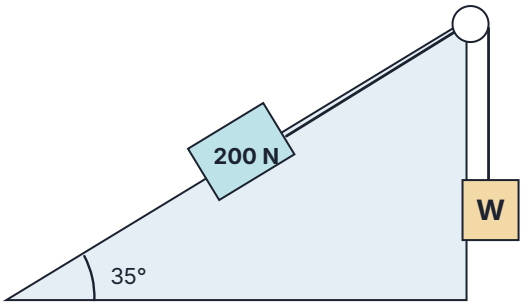
$\Sigma F_x = 0$: $T_2 \cos 60° - T_1 \cos 50° = 0$ (1)

$\Sigma F_y = 0$: $T_2 \sin 60° + T_1 \sin 50° - 80 = 0$ (2)

Solving: $T_1 = 42.5$ N, $T_2 = 54.8$ N

Example 8.1.5

The pulley is frictionless and the friction between the block and the incline is negligible. Find the weight W needed to keep the 200 N block at rest on the 35° incline.



Solution

For the 200 N block, along the incline: $T - 200 \sin 35° = 0 \Rightarrow T = 114.7$ N

For W: $T - W = 0 \Rightarrow$ **$W = 114.7$ N**

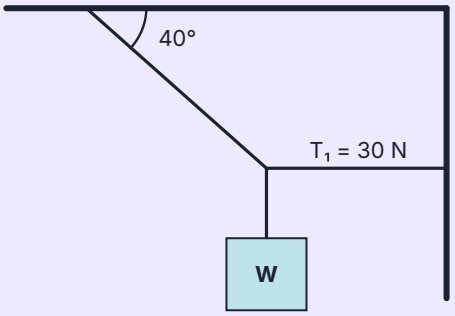
Exercises: equilibrium of a particle

1. A wooden block of weight 7.2 N rests on a slope at 20° to the horizontal. Calculate (a) the frictional force on the block, (b) the normal force on the block, (c) the coefficient of static friction.

   Answer

   (a) 2.46 N (b) 6.77 N (c) 0.36

2. An object hangs from a knot. One rope pulls horizontally with $T_1 = 30$ N; the other goes up to the ceiling at 40° to the ceiling. Find the weight of the object.



   Answer

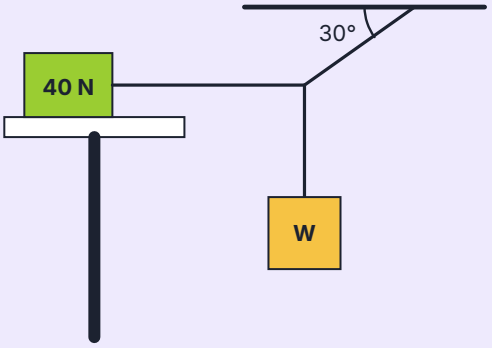
   $W = 30 \tan 40° =$ 25.2 N. (The slide answer, 30 N, would need the angle to be 45°.)

3. A 40 N weight hangs from a knot held by a rope at 60° to the ceiling and a second rope at 70° to the vertical (going down to the wall). Find the two tensions.

Answer

58 N in the rope at 60° to the ceiling; 31 N in the rope at 70° to the vertical

4. The system shown is in equilibrium. (a) What is the maximum value of W if the friction force on the 40 N block cannot exceed 12.0 N? (b) What is the coefficient of static friction between the block and the table?



Answer

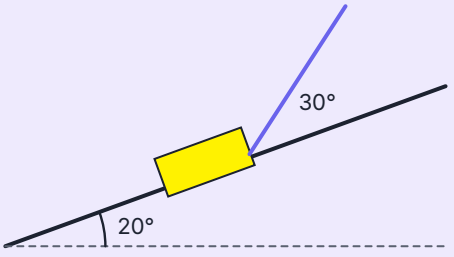
(a) 6.9 N (b) 0.30

5. The same system is just on the point of slipping when W = 8.0 N. What is the coefficient of static friction?

Answer

0.35

6. A 2 kg body rests on a smooth slope at 20° to the horizontal, held by a string at 30° to the slope. Find the tension in the string.



Answer

7.74 N

Example 8.1.6

Dynamic equilibrium. A hanging 8.0 N block pulls a 20 N block along a table at constant velocity. Find μ_k between the 20 N block and the table. (Frictionless pulley.)

Solution

Constant velocity, so each block is in equilibrium. Hanging block: $T = 8.0$ N

20 N block: $T = f_k = \mu_k R$ and $R = 20$ N

$$\mu_k = \frac{8.0}{20} = \mathbf{0.40}$$

Practice question · equilibrium on an incline

A box of mass 10.0 kg rests in equilibrium on a smooth plane inclined at 35° to the horizontal, held by a string parallel to the incline. Calculate the tension in the string. ($g = 9.81$ m s^{-2})

Answer

Along the incline: $T = mg \sin 35° = (10.0)(9.81)(0.574) = \mathbf{56.2\ N}$

8.1 Equilibrium of a rigid body

Rigid body

is a body with a **definite and unchanging shape and size**. It can have translational motion and rotational motion.

Torque (moment of a force)

Torque, τ

is the **vector product of r and F**, where **r** is the vector from the axis of rotation to the point where the force acts. It measures the tendency of a force to rotate a body about the axis.

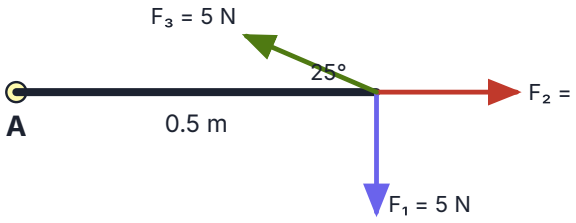
$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} \quad \tau = rF\sin\theta = Fd$$

θ = angle between **r** and **F**; $d = r\sin\theta$ = lever arm (moment arm); unit N m

- The **lever arm** d is the perpendicular distance from the axis to the line of action of the force.
- Torque is a vector; its direction is given by the right-hand rule (the thumb), along the axis.
- Sign convention: **clockwise negative**, **anticlockwise positive**.

Example 8.2.1

Calculate the torque about axis A due to each force shown: F_1 = 5 N downward at 0.5 m from A, F_2 = 5 N along the rod, and F_3 = 5 N at 25° to the rod.



Solution

$\tau = rF\sin\theta$

$\tau_{1A} = -(0.5)(5\sin 90°) = \mathbf{-2.5\ N\,m}$

$\tau_{2A} = (0.5)(5\sin 0°) = \mathbf{0}$ (the line of action passes through A)

$\tau_{3A} = +(0.5)(5\sin 25°) = \mathbf{+1.1\ N\,m}$

Example 8.2.2

Calculate the torque about A due to each force: 10 N downward, 20 N along the rod and 25 N at 65° to the rod, all acting 0.8 m from A.

Solution

$\tau_{1A} = -(0.8)(10 \sin 90°) = \mathbf{-8.0\ N\ m}$

$\tau_{2A} = (0.8)(20 \sin 0°) = \mathbf{0}$

$\tau_{3A} = +(0.8)(25 \sin 65°) = \mathbf{+18.1\ N\ m}$

Example 8.2.3

The torque about X is 25 N m. Find the angle θ when (a) an 8 N force acts 4 m from X, (b) a 60 N force acts 0.5 m from X.

Solution

(a) $25 = (4)(8 \sin \theta) \Rightarrow \sin \theta = 0.781 \Rightarrow \mathbf{\theta = 51.4°}$

(b) $25 = (0.5)(60 \sin \theta) \Rightarrow \sin \theta = 0.833 \Rightarrow \theta = 56.4°$ or **124°** (the angle in the figure is obtuse)

Conditions for the equilibrium of a rigid body (coplanar forces)

$\Sigma \mathbf{F} = 0 \quad (\Sigma F_x = 0, \Sigma F_y = 0)$

1st condition (force condition): no translational motion

$\Sigma \tau = 0$ about any axis

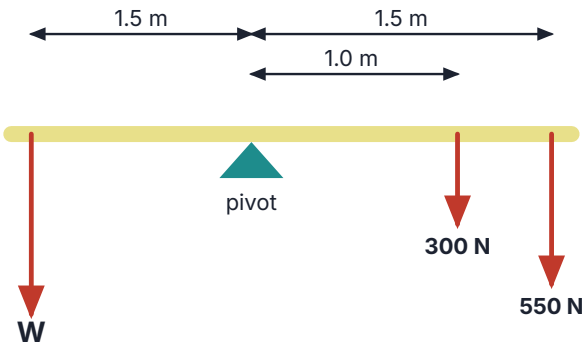
2nd condition (torque condition): no rotational motion

Teacher's note

Choose the axis for $\Sigma \tau = 0$ at the point where an unknown force acts (a hinge, a pivot, the foot of a ladder): that force then has zero torque and disappears from the equation.

Example 8.2.4

The see-saw shown is balanced. Calculate the weight W.



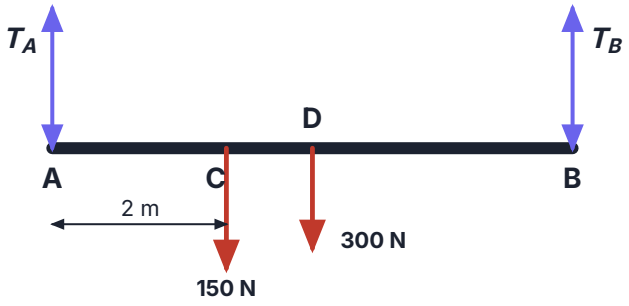
Solution

$\Sigma\tau = 0$ about the pivot: $(1.5)(W \sin 90°) - (1.0)(300 \sin 90°) - (1.5)(550 \sin 90°) = 0$

$\boldsymbol{W = 750\ \text{N}}$

Example 8.2.5

A uniform plank AB, 6 m long and of weight 300 N, is held horizontal by two vertical ropes at A and B. A 150 N weight rests on the plank at C, where AC = 2 m. Find the tension in each rope.



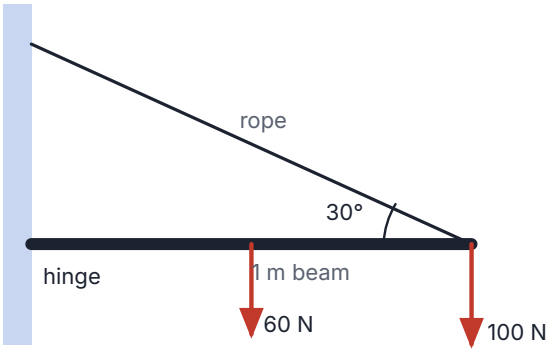
Solution

Take A as the axis: $\Sigma\tau_A = 0$: $(0)T_A - (2)(150) - (3)(300) + (6)T_B = 0 \Rightarrow \boldsymbol{T_B = 200\ \text{N}}$

$\Sigma F_y = 0$: $T_A + T_B - 150 - 300 = 0 \Rightarrow \boldsymbol{T_A = 250\ \text{N}}$

Example 8.2.6

A uniform 1 m beam of weight 60 N is hinged to a wall and held horizontal by a rope at 30° to the beam, attached at its end. A 100 N load hangs from the end. Calculate (a) the tension in the rope, (b) the force of the hinge on the beam.



Solution

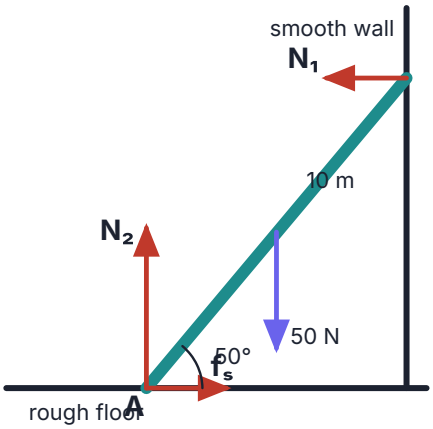
(a) $\Sigma\tau = 0$ about the hinge: $-(0.5)(60) - (1)(100) + (1)(T\sin 30°) = 0 \Rightarrow$ **$T = 260$ N**

(b) $\Sigma F_x = 0$: $R_x = T\cos 30° =$ **225 N**; $\Sigma F_y = 0$: $R_y = 160 - T\sin 30° =$ **30 N**

Hinge force = $\sqrt{225^2 + 30^2}$ = **227 N** at $\tan^{-1}(30/225) = 7.6°$ above the horizontal

Example 8.2.7

A 10 m ladder of weight 50 N leans against a smooth wall at 50° to the rough floor. (a) How large a friction force must act at the foot so that it does not slip? (b) What coefficient of static friction is needed?



Solution

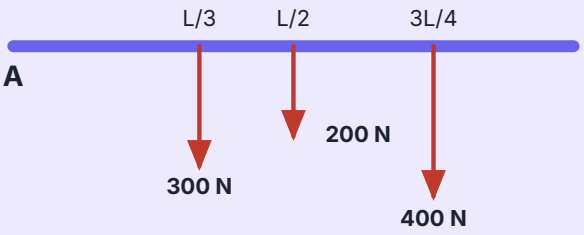
$\Sigma F_x = 0$: $f_s = N_1$ (1) ; $\Sigma F_y = 0$: $N_2 = 50$ N (2)

$\Sigma\tau = 0$ about the foot A: $-5(50 \sin 40°) + 10(N_1 \sin 50°) = 0 \Rightarrow N_1 = 21$ N

(a) f_s **= 21 N** (b) $f_s = \mu_s N_2 \Rightarrow \mu_s = \dfrac{21}{50}$ = **0.42**

Exercises: equilibrium of a rigid body

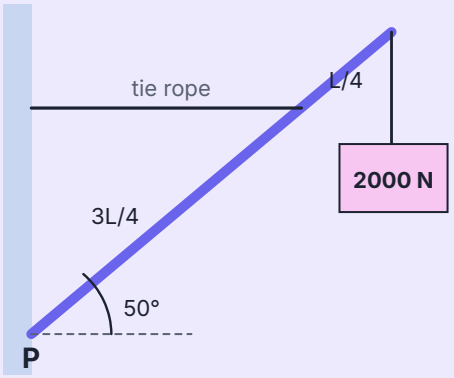
1. A uniform board of weight 0.20 kN and length L has 300 N hanging at $L/3$ from end A and 400 N at $3L/4$ from A. What single additional force will keep the board in equilibrium?



Answer

0.90 kN upward, at 0.56L from A

2. A uniform 0.40 kN beam is hinged at P and held at 50° to the horizontal by a horizontal tie rope attached $3L/4$ along it; a 2000 N load hangs from its end. Calculate the tension in the rope and the force of the hinge on the beam.



Answer

T = 2460 N; hinge force 3.4 kN at 44° to the horizontal

Practice question · centre of gravity and toppling

A rectangular block 4 cm wide and 8 cm long stands on a plane inclined at angle θ. Which statement is true?

A. The block topples if θ = 25°.

B. The block does not topple because it has a higher centre of gravity.

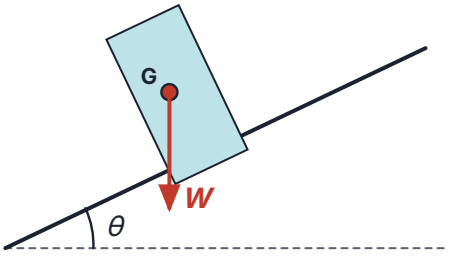
C. The block does not topple while the line of action of its weight passes inside its base (through or before the pivot).

D. The block topples if its length lies along the plane at the same angle.

Answer

Answer: C

The block tips over the lower edge when the vertical line through its centre of gravity passes outside the base, i.e. when $\tan\theta > \frac{4}{8}$, $\theta > 26.6°$. While the weight acts inside the base, its moment about the pivot edge restores the block. At θ = 25° it does not topple.



Chapter 9 · Semester 1

DEFORMATION OF SOLIDS

Notes by Cikgu Suhaili

2 worked examples 1 simulation Exercises with answers

Deformation

Deformation

is any **change in the shape or size of an object** caused by an external force.

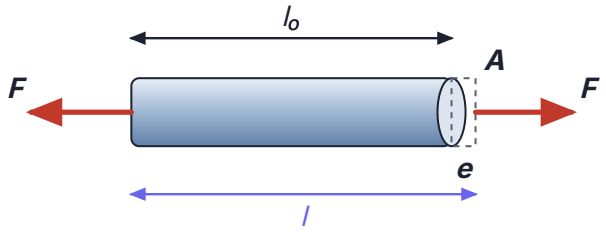
Elastic deformation

The object **regains its original shape and size** when the force is removed. It occurs when the material is deformed **within its elastic limit**.

Plastic deformation

The object **does not regain its original shape** when the force is removed. It occurs when the material is deformed **beyond its elastic limit**.

9.1 Stress and strain



A wire of original length l_o and cross-sectional area A, stretched by a force F perpendicular to A, extends by e.

Tensile stress, σ

is the **force per unit cross-sectional area** acting on a body that tends to deform it.

$$\sigma = \frac{F}{A}$$

- Scalar quantity.
- Dimension $\mathrm{M\,L^{-1}\,T^{-2}}$; unit N m^{-2} = Pa (kg m^{-1} s^{-2}).

Tensile strain, ε

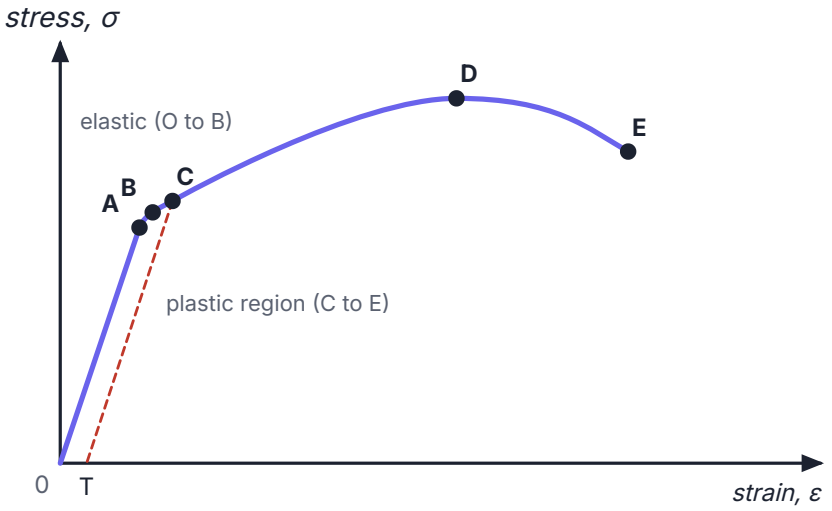
is the **ratio of the extension to the original length**.

$$\varepsilon = \frac{e}{l_0} = \frac{l - l_0}{l_0}$$

- Scalar quantity, no unit (dimensionless).
- l_0 = original length, l = final length, e = extension.

9.2 Force–extension and stress–strain graphs

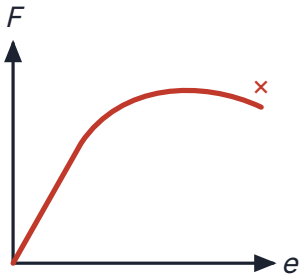
Stress–strain graph for a ductile metal



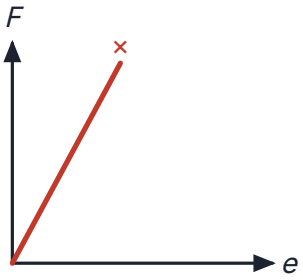
Point or region	Meaning
OA, A = **limit of proportionality**	Stress increases linearly with strain up to A. The straight line OA obeys **Hooke's law**: the restoring force is directly proportional to the extension.
B = **elastic limit**	Beyond B the material is permanently stretched and never regains its original length. In the region OB the material undergoes **elastic deformation**.
C = **yield point**	A small increase in stress produces a large increase in strain. When the stress is removed the material keeps a permanent strain (OT). Planes of atoms slide across each other and the wire thins.
CD = **plastic deformation**	As the stress increases, the strain increases rapidly.
D = maximum stress	The **breaking stress**, also called the **ultimate tensile strength** (UTS).
E = fracture point	The material breaks.

Unloading from a point beyond the elastic limit follows a line parallel to OA, so the material is left with a permanent strain OT.

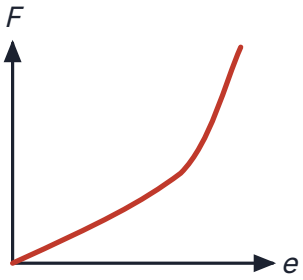
Force–extension graphs for ductile, brittle and polymeric materials



Ductile



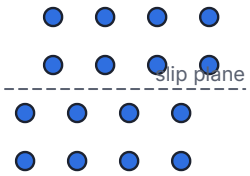
Brittle



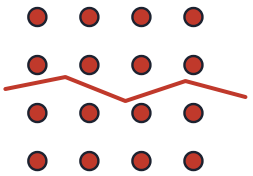
Polymer (rubber)

Material	Examples	Shape of the F–e graph
Ductile	copper, mild steel, aluminium	straight line, then a long plastic region before fracture
Brittle	glass, cast iron, ceramics	straight line almost up to fracture; snaps suddenly with almost no plastic extension
Polymeric	rubber	non-linear, S-shaped; very large extension, much of it recoverable

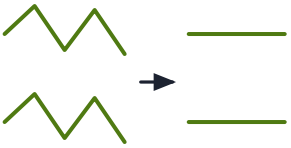
The gradient of the straight part is the stiffness. A steeper line (e.g. brittle glass) means a larger Young's modulus, but the material still fractures suddenly with little or no plastic deformation.



Ductile: planes slip



Brittle: crack runs through



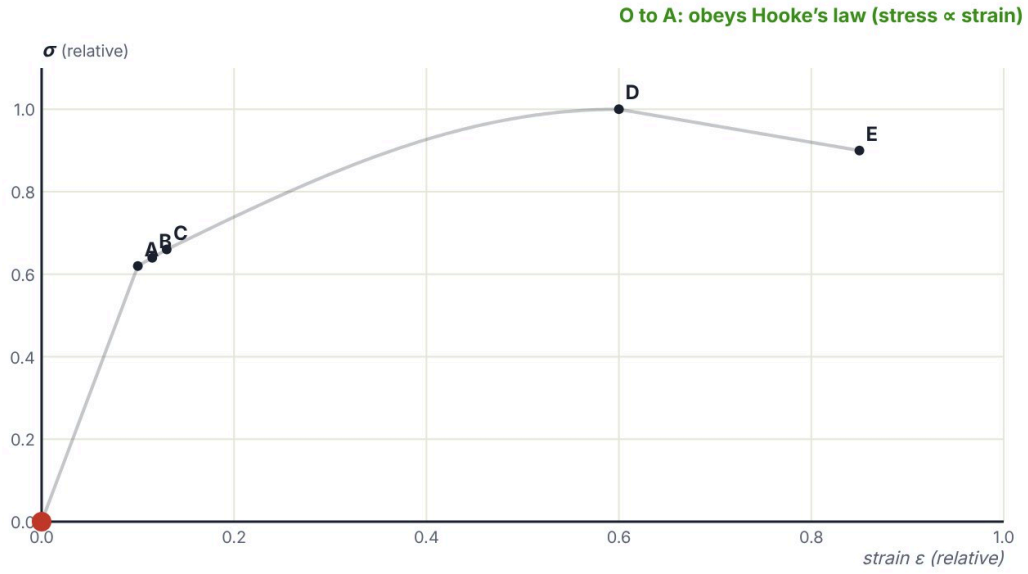
Polymer: chains uncoil

- **Ductile metals**: atoms are held by a "sea" of shared electrons that bond equally in every direction. Under tension whole planes of atoms slide past one another (dislocation motion) without breaking the metallic bond, so the metal stretches a long way before fracture.
- **Brittle materials**: strong ionic or covalent bonds hold atoms rigidly and do not allow planes to slide. The bonds stretch only slightly; when those along one plane reach their limit they all snap together and a crack runs straight through.
- **Polymers**: long tangled chain molecules held together by weak van der Waals forces. A small force uncoils and straightens the chains (large extension for little force). Once the chains are aligned, the force acts on the strong covalent bonds along the backbone, so the graph becomes steep: the S-shaped curve.

Simulation

Stretch a wire: stress–strain explorer

Choose a material and slowly increase the load. The point moves along the curve and the region is named. Release the load after passing the elastic limit to see the permanent extension.



material	stress now	strain now	permanent strain
Ductile metal (copper)	**0.00**	**0.000**	**0.000**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 9 to use the simulation.

Young's modulus

Young's modulus, Y (or E)

is the **ratio of tensile stress to tensile strain**, provided the limit of proportionality is not exceeded.

$$Y = \frac{\sigma}{\varepsilon} = \frac{F/A}{e/l_0} = \frac{Fl_0}{Ae}$$

dimension $\mathrm{M\,L^{-1}\,T^{-2}}$; unit $\mathrm{N\,m^{-2}}$ = Pa

- Young's modulus describes the elastic properties of a material.
- The larger Y, the harder it is to stretch the material.

Material	Y / 10^{10} Pa
Aluminium	6.9
Copper	11.7
Steel	20
Nylon	0.37
Glass	7.0

Young's modulus and Hooke's law for a wire

Young's modulus: $Y = \frac{Fl_0}{Ae} \Rightarrow F = (\frac{YA}{l_0})e$ (i)

Hooke's law: $F = ke$ (ii)

Comparing (i) and (ii):

$$k = \frac{YA}{l_0}$$

force constant (spring constant) of the wire

9.3 Strain energy

Strain energy

is the **elastic potential energy stored in a stretched (or compressed) material**: the work done on the wire in stretching it.

Up to the limit of proportionality, F increases linearly from 0 to F as the extension goes from 0 to e.

Work done = area under the F–e graph = $\frac{1}{2}Fe$, or $W = \int_0^e kx \, dx = \frac{1}{2}ke^2$

$$U = \tfrac{1}{2}Fe = \tfrac{1}{2}ke^2$$

strain energy = area under the force–extension graph

Divide by the volume Al_0: $\frac{U}{Al_0} = \frac{1}{2}\,\frac{F}{A}\,\frac{e}{l_0} = \frac{1}{2}\sigma\varepsilon$

Strain energy per unit volume = $\frac{1}{2}\sigma\varepsilon$ = area under the stress–strain graph

Example 9.1

A wire of length 1.0 m stretches by 0.6 mm when a 20 kg mass hangs on it. The radius of the wire is 1 mm. Calculate (a) the stress, (b) the strain, (c) the Young's modulus, (d) the force constant of the wire. (g = 9.81 m s^{-2})

Solution

l_0 = 1.0 m, e = 0.6 × 10^{-3} m, m = 20 kg, r = 1 × 10^{-3} m

(a) $\sigma = \dfrac{mg}{\pi r^2} = \dfrac{(20)(9.81)}{\pi(1 \times 10^{-3})^2} =$ **6.24 × 10^7 Pa**

(b) $\varepsilon = \dfrac{e}{l_0} =$ **6.0 × 10^{-4}**

(c) $Y = \dfrac{\sigma}{\varepsilon} =$ **1.04 × 10^{11} Pa**

(d) $k = \dfrac{Y(\pi r^2)}{l_0} =$ **3.27 × 10^5 N m^{-1}**

Example 9.2

A bridge support cable has cross-sectional area 0.0085 m^2 and length 35 m. It is made of high-tensile steel with Young's modulus 2.8 × 10^{11} Pa. The tension in the cable is 720 kN. Calculate (a) the extension, (b) the strain energy stored.

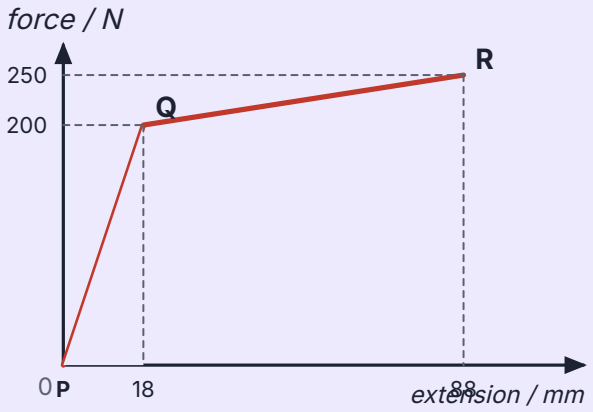
Solution

(a) $Y = \dfrac{Tl_0}{Ae} \Rightarrow e = \dfrac{Tl_0}{YA} = \dfrac{(720 \times 10^3)(35)}{(2.8 \times 10^{11})(0.0085)} =$ **0.0106 m**

(b) $U = \frac{1}{2}Te = \frac{1}{2}(720 \times 10^3)(0.0106) =$ **3.82 kJ**

Exercises: deformation of solids

1. A wire of unstretched length 2.00 m is stretched by a tensile force. Its force–extension graph has two straight sections: PQ from 0 to 200 N at 18 mm, and QR from 200 N to 250 N at 88 mm. Calculate (a) the strain in the wire at Q and at R, (b) the strain energy stored in the wire at Q, (c) the work done in stretching the wire from P to R.



Answer

(a) 0.009 at Q; 0.044 at R (b) ½(200)(0.018) = 1.8 J (c) 1.8 + ½(200 + 250)(0.070) = 17.55 J

The slide asks for "strain energy" in (a); the answers 0.009 and 0.044 are the strains.

2. A metal wire 75.0 cm long and 0.13 cm in diameter stretches 0.0350 cm when an 8.00 kg load hangs on it. Find the stress, the strain and the Young's modulus.

Answer

5.91×10^{7} N m^{-2}; 4.67×10^{-4}; 1.27×10^{11} N m^{-2}

Chapter 10 · Semester 1

KINETIC THEORY OF GASES

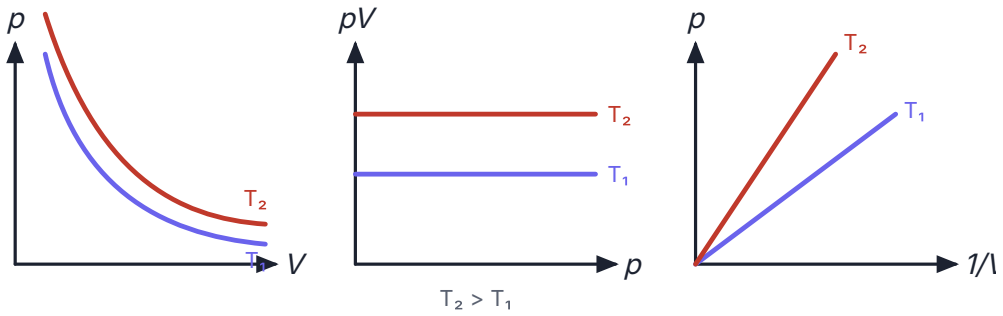
Notes by Cikgu Suhaili

11 worked examples 1 simulation Exercises with answers

10.1 Ideal gas equation

Boyle's law

The pressure of a fixed mass of gas at constant temperature is **inversely proportional to its volume**: $p \propto \frac{1}{V}$, so pV = constant and $p_1V_1 = p_2V_2$.



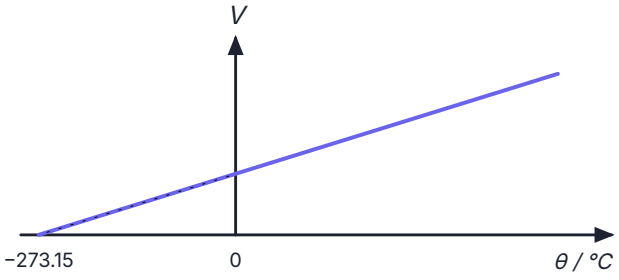
Graphs of Boyle's law at two temperatures ($T_2 > T_1$).

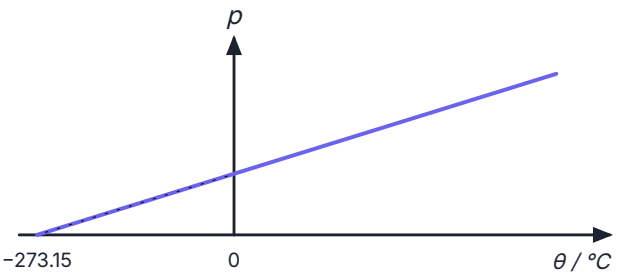
Charles's law

The volume of a fixed mass of gas at constant pressure is **directly proportional to its absolute temperature**: $V \propto T$, so $\frac{V_1}{T_1} = \frac{V_2}{T_2}$.

Pressure (Gay-Lussac's) law

The pressure of a fixed mass of gas at constant volume is **directly proportional to its absolute temperature**: $p \propto T$, so $\frac{p_1}{T_1} = \frac{p_2}{T_2}$.





Plotted against temperature in °C, both lines extrapolate to −273.15 °C (0 K, absolute zero); plotted against kelvin they pass through the origin.

Ideal gas

is a gas that **obeys the ideal gas equation $pV = nRT$ exactly** at all pressures and temperatures.

Combining the three gas laws gives the **ideal gas equation** (no derivation needed):

$$pV = nRT = NkT$$

n = number of moles, p in Pa, V in m^3, T in K, R = 8.31 J mol^{-1} K^{-1} (molar gas constant), N = number of molecules, $k = R/N_A$ = 1.38 × 10^{-23} J K^{-1} (Boltzmann constant)

- $n = \frac{N}{N_A} = \frac{m}{M}$, where N_A = 6.02 × 10^{23} mol^{-1} and M is the molar mass.
- For a fixed mass of gas: $\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$.

Example 10.1

Estimate the number of molecules in a 5 × 10^{-4} m^3 flask of oxygen at 2 × 10^5 Pa and 300 K. (R = 8.31 J mol^{-1} K^{-1}, N_A = 6 × 10^{23} mol^{-1})

Solution

$$N = nN_A = \frac{pV}{RT}N_A = \frac{(2 \times 10^5)(5 \times 10^{-4})}{(8.31)(300)}(6 \times 10^{23}) = \mathbf{2.41 \times 10^{22}\ molecules}$$

Example 10.2

Bulbs X and Y, kept at constant temperature, contain an ideal gas at 100 kPa and 300 kPa. The volume of X is three times that of Y. When tap T is opened the gases mix. Find the final pressure.

Solution

The total number of moles is constant: $p_XV_X + p_YV_Y = p(V_X + V_Y)$, with $V_X = 3V_Y$

$(100)(3V_Y) + (300)V_Y = p(4V_Y) \Rightarrow \mathbf{p = 150\ kPa}$

Example 10.3

At constant pressure an ideal gas has volume 0.00308 m^3 at 273 K. Find its temperature when its volume is 0.00783 m^3.

Solution

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \Rightarrow T = 273 \times \frac{0.00783}{0.00308} = \mathbf{694\ K}$$

Exercises: gas laws

1. Vessel X has twice the volume of vessel Y. Both contain an ideal gas at the same pressure; X is at 200 K and Y at 400 K. If the mass of gas in X is m, find the mass of gas in Y.

   Answer

   $m \propto \frac{pV}{T}: \frac{m_Y}{m_X} = (½)(\frac{200}{400}) = ¼$, so $m/4$

2. A cylinder with a movable piston contains gas at 27 °C, 0.50 × 10^5 Pa and 1.25 m^3. Find the final temperature if it is compressed to 0.80 m^3 and the pressure rises to 0.82 × 10^5 Pa.

   Answer

   $T_2 = 300 \times \frac{(0.82)(0.80)}{(0.50)(1.25)} = 314.9\ K = 41.9\ °C$

3. A 24 L tank contains helium at 27 °C and 22.0 atm. How many moles are in the tank? (R = 0.0821 L atm mol^{-1} K^{-1})

   Answer

   $n = \frac{(22.0)(24)}{(0.0821)(300)} = 21.5\ mol$

4. Oxygen occupies 0.0200 m^3 at 101 kPa and 5.0 °C. Find its volume at 108 kPa and 30 °C.

   Answer

   $V_2 = 0.0200 \times \frac{101}{108} \times \frac{303}{278} = 0.0204\ m^3$

10.2 Pressure of a gas

The kinetic theory explains the large-scale (macroscopic) properties of a gas from the behaviour of its molecules.

Assumptions of the kinetic theory for an ideal gas

1. A gas is made up of a large number of identical atoms or molecules.
2. The molecules move randomly in all directions.
3. The volume of the molecules is negligible compared with the volume occupied by the gas.
4. Intermolecular forces are negligible except during collisions.
5. Collisions between molecules, and with the walls, are perfectly elastic.
6. The duration of a collision is negligible compared with the time between collisions.
7. Between collisions molecules move with constant velocity; gravity has no effect on their motion.

Ideal gas and real gas

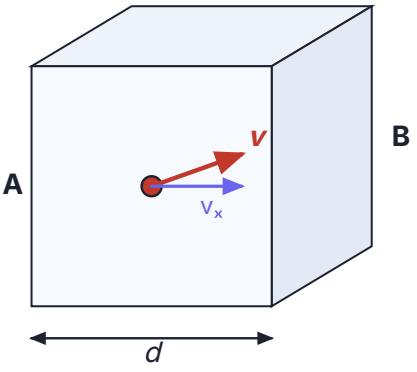
Ideal gas	Real gas
no intermolecular forces	intermolecular forces exist (attraction at moderate range, repulsion at very short range)
molecules have negligible volume	molecules have a finite volume
obeys $pV = nRT$ exactly at all pressures and temperatures	obeys $pV = nRT$ only approximately, at **low pressure and high temperature**
collisions perfectly elastic	collisions may lose a little kinetic energy because of intermolecular forces

Derivation of $p = \frac{1}{3}\rho\langle c^2\rangle$

A cubic box of side d contains N molecules, each of mass m. A molecule has velocity c with components c_x, c_y, c_z.

It hits wall A and rebounds elastically with $-c_x$. Change in momentum: $\Delta p = -mc_x - mc_x = -2mc_x$ (the wall receives $+2mc_x$).

Time to travel to wall B and back to A: $\Delta t = \dfrac{2d}{c_x}$



Average force of one molecule on wall A (impulse = change of momentum): $F_{x1} = \frac{2mc_x}{2d/c_x} = \frac{mc_x^2}{d}$

For N molecules: $F_x = \frac{m}{d}(c_{x1}^2 + c_{x2}^2 + \ldots + c_{xN}^2) = \frac{Nm}{d}\langle c_x^2\rangle$, where $\langle c_x^2\rangle$ is the mean of c_x^2

Since $c^2 = c_x^2 + c_y^2 + c_z^2$ and the motion is random, $\langle c_x^2\rangle = \langle c_y^2\rangle = \langle c_z^2\rangle = \frac{1}{3}\langle c^2\rangle$

$F = \frac{Nm}{3d}\langle c^2\rangle$; pressure $p = \frac{F}{d^2} = \frac{Nm}{3d^3}\langle c^2\rangle = \frac{Nm}{3V}\langle c^2\rangle$

Density $\rho = \frac{Nm}{V}$, so:

$$p = \tfrac{1}{3}\rho\langle c^2\rangle = \frac{Nm}{3V}\langle c^2\rangle$$

$\langle c^2\rangle$ = mean square speed; the root mean square speed is $c_{rms} = \sqrt{\langle c^2\rangle}$

Compare $pV = \frac{1}{3}Nm\langle c^2\rangle$ with $pV = nRT$, and use Nm = total mass = nM:

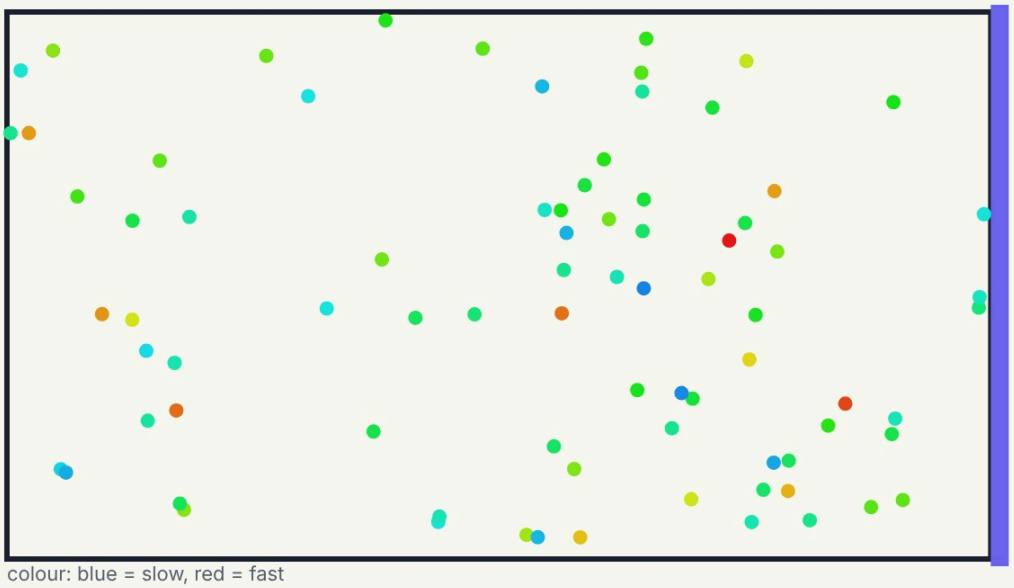
$$c_{rms} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3kT}{m}}$$

M = molar mass (kg mol^{-1}), m = mass of one molecule

Simulation

Gas in a box: molecules, pressure and temperature

Each dot is a molecule. Raise the temperature: the molecules move faster, hit the walls harder and more often, so the pressure rises. Halve the volume at constant temperature: the pressure doubles (Boyle's law).



colour: blue = slow, red = fast

temperature T	c_rms compared with 300 K	pressure measured from wall hits
300 K	**1.00 ×**	**1.02 p_o**

ideal gas law: $p = (N/80)(T/300)(V_o/V)\,p_o$

1.00 p_o

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 10 to use the simulation.

Example 10.4

Ten gas molecules have speeds of 1, 1, 2, 2, 3, 3, 4, 4, 5 and 5 m s^{-1}. Find (a) the mean square speed, (b) the root mean square speed.

Solution

(a) $\langle c^2 \rangle = \dfrac{2(1^2 + 2^2 + 3^2 + 4^2 + 5^2)}{10} = \dfrac{110}{10} = \mathbf{11\ m^2\ s^{-2}}$

(b) $c_{rms} = \sqrt{11} = \mathbf{3.32\ m\ s^{-1}}$ (note the mean speed is 3.0 m s^{-1}: the rms speed is larger)

Example 10.5

Air at room temperature has density about 1.29 kg m^{-3}. Treating it as one gas at 100 kPa, find c_{rms} for its molecules.

Solution

$$c_{\text{rms}} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3(1.00 \times 10^5)}{1.29}} = \mathbf{482\ m\ s^{-1}}$$

Example 10.6

A gas is at 20 °C. To what temperature must it be raised to double the rms speed of its molecules?

Solution

$c_{\text{rms}} \propto \sqrt{T}$, so doubling c_{rms} needs $4T$

$T = 4(293.15) = \mathbf{1172.6\ K = 899.5\ °C}$

Example 10.7

A 0.08 m^3 cylinder contains oxygen at 280 K and 90 kPa. Find (a) the mass of oxygen, (b) the number of molecules, (c) the rms speed. (R = 8.31 J mol^{-1} K^{-1}, M = 0.032 kg mol^{-1}, N_A = 6.02 × 10^{23} mol^{-1})

Solution

$$n = \frac{pV}{RT} = \frac{(90 \times 10^3)(0.08)}{(8.31)(280)} = 3.09\ \text{mol}$$

(a) $m = nM = (3.09)(0.032) = \mathbf{0.099\ kg}$

(b) $N = nN_A = \mathbf{1.86 \times 10^{24}}$

(c) $c_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.31)(280)}{0.032}} = \mathbf{467\ m\ s^{-1}}$

Exercises: pressure and rms speed

1. What is the rms speed of nitrogen molecules in an 8.0 m^3 volume at 2.1 atm if there are 1300 mol of nitrogen? (M = 28 g mol^{-1})

   Answer

   3.7×10^2 m s^{-1}

2. An object needs at least 11.2 km s^{-1} to escape from the Earth. At what temperature does c_{rms} for H_2 molecules equal this speed? Repeat for N_2. (M_{H_2} = 2 g mol^{-1}, M_{N_2} = 28 g mol^{-1})

   Answer

   1.0×10^4 K; 1.4×10^5 K

3. Outer space has about one hydrogen atom per cm^3 at about 3.5 K. Calculate the rms speed of these atoms and the pressure they exert. (M = 1 g mol^{-1})

   Answer

   295 m s^{-1}; $p = NkT/V \approx 5 \times 10^{-17}$ Pa

10.3 Molecular kinetic energy

From $pV = \frac{1}{3}Nm\langle c^2\rangle$ and $pV = NkT$: $\frac{1}{3}m\langle c^2\rangle = kT$

Multiply by $\frac{3}{2}$: $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$

$$\langle K\rangle = \tfrac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$$

mean translational kinetic energy of one molecule; it depends only on the absolute temperature

- $k = \frac{R}{N_A} = \frac{8.31}{6.02 \times 10^{23}} = 1.38 \times 10^{-23}$ J K^{-1} (Boltzmann constant: the gas constant per molecule).
- Total translational kinetic energy of N molecules: $E = \frac{3}{2}NkT = \frac{3}{2}nRT$.

Example 10.8

What is the average translational kinetic energy of a molecule of an ideal gas at 27 °C?

Solution

$$\langle K \rangle = \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23})(300) = \mathbf{6.21 \times 10^{-21}\ J}$$

Example 10.9

One mole of oxygen (32 g) behaves as an ideal gas at 273 K and 1.01×10^5 Pa. Calculate (a) its volume, (b) its density, (c) the rms speed of its molecules, (d) the average translational kinetic energy of a molecule.

Solution

(a) $V = \frac{nRT}{p} = \frac{(1)(8.31)(273)}{1.01 \times 10^5} = \mathbf{2.25 \times 10^{-2}\ m^3}$

(b) $\rho = \frac{0.032}{2.25 \times 10^{-2}} = \mathbf{1.42\ kg\ m^{-3}}$

(c) $c_{rms} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3(1.01 \times 10^5)}{1.42}} = \mathbf{461\ m\ s^{-1}}$

(d) $\langle K \rangle = \frac{3}{2}(1.38 \times 10^{-23})(273) = \mathbf{5.65 \times 10^{-21}\ J}$

The slide answer for (c), 5.94×10^{-10} m s^{-1}, is a slip; the rms speed of oxygen at 273 K is about 461 m s^{-1}.

10.4 Degrees of freedom and equipartition of energy

Degrees of freedom, f

of a molecule is the **number of independent ways (modes of motion) in which it can store energy**: translational, rotational and vibrational.

Law of equipartition of energy

Each degree of freedom of a molecule has an **average energy of $\frac{1}{2}kT$**.

$$\langle K \rangle = \frac{f}{2} kT$$

average kinetic energy per molecule

Molecule (at about 300 K)	Translational	Rotational	Total f	Average energy per molecule
monatomic (He, Ne, Ar)	3	0	3	$\frac{3}{2}kT$
diatomic (H_2, O_2, N_2)	3	2	5	$\frac{5}{2}kT$
polyatomic, non-linear (H_2O, NH_3)	3	3	6	$3kT$

- A monatomic molecule moves in three independent directions (x, y, z): 3 translational degrees of freedom.
- A diatomic molecule also rotates about two axes perpendicular to its bond (rotation about the bond itself has negligible energy): 3 + 2 = 5.
- A non-linear polyatomic molecule rotates about three axes: 3 + 3 = 6.

Variation of the degrees of freedom of a diatomic molecule with temperature

Temperature	Modes of motion active	f
very low (about 30 K)	translational only	3
room temperature (about 300 K)	translational + rotational	5
very high (about 3000 K)	translational + rotational + vibrational	7

Rotational and vibrational modes are "frozen out" at low temperature because the energy gaps between their quantum levels are large compared with kT. As the temperature rises, kT becomes large enough to excite them, so the degrees of freedom, and the molar heat capacity, increase. (Vibration adds 2: kinetic and potential energy of the vibration.)

10.5 Internal energy of an ideal gas

> **Internal energy, U**
>
> of an ideal gas is the **total kinetic energy of all its molecules**. There are no intermolecular forces, so there is no potential energy between molecules.

For N molecules each with f degrees of freedom: $U = N \times \frac{f}{2} kT$

With $Nk = nR$:

$$U = \frac{f}{2} NkT = \frac{f}{2} nRT$$

monatomic: $U = \frac{3}{2} nRT$; diatomic at room temperature: $U = \frac{5}{2} nRT$

The internal energy of an ideal gas depends only on its temperature (and the amount of gas), not on its pressure or volume.

Example 10.10

Neon is a monatomic gas. Find the internal energy of 1 kg of neon at 293 K. (Molar mass 20 g mol^{-1})

Solution

$$n = \frac{1000}{20} = 50 \text{ mol}$$

$$U = \frac{3}{2} nRT = \frac{3}{2}(50)(8.31)(293) = \mathbf{1.83 \times 10^5 \text{ J}}$$

Example 10.11

A 0.30 m^3 tank contains 1.50 mol of neon at 22 °C. Find (a) the total internal energy, (b) the total translational kinetic energy, (c) the average translational kinetic energy per molecule, (d) the rms speed. (M = 0.020 kg mol^{-1})

Solution

T = 295 K. (a) $U = \frac{3}{2}nRT = \frac{3}{2}(1.50)(8.31)(295)$ = **5.52 kJ**

(b) Neon is monatomic, so all its kinetic energy is translational: **5.52 kJ**

(c) $\frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23})(295)$ = **6.10×10^{-21} J**

(d) $c_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.31)(295)}{0.020}}$ = **606 m s^{-1}**

Chapter 11 · Semester 1

THERMODYNAMICS OF GASES

Notes by Cikgu Suhaili

16 worked examples 1 simulation Exercises with answers

11.1 Molar heat capacity

Molar heat capacity at constant pressure, $C_{p,m}$

is the **heat required to raise the temperature of 1 mole of gas by 1 K** (or 1 °C) at **constant pressure**.

Molar heat capacity at constant volume, $C_{V,m}$

is the **heat required to raise the temperature of 1 mole of gas by 1 K** (or 1 °C) at **constant volume**.

$$C_{p,m} = \frac{Q}{n\Delta T} \quad \Rightarrow \quad Q = nC_{p,m}\Delta T$$

$$C_{V,m} = \frac{Q}{n\Delta T} \quad \Rightarrow \quad Q = nC_{V,m}\Delta T$$

$$C_{p,m} - C_{V,m} = R \qquad \gamma = \frac{C_{p,m}}{C_{V,m}}$$

γ is dimensionless and always greater than 1: monatomic 1.67, diatomic 1.40, polyatomic 1.33

Example 11.1

440.0 J of heat raises the temperature of 2.0 mol of hydrogen by 10 K at constant volume. Treating hydrogen as an ideal gas, find its molar heat capacity at constant pressure.

Solution

$$C_{V,m} = \frac{Q}{n\Delta T} = \frac{440.0}{2.0(10)} = 22.0 \text{ J mol}^{-1}\text{ K}^{-1}$$

Hydrogen is diatomic, $\gamma = 1.40$: $C_{p,m} = \gamma C_{V,m} = (1.40)(22.0) = \mathbf{30.8\ J\ mol^{-1}\ K^{-1}}$

Example 11.2

One mole of an ideal gas in a closed container at constant volume is heated until its pressure doubles. The heat supplied is 3735 J and the final temperature is 600 K. Find the molar heat capacity at constant pressure. (R = 8.31 J K^{-1} mol^{-1})

Solution

Constant volume: $\frac{p_1}{T_1} = \frac{p_2}{T_2}$ with $p_2 = 2p_1 \Rightarrow T_1 = 300$ K

$Q = nC_{V,m}(T_2 - T_1) \Rightarrow C_{V,m} = \frac{3735}{(1)(300)} = 12.45$ J K^{-1} mol^{-1}

$C_{p,m} = C_{V,m} + R = 12.45 + 8.31 =$ **20.76 J K^{-1} mol^{-1}**

11.2 Work done by a gas

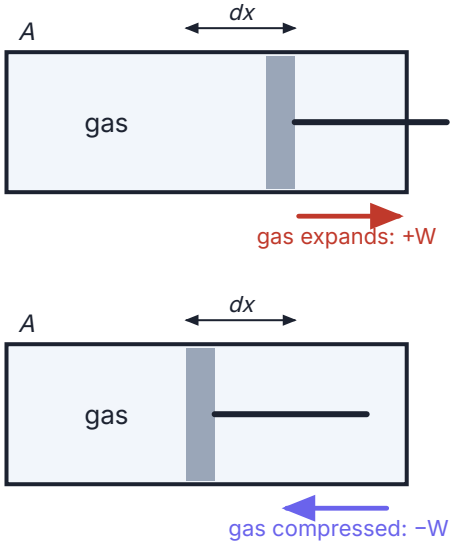
A gas at pressure p in a cylinder of cross-sectional area A pushes the piston with force $F = pA$.

When the piston moves out a small distance dx, the work done by the gas is d$W = F\,\mathrm{d}x = pA\,\mathrm{d}x = p\,\mathrm{d}V$

When the volume changes from V_1 to V_2:

$$W = \int_{V_1}^{V_2} p\,\mathrm{d}V$$

Work done by the gas = area under the p–V graph



- Gas **expands** (volume increases): work is done **by** the gas, W positive.
- Gas is **compressed** (volume decreases): work is done **on** the gas, W negative.

Work done in each process

Process	Work done by the gas
constant temperature (isothermal): $pV = nRT$ = constant	$W = \int \frac{nRT}{V}\,dV = nRT\ln\frac{V_2}{V_1} = nRT\ln\frac{p_1}{p_2}$
constant volume (isochoric)	$W = 0$ ($dV = 0$)
constant pressure (isobaric)	$W = p(V_2 - V_1) = p\Delta V = nR\Delta T$
adiabatic: pV^γ = constant	$W = \frac{p_1V_1 - p_2V_2}{\gamma - 1}$

Adiabatic work. $pV^\gamma = p_1V_1^\gamma$, so $p = \frac{p_1V_1^\gamma}{V^\gamma}$

$$W = p_1V_1^\gamma \int_{V_1}^{V_2} V^{-\gamma}\,dV = p_1V_1^\gamma\,\frac{V_2^{1-\gamma} - V_1^{1-\gamma}}{1-\gamma}$$

Using $p_1V_1^\gamma = p_2V_2^\gamma$: $W = \frac{p_2V_2 - p_1V_1}{1-\gamma} = \frac{p_1V_1 - p_2V_2}{\gamma - 1}$

Example 11.3

How much work is done by an ideal gas expanding at constant temperature (isothermally) from 3.00 L at 20.0 atm to 24.0 L? (1 L = 1×10^{-3} m^3, 1 atm = 1.013×10^5 Pa)

Solution

$$W = p_1V_1\ln\frac{V_2}{V_1} = (20.0 \times 1.013 \times 10^5)(3.00 \times 10^{-3})\ln\frac{24.0}{3.00} = \mathbf{1.26 \times 10^4\ J}$$

Example 11.4

2.00 L of an ideal gas at 300 K and 20.0 atm expands at constant pressure (isobaric) while its temperature rises to 500 K. What work is done by the gas?

Solution

$$V_2 = V_1\frac{T_2}{T_1} = 2.00 \times \frac{500}{300} = 3.33\ \text{L}$$

$$W = p(V_2 - V_1) = (20.0 \times 1.013 \times 10^5)(1.33 \times 10^{-3}) = \mathbf{2.70 \times 10^3\ J}$$

11.3 First law of thermodynamics

- Thermodynamics is the study of energy relationships involving heat, mechanical work and other forms of energy transfer.
- The first law extends the principle of conservation of energy to include heat as well as mechanical energy.
- Three quantities: heat Q, change in internal energy ΔU, and work W.

> **First law of thermodynamics**
>
> The **heat supplied to a system** is equal to the **increase in its internal energy plus the work done by the system**.

$$Q = \Delta U + W \quad \text{or} \quad \Delta U = Q - W$$

Quantity	Positive (+)	Negative (−)
Q	heat supplied to the gas	heat released by the gas
W	work done **by** the gas (expansion)	work done **on** the gas (compression)
ΔU	internal energy increases (temperature rises)	internal energy decreases (temperature falls)

When heat is supplied to a gas its temperature rises, so its internal energy increases; at the same time it may expand and do work on its surroundings.

$\Delta U = nC_{V,m}\Delta T$ for any process

At constant volume (isochoric), $W = 0$, so $\Delta U = Q = nC_{V,m}\Delta T$

The internal energy of an ideal gas depends only on its temperature, not on its pressure or volume. So the same ΔT always gives the same ΔU:

$$\Delta U = nC_{V,m}\Delta T$$

valid for any process of an ideal gas

Additional derivation (syllabus 11.3(e)) · $C_{p,m} - C_{V,m} = R$

Derive the relationship between the two molar heat capacities.

Solution

Heat the gas at constant pressure (isobaric) through ΔT: $Q = nC_{p,m}\Delta T$, $W = p\Delta V = nR\Delta T$, $\Delta U = nC_{V,m}\Delta T$

First law: $nC_{p,m}\Delta T = nC_{V,m}\Delta T + nR\Delta T$

$\boldsymbol{C_{p,m} - C_{V,m} = R}$. More heat is needed at constant pressure because part of it is used to do work as the gas expands.

Additional derivation (syllabus 11.3(f)) · $C_{V,m} = (f/2)R$

Use the law of equipartition of energy.

Solution

$U = \frac{f}{2}nRT$ (Chapter 10), so $\Delta U = \frac{f}{2}nR\Delta T$

$\Delta U = nC_{V,m}\Delta T \Rightarrow \boldsymbol{C_{V,m} = \frac{f}{2}R}$, and $C_{p,m} = C_{V,m} + R = \frac{f+2}{2}R$

$\gamma = \frac{C_{p,m}}{C_{V,m}} = \frac{f+2}{f}$: monatomic ($f = 3$) 5/3 = 1.67; diatomic ($f = 5$) 7/5 = 1.40; polyatomic ($f = 6$) 8/6 = 1.33. So γ identifies the type of molecule.

Example 11.5

2500 J of heat is added to a system and 1800 J of work is done on the system. Calculate the change in internal energy.

Solution

$Q = +2500$ J, $W = -1800$ J: $\Delta U = Q - W = 2500 - (-1800) = \mathbf{4300\ J}$

Example 11.6

1 mol of a monatomic ideal gas is compressed with 6200 J of work done on it; its temperature changes from 350 K to 550 K. (a) How much heat flows between the gas and its surroundings? (b) Does the heat flow into or out of the gas?

Solution

$\Delta U = nC_{V,m}\Delta T = (1)(\frac{3}{2})(8.31)(200) = 2493$ J; $W = -6200$ J

(a) $Q = \Delta U + W = 2493 - 6200 =$ **-3.71×10^3 J**

(b) Q is negative: **heat flows out of the gas**

Example 11.7

A gas in a cylinder expands from 0.400 m^3 to 0.700 m^3 at a constant pressure of 2.00×10^5 Pa. The heat added is 1.40×10^5 J. Calculate the work done by the gas and the change in its internal energy.

Solution

$W = p\Delta V = (2.00 \times 10^5)(0.300) =$ **6.00×10^4 J**

$\Delta U = Q - W = 1.40 \times 10^5 - 6.00 \times 10^4 =$ **8.00×10^4 J**

Exercises: first law

1. A system absorbs 200 J of heat while its internal energy increases by 150 J. What work is done by the gas?

   Answer

   $W = Q - \Delta U = 50$ J

2. A technician supplies 340 J of heat to a gas while the surroundings do 140 J of work on the gas. What is the change in internal energy?

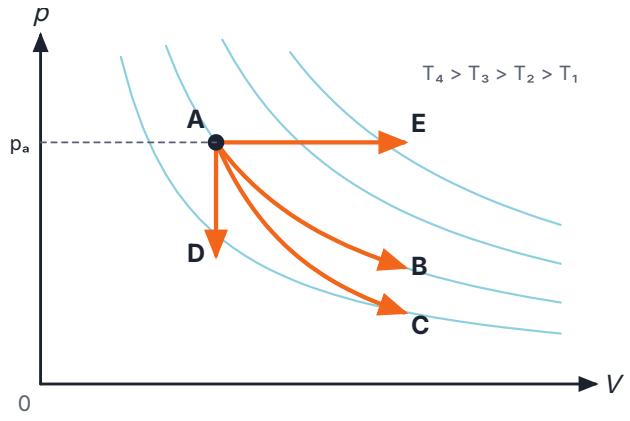
   Answer

   $\Delta U = 340 - (-140) = 480$ J

11.4 Constant-temperature (isothermal) and adiabatic changes

Process	Condition	First law, $\Delta U = Q - W$
constant temperature (isothermal)	$T_1 = T_2$; pV = constant (Boyle's law)	$\Delta U = 0 \Rightarrow Q = W$
constant volume (isochoric)	$V_1 = V_2$	$W = 0 \Rightarrow \Delta U = Q$
constant pressure (isobaric)	$p_1 = p_2$	$\Delta U = Q - p(V_2 - V_1)$
adiabatic	no heat transfer into or out of the system, $Q = 0$	$\Delta U = -W$

- A **constant-temperature (isothermal)** process occurs at constant temperature: slowly, in a container with good thermal contact with its surroundings.
- An **adiabatic** process occurs without heat transfer: quickly, or in a well-insulated container. Adiabatic expansion cools the gas; adiabatic compression heats it.



Path from A	Process	Temperature
A → B	constant temperature (isothermal)	$T_B = T_A$
A → C	adiabatic	$T_C < T_A$
A → D	constant volume (isochoric)	$T_D < T_A$
A → E	constant pressure (isobaric)	$T_E > T_A$

The adiabatic curve is steeper than the isotherm through the same point.

Equations for an adiabatic process

$\Delta U = nC_{V,m}\mathrm{d}T$ for any process (1) . Adiabatic, $Q = 0$: $\Delta U = -W = -p\,\mathrm{d}V$ (2)

(1) = (2): $nC_{V,m}\mathrm{d}T = -p\,\mathrm{d}V = -\dfrac{nRT}{V}\mathrm{d}V \Rightarrow \dfrac{\mathrm{d}T}{T} + \dfrac{R}{C_{V,m}}\dfrac{\mathrm{d}V}{V} = 0$

With $R = C_{p,m} - C_{V,m}$: $\frac{R}{C_{V,m}} = \gamma - 1$, so $\frac{dT}{T} + (\gamma - 1)\frac{dV}{V} = 0$

Integrating: $\ln T + (\gamma - 1) \ln V = \text{constant} \Rightarrow \ln(TV^{\gamma-1}) = \text{constant} \Rightarrow$ **$TV^{\gamma-1}$ = constant**

Substitute $T = \frac{pV}{nR}$ (n, R constant): $\frac{pV}{nR} V^{\gamma-1} = \text{constant} \Rightarrow$ **pV^{γ} = constant**

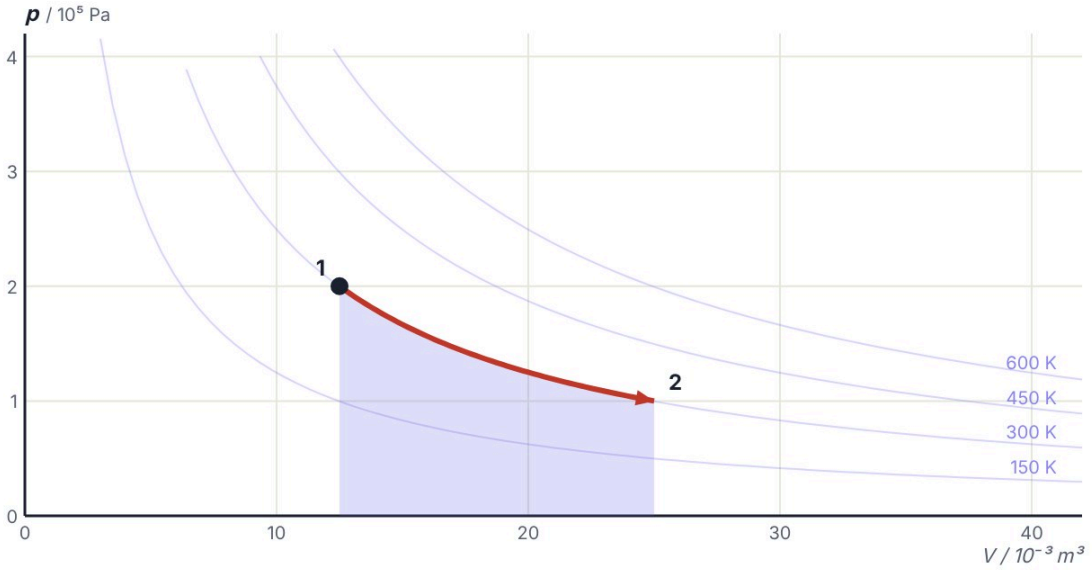
pV^{γ} = constant: $p_1 V_1^{\gamma} = p_2 V_2^{\gamma}$

$TV^{\gamma-1}$ = constant: $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$

adiabatic change; compare $p_1 V_1 = p_2 V_2$ for a constant-temperature change

Simulation p–V explorer: four thermodynamic processes

Start from the same state and choose a process. Drag the final volume (or, for constant volume, the final pressure). The shaded area is the work done by the gas; the bars show Q, W and ΔU, which always satisfy Q = ΔU + W.





Q (heat in) ΔU W (by gas)

process
Constant temperature (isothermal)

$T_1 \rightarrow T_2$
301 K → 301 K

p_2
1.00 × 10⁵ Pa

γ = (f + 2)/f
1.67

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 11 to use the simulation.

Example 11.8

1 mol of an ideal gas at 400 K and 8 atm occupies 4 m^3. It expands adiabatically until its pressure is 1 atm. Calculate (a) the final volume, (b) the final temperature. ($C_{V,m}$ = 12.5 J mol^{-1} K^{-1})

Solution

$\gamma = 1 + \frac{R}{C_{V,m}} = 1 + \frac{8.31}{12.5} = 1.67$

(a) $p_1V_1^\gamma = p_2V_2^\gamma \Rightarrow V_2 = 4(8)^{1/1.67} =$ **13.9 m^3**

(b) $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1} \Rightarrow T_2 = 400(\frac{4}{13.9})^{0.67} =$ **174 K**

Example 11.9

A fixed mass of gas at 7 °C and 1.00 × 10^5 Pa is compressed at constant temperature (isothermally) to one third of its volume, then expanded adiabatically back to its original volume. Find the final temperature and pressure (γ = 1.40).

Solution

Constant temperature: $p_1V = p(V/3) \Rightarrow p = 3.00 \times 10^5$ Pa, $T = 280.15$ K

Adiabatic: $p_2 = (3.00 \times 10^5)(\frac{1}{3})^{1.40} =$ **6.44 × 10^4 Pa**

$T_2 = 280.15(\frac{1}{3})^{0.40} =$ **181 K** (the slide gives 180.1 K)

Exercises: adiabatic changes

1. 20 cm^3 of a monatomic gas at 12 °C and 100 kPa is suddenly (adiabatically) compressed to 0.50 cm^3. Find its new pressure and temperature. (γ = 1.67)

   Answer

   4.74 × 10^7 Pa; 3.38 × 10^3 K

2. A monatomic ideal gas at 1.50×10^5 Pa and 0.0800 m^3 is compressed adiabatically to 0.0400 m^3. Find (a) the final pressure, (b) the ratio of the final temperature to the initial temperature.

Answer

(a) $1.50 \times 10^5 \times 2^{5/3} = 4.76 \times 10^5$ Pa (b) $2^{2/3} = 1.59$

Examination-style questions

Example 11.10

(a) Write (i) the first law of thermodynamics, stating the meaning of each symbol, (ii) the work done by an ideal gas at variable pressure. [3] (b) Sketch a p–V graph for 1 mol of an ideal gas showing the four thermodynamic processes. [5] (c) A monatomic ideal gas at pressure P and volume V is compressed at constant temperature until its pressure is $3P$. It then expands adiabatically until its volume is $9V$. With $P = 1.2 \times 10^5$ Pa, $V = 1 \times 10^{-2}$ m^3 and γ = 1.67, find (i) the final pressure, (ii) the work done on the gas during the compression. [7]

Solution

(a)(i) $Q = \Delta U + W$: Q = heat supplied to the gas, ΔU = increase in internal energy, W = work done by the gas. (ii) $W = \int_{V_1}^{V_2} p \, dV$

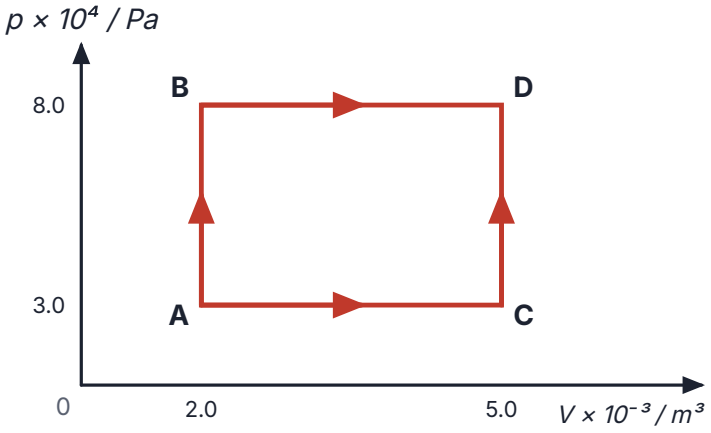
(b) See the p–V diagram above with the four labelled paths from A.

(c)(i) Constant temperature: $PV = 3PV' \Rightarrow V' = V/3$. Adiabatic: $p_f = 3P\left(\frac{V/3}{9V}\right)^{1.67} = (3.6 \times 10^5)\left(\frac{1}{27}\right)^{1.67}$ = **1.47×10^3 Pa**

(ii) Work done on the gas = $nRT \ln \frac{V}{V/3} = PV \ln 3 = (1.2 \times 10^5)(1 \times 10^{-2})(1.099)$ = **1.32×10^3 J**

Example 11.11

A system changes state as shown. In process AB, 150 J of heat is added; in process BD, 600 J is added. Find (a) ΔU in AB, (b) ΔU in ABD, (c) the total heat added in ACD.



Solution

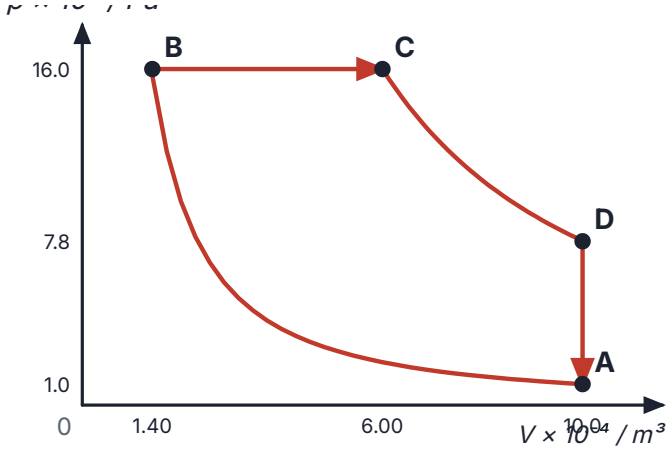
(a) AB is at constant volume: $W_{AB} = 0$, $\Delta U_{AB} = Q_{AB} =$ **150 J**

(b) $W_{ABD} = p_B(V_D - V_B) = (8 \times 10^4)(3.0 \times 10^{-3}) = 240$ J; $Q_{ABD} = 750$ J $\Rightarrow \Delta U_{ABD} = 750 - 240 =$ **510 J**

(c) Same end points, so $\Delta U_{ACD} = 510$ J; $W_{ACD} = p_A(V_C - V_A) = (3 \times 10^4)(3.0 \times 10^{-3}) = 90$ J $\Rightarrow$ $Q_{ACD} = 510 + 90 =$ **600 J**

Example 11.12

A gas in a diesel engine goes round the cycle ABCDA shown. T_A = 300 K and T_B = 660 K. (a) Find the temperatures at C and D. (b) Find the work done by the gas in process BC.



Solution

(a) $\frac{pV}{T}$ = constant. B → C at constant pressure: $T_C = 660 \times \frac{6.00}{1.40} =$ **2829 K**

D → A at constant volume: $T_D = 300 \times \frac{7.8}{1.0} =$ **2340 K**

(b) $W_{BC} = p_B(V_C - V_B) = (16.0 \times 10^5)(4.60 \times 10^{-4}) =$ **736 J**

Example 11.13

(a) Define (i) an adiabatic compression, (ii) a reversible process. [2] (b) 1 mol of an ideal monatomic gas at 650 K, pressure P_0 and volume V_0 expands at constant temperature to $2V_0$, then changes at constant volume back to pressure P_0, then is compressed at constant pressure back to its initial state. (i) Sketch the p–V graph. [4] (ii) Using the first law, show that the total heat is $Q = nRT_0 \ln 2 - P_0V_0$, and calculate it. [8] (iii) State whether heat is absorbed or released. [1]

Solution

(a)(i) A process in which the volume of the system decreases without any heat transfer into or out of the system. (ii) A process in which the variables that define the state of the system can be made to change so that they pass through the same values in reverse order when the process is reversed.

(b)(ii) A → B (constant temperature): $\Delta U = 0$, $Q_{AB} = nRT_0 \ln 2$

B → C (constant volume): pressure doubles from $P_0/2$ to P_0, so T rises from T_0 to $2T_0$: $Q_{BC} = n(\frac{3}{2}R)T_0$

C → A (constant pressure): T falls from $2T_0$ to T_0: $Q_{CA} = n(\frac{5}{2}R)(-T_0)$

Total: $Q = nRT_0 \ln 2 + \frac{3}{2}nRT_0 - \frac{5}{2}nRT_0 = nRT_0 \ln 2 - nRT_0 = \mathbf{nRT_0 \ln 2 - P_0V_0}$ (proved)

$Q = (1)(8.31)(650)(\ln 2 - 1) = \mathbf{-1.66 \times 10^3\ J}$

(iii) Q is negative: heat is **released** to the surroundings.

Test on thermodynamics

(a) State (i) a constant-pressure (isobaric) process, (ii) a constant-temperature (isothermal) process, (iii) an adiabatic process. [3] (b) State the first law of thermodynamics. [2] (c) 2 mol of helium at 27 °C occupy 0.02 m^3. The gas expands at constant pressure to twice its volume, then expands adiabatically until its temperature returns to its initial value. ($C_{V,m}$ = 12.47, $C_{p,m}$ = 20.78, R = 8.31 J mol^{-1} K^{-1}) (i) Sketch both processes on one p–V graph. (ii) Find the temperature at the end of the constant-pressure expansion. (iii) Find the total heat supplied. (iv) Find the total work done by the gas. [10]

Solution

(a)(i) A process that occurs at constant pressure. (ii) A process that occurs at constant temperature. (iii) A process that occurs without heat transfer into or out of the system.

(b) The heat supplied to a system equals the increase in its internal energy plus the work done by the system: $Q = \Delta U + W$.

(c)(ii) Charles's law: T_1 = 300.15 × 2 = **600.3 K**

(iii) $Q_{AB} = nC_{p,m}\Delta T$ = 2(20.78)(300.15) = 1.25 × 10^4 J; Q_{BC} = 0 (adiabatic). Total **1.25 × 10^4 J**

(iv) $W_{AB} = nR\Delta T$ = 2(8.31)(300.15) = 4.99 × 10^3 J; $W_{BC} = -\Delta U = nC_{V,m}(T_1 - T_0)$ = 2(12.47)(300.15) = 7.49 × 10^3 J. Total **1.25 × 10^4 J** (equal to Q, since the gas ends at its initial temperature, ΔU = 0)

Chapter 12 · Semester 1

HEAT TRANSFER

Notes by Cikgu Suhaili

4 worked examples 2 simulations Exercises with answers

Heat

Heat

is the **energy transferred from one body to another because of a temperature difference**.

- Heat always flows from a region of higher temperature to a region of lower temperature, until thermal equilibrium is reached.
- Heat is a scalar; unit joule (J).
- Heat is transferred by three mechanisms: **conduction**, **convection** and **radiation**.

12.1 Conduction

Conduction

is the process by which **heat is transferred through a substance from a region of high temperature to a region of lower temperature**, without any movement of the substance as a whole.

Mechanism of conduction

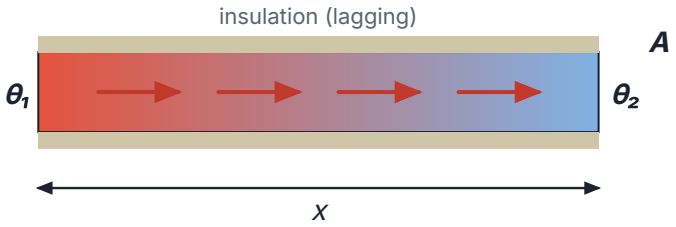
In a non-metal (insulator) rod

- At the hot end A, the molecules receive heat and vibrate faster, with larger amplitude.
- They collide with neighbouring, colder molecules and pass energy to them.
- The energy is passed along molecule by molecule (lattice vibrations) until the cold end B becomes hot.

In a metal rod

- A metal has many **free electrons**.
- At the heated end A the free electrons gain thermal energy and move faster.
- They travel quickly through the metal and collide with atoms in the colder regions, transferring energy to them (in addition to the lattice vibrations).
- Free electrons are light and fast, so this transfer is much quicker: **metals conduct heat much faster than insulators**.

Rate of heat flow



A lagged rod of cross-sectional area A and length x, ends at $\theta_1 > \theta_2$. No heat is lost from the sides.

Steady state

is reached when the **temperature at every point of the rod no longer changes with time**; the rate of heat flow is then the same through every cross-section.

The rate of heat flow through the rod depends on (a) the cross-sectional area A, (b) the temperature gradient, and (c) the material of the rod (e.g. copper or iron).

Temperature gradient

is the **change in temperature per unit length** along the direction of heat flow: $\frac{d\theta}{dx}$ (unit K m^{-1} or °C m^{-1}).

$$\frac{dQ}{dt} \propto A\frac{d\theta}{dx} \Rightarrow \frac{dQ}{dt} = -kA\frac{d\theta}{dx}$$

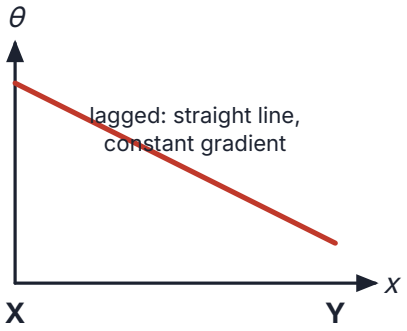
the minus sign: heat flows in the direction of decreasing temperature. Unit of the rate of heat flow: J s^{-1} = W

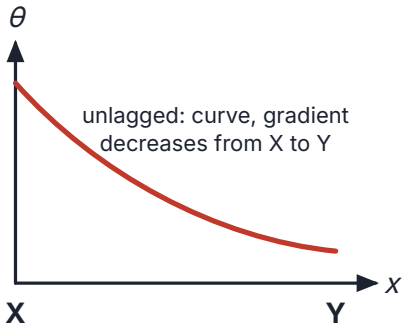
Thermal conductivity, k

is the **rate of heat flow per unit cross-sectional area per unit temperature gradient**, perpendicular to the area: $k = -\frac{dQ/dt}{A(d\theta/dx)}$. Unit W m^{-1} K^{-1}. Good conductors have large k.

Substance	k / W m^{-1} K^{-1}
Silver	406.0
Copper	385.0
Steel	50.2
Glass	0.8
Wood	0.08

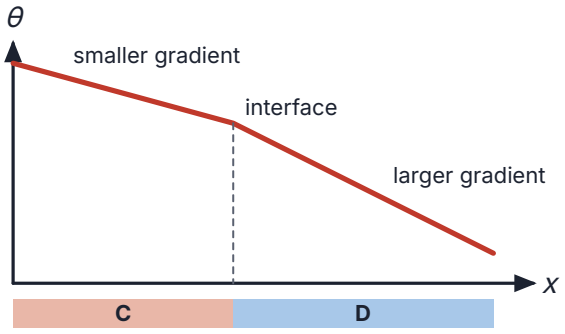
Insulated (lagged) and non-insulated (unlagged) rods





- **Unlagged** bar XY: heat is lost from the sides, so less heat reaches the far end. The rate of heat flow decreases along the bar and the temperature gradient decreases from X to Y: the θ–x graph is a **curve**.
- **Lagged** bar: no heat is lost from the sides. The rate of heat flow is the same everywhere and, with A and k constant, the temperature gradient is **constant**: the θ–x graph is a **straight line**, $\frac{d\theta}{dx} = \frac{\theta_2 - \theta_1}{x}$.

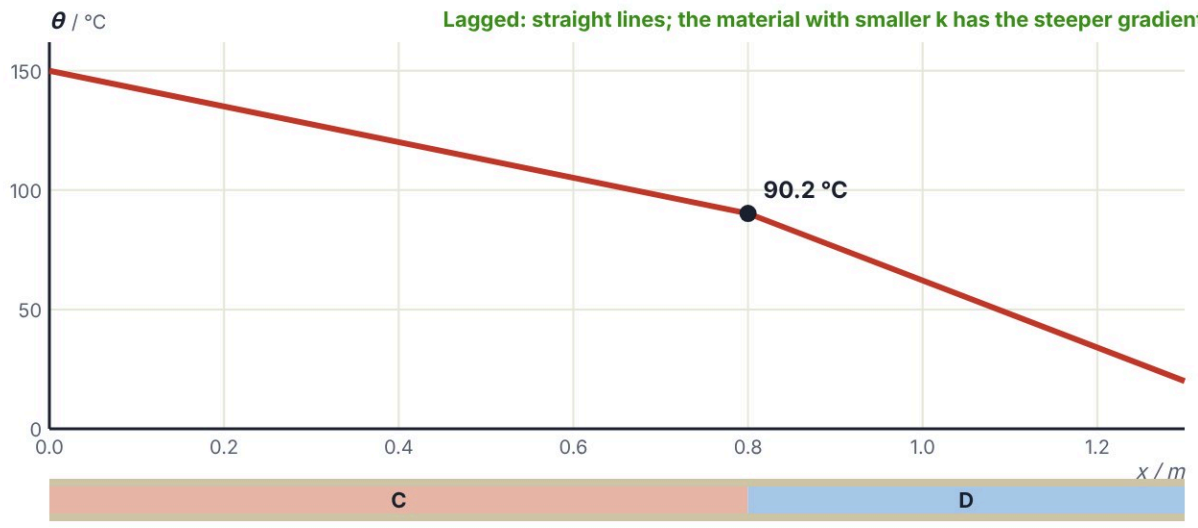
Heat conduction through layers of different materials



- In a lagged composite bar of materials C and D at steady state, the **rate of heat flow is the same through both**: $k_C A \frac{\Delta\theta_C}{x_C} = k_D A \frac{\Delta\theta_D}{x_D}$.
- With the same A: $k_C(d\theta/dx)_C = k_D(d\theta/dx)_D$, so the material with the **smaller k has the larger temperature gradient** (the steeper line).

Simulation — Conduction through a composite bar

Choose the two materials and their lengths, and set the end temperatures. The graph shows the steady-state temperature along the bar, the interface temperature and the rate of heat flow. Switch off the lagging to see the curve of an unlagged bar.



interface temperature	gradient in C	gradient in D	rate of heat flow (A = 1 cm²)
90.2 °C	**−74.7 °C/m**	**−140.5 °C/m**	**1.53 W**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 12 to use the simulation.

Example 12.1

A metal cube of side 8 cm has thermal conductivity 250 W m^{-1} K^{-1}. Two opposite faces are at 90 °C and 10 °C. Calculate (a) the temperature gradient in the cube, (b) the heat that flows through it in 10 minutes. (Steady flow, no heat lost to the surroundings.)

Solution

$A = (8 \times 10^{-2})^2 = 64 \times 10^{-4}$ m^2

(a) $\dfrac{d\theta}{dx} = \dfrac{10 - 90}{8 \times 10^{-2}} = \mathbf{-1.0 \times 10^3\ K\ m^{-1}}$

(b) $\dfrac{dQ}{dt} = -kA\dfrac{d\theta}{dx} = (250)(64 \times 10^{-4})(1.0 \times 10^3) = 1600$ W; $Q = 1600 \times 600 = \mathbf{9.6 \times 10^5\ J}$

Example 12.2

A 5 mm copper plate is sealed to a 10 mm aluminium plate; both have area 1 m^2. The outer face of the copper is at 100 °C and that of the aluminium at 80 °C. (a) Find the temperature of the copper–aluminium interface. (b) Calculate the rate of heat flow. (k_{Cu} = 400, k_{Al} = 200 W m^{-1} $°C^{-1}$)

Solution

(a) Same rate through both: $\frac{400(1)(100-\theta)}{5\times10^{-3}} = \frac{200(1)(\theta-80)}{10\times10^{-3}} \Rightarrow 4(100-\theta) = \theta - 80 \Rightarrow$ **θ = 96 °C**

(b) $\frac{dQ}{dt} = \frac{400(1)(100-96)}{5\times10^{-3}} =$ **3.2×10^5 W**

Exercises: conduction

1. A metal plate 5.0 cm thick has area 300 cm^2. One face is kept at 100 °C (steam) and the other at 30 °C (flowing water). The rate of heat flow is 9 kW. Find the thermal conductivity of the metal.

   Answer

   214 W m^{-1} K^{-1}

2. A 1.300 m rod is made of 0.800 m of aluminium joined end to end to 0.500 m of brass. The free end of the aluminium is at 150.0 °C and that of the brass at 20.0 °C; no heat is lost from the sides. Find the temperature of the junction. (k_{Al} = 205, k_{brass} = 109 W m^{-1} $°C^{-1}$)

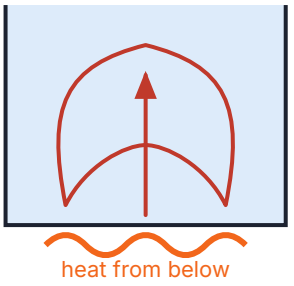
   Answer

   90.2 °C

12.2 Convection

Convection

is the process by which **heat is transferred from one part of a fluid to another by the movement of the fluid itself**. It occurs in liquids and gases.



- The fluid moves because its density changes when it is heated.
- In a pot of water heated from below, the heated water at the bottom expands, its density falls and it rises; cooler, denser water sinks to take its place. The water circulates in a **convection current**.

Natural (free) convection	Forced convection
The fluid moves because of density differences caused by heating.	The fluid is made to move by a pump or a fan.
e.g. warm and cold ocean currents, land and sea breezes, water heated in a pot	e.g. air heated and blown into a room by a fan; car radiator cooling

12.3 Radiation

Radiation

is the process by which **heat is transferred by electromagnetic waves** (infrared, visible light, ultraviolet). It can take place in a vacuum.

- All objects radiate energy continuously as electromagnetic waves produced by the thermal motion of their particles.
- The wavelengths emitted depend on the temperature: the lower the temperature, the longer the wavelength at which most energy is emitted.

Stefan–Boltzmann law

The **rate of energy radiated per unit surface area of a body is proportional to the fourth power of its absolute temperature**.

$$\frac{dQ}{dt} = P = e\sigma A T^4$$

$\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4} (Stefan–Boltzmann constant); A = surface area; T in kelvin; e = emissivity, $0 \le e \le 1$ (dimensionless)

- $e = 0$: ideal reflector; $e = 1$: ideal absorber (a black body). A good absorber is also a good emitter.
- An object at T_0 in surroundings at T_s also absorbs radiation from them. The **net** rate of energy radiated is $P_{net} = e\sigma A(T_0^4 - T_s^4)$.
- P_{net} positive: net heat flows out and the body cools. Negative: net heat flows in and the body warms.

Example 12.3

Tungsten has emissivity 0.35. A tungsten sphere of radius 3.00 cm hangs in a large evacuated enclosure whose walls are at 300 K. What power input keeps the sphere at 2500 K? (Ignore conduction along the supports; $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4})

Solution

$A = 4\pi r^2 = 4\pi(3.00 \times 10^{-2})^2 = 1.13 \times 10^{-2}$ m^2

$P_{net} = e\sigma A(T_0^4 - T_s^4) = (0.35)(5.67 \times 10^{-8})(1.13 \times 10^{-2})(2500^4 - 300^4) =$ **8.77 kW**

Exercise: radiation

1. (a) Find the power radiated by a tungsten sphere ($e = 0.35$) of radius 18.0 cm at 25 °C. (b) If the sphere is in a room whose walls are at −5 °C, find the net rate of energy flow out of the sphere.

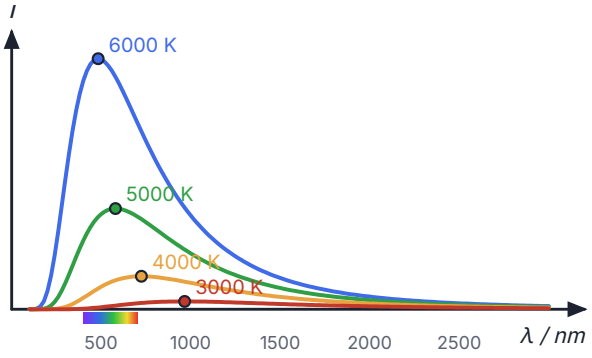
Answer

(a) 64 W (b) 22 W

Black body and its emission spectra

Black body

is an **ideal body that absorbs all the radiation incident on it** ($e = 1$); it is also the best possible emitter of thermal radiation at every wavelength.



Black-body emission spectra at four temperatures (same vertical scale).

- At a given temperature a black body emits a **continuous spectrum** over a range of wavelengths, with a peak at λ_{max}.
- As the temperature increases, the **total energy radiated** (the area under the curve) increases ($\propto T^4$).
- The **peak shifts to shorter wavelengths**, and the curve becomes taller and narrower.

Wien's displacement law

The **wavelength at which a black body emits most strongly is inversely proportional to its absolute temperature**.

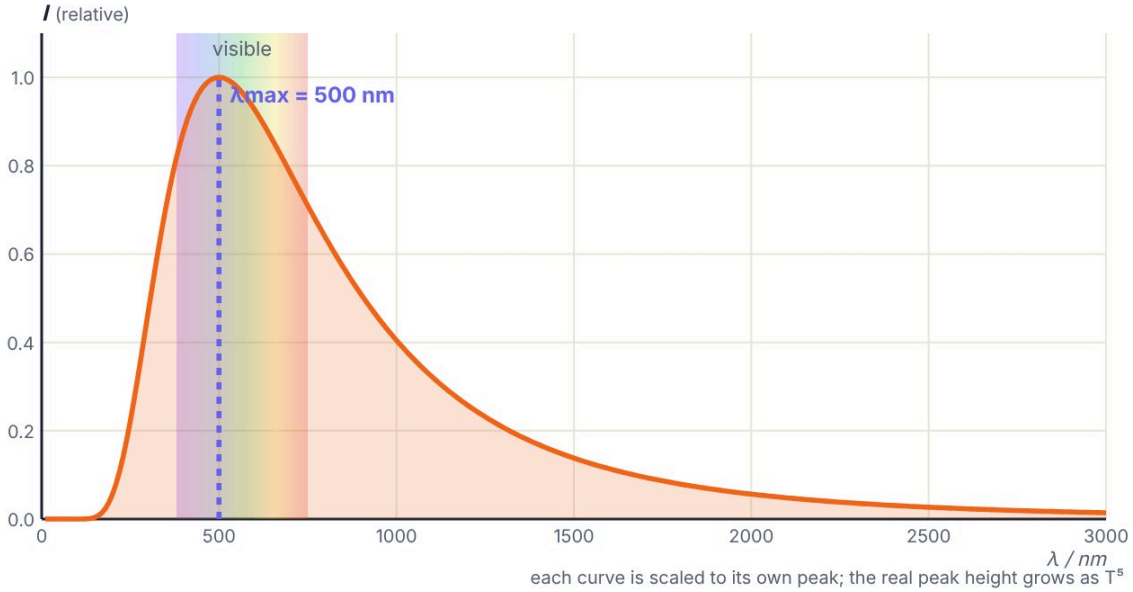
$$\lambda_{max} T = b$$

$b = 2.898 \times 10^{-3}$ m K (Wien's constant)

Simulation

Black-body spectrum and Wien's law

Change the temperature. The peak wavelength follows λmax = b/T and the total power grows as T^4. Compare a hot star, the Sun, a light bulb filament and a human body.



each curve is scaled to its own peak; the real peak height grows as T^5

temperature	λmax = b/T	power per m^2 σT^4	compared with 5800 K
5800 K	**500 nm (visible)**	**64.165 MW/m²**	**1.000 ×**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 12 to use the simulation.

Example · Wien's law

The Sun's surface is at about 5800 K. Estimate the wavelength at which it radiates most strongly.

Solution

$$\lambda_{max} = \frac{b}{T} = \frac{2.898 \times 10^{-3}}{5800} \approx \mathbf{5.0 \times 10^{-7}\ m = 500\ nm}\ \text{(visible light)}$$

12.4 Global warming

The greenhouse effect

- The Sun sends energy as light and heat. Some is reflected back into space; the rest passes through the atmosphere and warms the Earth.
- The Earth radiates heat back towards space as **infrared** radiation (longer wavelength, because the Earth is much cooler than the Sun).
- Some gases in the atmosphere (**greenhouse gases**) absorb this infrared radiation and re-emit part of it back towards the Earth, trapping heat like a blanket. This is the **natural greenhouse effect**; without it the Earth would be much colder and could not support life.

> **Global warming**
> is the increase in the average temperature of the Earth caused by the **enhanced greenhouse effect**: human activities increase the amount of greenhouse gases, so more heat is trapped and less escapes to space.

Natural greenhouse effect	Enhanced greenhouse effect
The atmosphere traps some of the heat radiated by the Earth.	More greenhouse gases trap more heat.
Some heat still escapes to space.	Less heat escapes to space.
The Earth stays warm enough for life.	The Earth heats up and the climate changes.

Greenhouse gases and human activity

- The atmosphere is 78% nitrogen and 21% oxygen, which do not trap heat. The greenhouse gases make up less than 1%: **carbon dioxide, methane, nitrous oxide, ozone, water vapour and halocarbons**.
- **Carbon dioxide**: burning fossil fuels (the largest source) and deforestation (fewer trees to absorb CO_2).
- **Methane**: livestock farming, flooded rice fields and decomposing rubbish in landfills; the amount has increased about two and a half times since the Industrial Revolution.

Ways to reduce global warming

- Switch to renewable energy: solar, wind and hydroelectric power instead of fossil fuels.
- Improve energy efficiency: energy-saving appliances, insulated buildings, less wasted electricity.
- Reduce, reuse and recycle: less waste in landfills, so less methane.
- Plant trees (reforestation and afforestation) to absorb carbon dioxide.
- Use public transport and electric vehicles to cut emissions.
- Support sustainable policies and international agreements to limit greenhouse gas emissions.